Sohag university
Faculty of medicine

General Physiology



CSF-103

2024/2025

1 Human Physiology

Human physiology is the study of the functioning of the normal body, and is responsible for describing *how* various systems of the human body work.

The fact that we remain alive is the result of complex control systems. Hunger makes us seek food, and fear makes us seek refuge. Sensations of cold make us look for warmth. Other forces cause us to seek fellowship and to reproduce.

1 Body fluids

Body is formed by solids and fluids. Fluid part is more than two third of the whole body. Water forms most of the fluid part of the body. Total quantity of body water in an average human being weighing about 70 kg is about 40 L.

In human beings, the total body water varies from 45% to 75% of body weight.

Significance of body fluids:

Body fluids are important in:

- Homeostasis.
- Transport mechanism.
- Metabolic reactions.
- Texture of tissues.
- Temperature regulation

Effect of different physiological factors on body fluid volume:

1- **Age**: in old age, water content is decreased due to increase in adipose tissue
2- **Sex**: in females, water is less because of more amount of subcutaneous adipose tissue (60% - 65% in a normal young adult male and 50% - 55% in a normal young adult female).
3- **Body mass index** (BMI): In thin persons, water content is more than that in obese persons.

Compartments of body fluids – distribution of body fluids:

Total water in the body is about 40 L. It is distributed into two major compartments:

I- Intracellular fluid (ICF):

Its volume is 22 L and it forms 55% of the total body water

II- Extracellular fluid (ECF):

Its volume is 18 L and it *forms* 45% of the total body water.

ECF is divided into 5 subunits:

1- Interstitial fluid (tissue fluids) and lymph (about 12 L, 20%).
2- Plasma (2.75L, 7.5%).
3- Fluid in bones (7.5%).
4- Fluid in dense connective tissues like cartilage (7.5%)
5- Transcellular fluid (2.5%) that includes:
   a- Cerebrospinal fluid.
   b- Intraocular fluid.
   c- Digestive juices.
   d- Serous fluid (intrapleural fluid, pericardial fluid and peritoneal fluid).
   e- Synovial fluid in joints.
   f- Fluid in urinary tract

Composition of body fluids

Body fluids contain water and solids. Solids are organic and inorganic substances.

- **Organic substances:**
  Glucose, amino acids and other proteins, fatty acids and other lipids, hormones and enzymes.
- **Inorganic substances:**
  Sodium, potassium, calcium, magnesium, chloride, bicarbonate, phosphate and

sulfate.

- ⇨ **ECF contains** large quantity of sodium, chloride, bicarbonate, glucose, fatty acids and oxygen.
- ⇨ **ICF contains** large quantities of potassium, magnesium, phosphates, sulfates and proteins.
- ⇨ The pH of ECF is 7.4. The pH of ICF is 7.0.

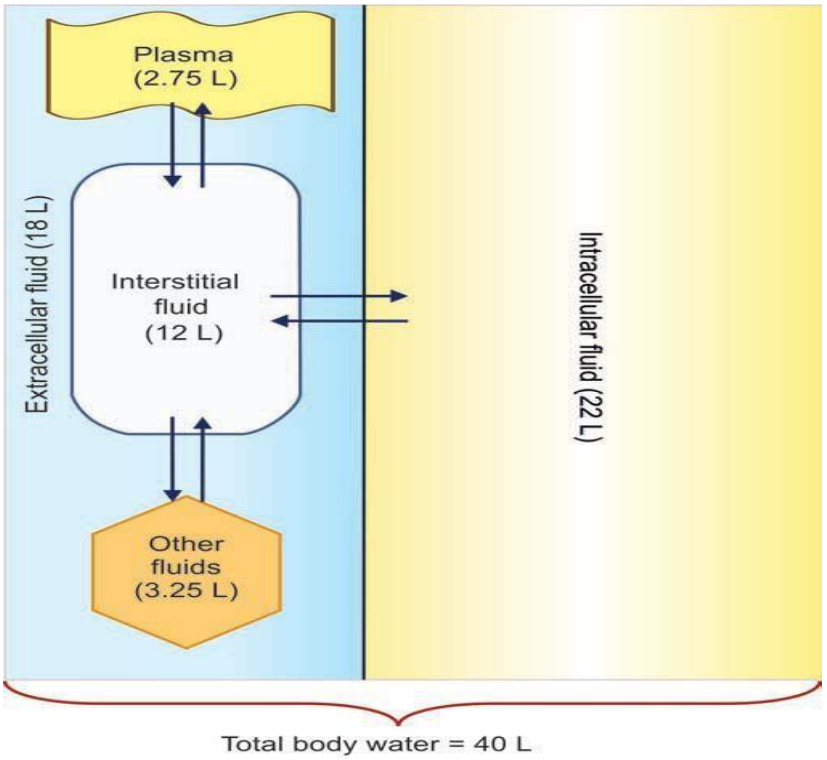


FIGURE 1: Body fluid compartments and movement of fluid between different compartments. Other fluids = Transcellular fluid, fluid in bones and fluid in connective

Concentration of body fluids:

Concentration of body fluids is expressed in three ways:

1. Osmolality.
2. Osmolarity.
3. Tonicity.

1- Osmolality:

- ➢ It is the concentration of osmotically active substance in the solution. Osmolality is expressed as the number of particles (osmoles) per kilogram of solution (osmoles/kg H_2O).

2- Osmolarity:

- ➢ It is the number of particles (osmoles) per liter of solution (osmoles/L).

Osmotic pressure in solutions depends upon osmolality.

Often, these two terms are used interchangeably. Change in osmolality of ECF affects the volume of both ECF and ICF. When osmolality of ECF increases, water moves from IC to ECF. When the osmolality decreases in ECF, water moves from ECF to ICF. Water movement continues until the osmolality of these two fluid compartments becomes equal.

3- Tonicity:

Usually, movement of water between the fluid compartments is not influenced by small molecules like urea and alcohol, which cross the cell membrane very rapidly. These small molecules are called ineffective osmoles. On the contrary, the larger molecules like sodium and glucose, which cross the cell membrane slowly, can influence the movement of water. Therefore, such molecules are called effective osmoles. Osmolality that causes the movement of water from one compartment to another is called effective osmolality and the effective osmoles are responsible for this.

Tonicity is the measure of effective osmolality. In terms of tonicity, the solutions are classified into three categories:

I. Isotonic fluid.
II. Hypertonic fluid
III. Hypotonic fluid

- Fluid which has the same effective osmolality (tonicity) as body fluids is called isotonic fluid. Examples are 0.9% sodium chloride solution (normal saline) and 5% glucose solution.
- Red blood cells or other cells placed in isotonic fluid (normal saline) neither gain nor lose water by osmosis (Fig.2). This is because of the **osmotic equilibrium** between inside and outside the cell across the cell membrane.

2- Hypertonic Fluid

- Fluid which has greater effective osmolality than the body fluids is called hypertonic fluid. Example is 2% sodium chloride solution.
- When red blood cells or other cells are placed in hypertonic fluid, water moves out of the cells (exosmosis) resulting in shrinkage of the cells. (Fig.2)

3- Hypotonic Fluid

- Fluid which has less effective osmolality than the body fluids is called hypotonic fluid. Example is 0.3% sodium chloride solution.
- When red blood cells or other cells are placed in hypotonic fluid, water moves into the cells (endosmosis) and causes swelling of the cells (Fig.2). Now the red blood cells become globular (sphereocytic) and get ruptured (hemolysis).

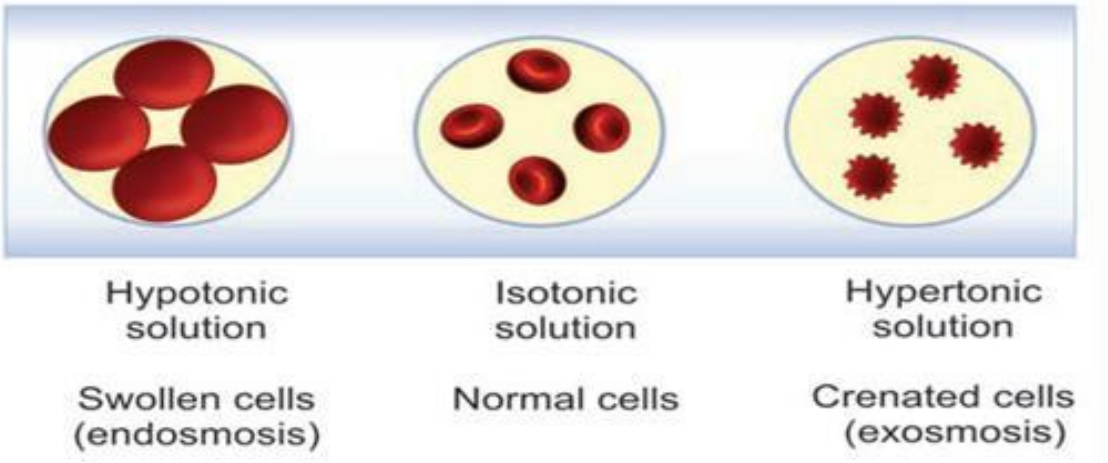


FIGURE 2: Effect of isotonic, hypertonic and hypotonic solutions on red blood cells

Maintenance of water balance:

Body has several mechanisms which work together to maintain the water balance. The important mechanisms involve hypothalamus and kidneys.

Common intravenous fluids used in practical:

1- 0.9% Normal Saline (also known as NS, 0.9NaCl):

This is one of the most common IV fluids because it works for most hydration needs due to vomiting, diarrhea, hemorrhage, or even shock. It is an isotonic crystalloid that contains 0.9% sodium chloride (salt) that is dissolved in sterile water.

2- Lactated Ringers (also known as LR, Ringers Lactate, or RL)

This solution is an isotonic crystalloid that contains sodium chloride, potassium chloride,

calcium chloride, and sodium lactate in sterile water. It is the most similar to the body's plasma and serum concentration, and is especially great for burn victims or those with hypovolemia due to fluid shifts.

3- 5% Dextrose in Water (also known as D5 or D5W)

Is an isotonic carbohydrate (sugar) solution that contains glucose (sugar) as the solute. When this is absorbed, the glucose is usually quickly grabbed up by cells and utilized for energy, leaving only water which is then a hypotonic solution.

4- 0.45% Normal Saline (also known as Half Normal Saline, 0.45NaCl)

This is a hypotonic crystalloid solution of sodium chloride dissolved in sterile water (unlike 0.9% Normal Saline which is isotonic, as mentioned above). This fluid is designed to treat cellular dehydration. Infusing half normal saline, causes water to shift from the extracellular fluid compartment to the ICF compartment.



2 Tissue fluids

Definition:

Tissue fluid is the medium in which cells are bathed. It is otherwise known as **interstitial fluid.** It forms about 20% of extracellular fluid (ECF).

Functions of tissue fluid:

Tissue fluid acts as a medium for exchange of various substances between the

cells and capillaries. Oxygen and nutritive substances diffuse from the arterial end of capillary through the tissue fluid and reach the cells. Carbon dioxide and waste materials diffuse from the cells into the venous end of capillary through this fluid.

Formation of tissue fluid:

Formation of tissue fluid involves two processes:

1. Filtration.
2. Reabsorption.

1- Filtration:

Tissue fluid is formed by the process of filtration. Normally, the blood pressure (also called **hydrostatic pressure**) in arterial end of the capillary is about 30 mm Hg. This hydrostatic pressure is the driving force for filtration of water and other substances from blood into tissue spaces. Along the course of the capillary, the pressure falls gradually and it is about 15 mm Hg at the venous end.

Capillary membrane is not permeable to the large molecules, particularly the plasma proteins. So, these proteins remain in the blood and exert a pressure called **oncotic pressure** or **colloidal osmotic pressure.** It is about 25 mm Hg.

Osmotic pressure is an opposing force for the filtration of water and other materials from capillary blood into the tissue space. However, the hydrostatic pressure in the arterial end of the capillary (30 mm Hg) is greater than the osmotic pressure. And, the net filtration pressure of 5 mm Hg is responsible for continuous filtration (Fig. 3).

2- Reabsorption:

Fluid filtered at the arterial end of capillaries is reabsorbed back into the blood at the venous end of capillaries. Here also, the pressure gradient plays an important role. At the venous end of capillaries, the hydrostatic pressure is less (15 mm Hg) and the oncotic pressure is more (25 mm Hg). Due to the pressure gradient of 10 mm Hg, the fluid is reabsorbed along with waste materials from the tissue fluid into the capillaries. (Figure 3)

Thus, the process of filtration at the arterial end of the capillaries helps in the formation of tissue fluids and the process of reabsorption at the venous end helps to maintain the volume of tissue fluid.

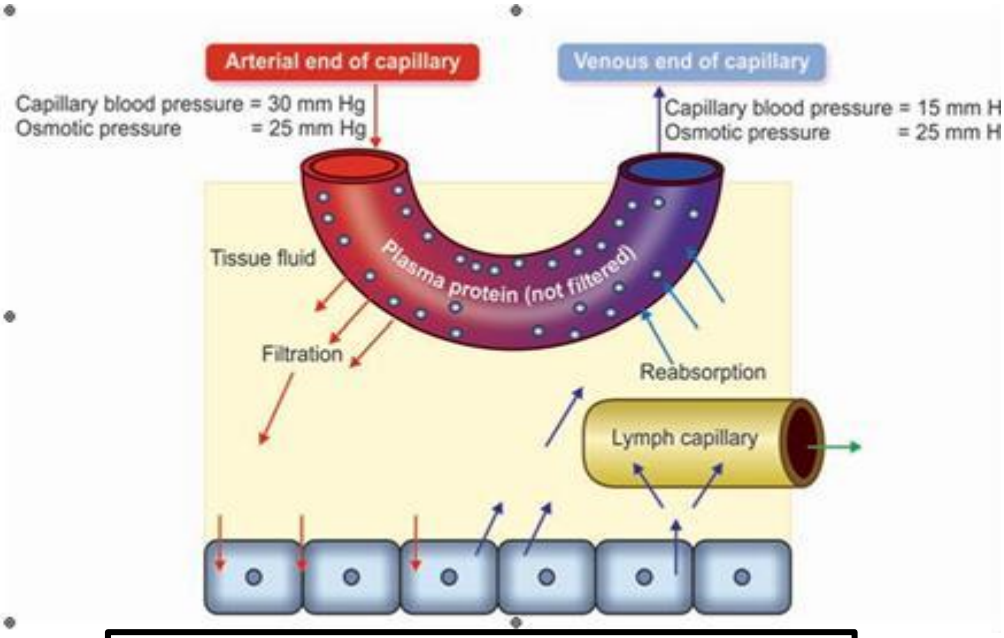


Figure 3 Formation of tissue fluid. Plasma proteins remain inside the blood capillary as the capillary membrane is not permeable to plasma

3 Edema

Definition:

Edema is defined as the swelling caused by excessive accumulation of fluid in the tissues. It may be generalized or local. Edema that involves the entire body is called generalized edema. Local edema is the one that occurs in specific areas of the body such as abdomen, lungs and extremities like feet, ankles and legs. Accumulation of fluid may be inside or outside the cell.

Types of edema:

Edema is classified into two types, depending upon the body fluid compartment where accumulation of excess fluid occurs:

1. Intracellular edema.
2. Extracellular edema.

Intracellular edema:

Intracellular edema is the accumulation of fluid inside the cell. It occurs because of three reasons:

1. Malnutrition.
2. Poor metabolism.
3. Inflammation of the tissues.

Extracellular edema:

Extracellular edema is defined as the accumulation of fluid outside the cell.

Causes for extracellular edema:

1. Abnormal leakage of fluid from capillaries into interstitial space.
2. Obstruction of lymphatic vessels that prevents fluid return from interstitium to blood.

Conditions which lead to extracellular edema:

1. Heart failure.
2. Renal disease.
3. Decreased amount of plasma proteins.
4. Lymphatic obstruction.
5. Increased endothelial permeability.

Pitting and non-pitting edema:

Interstitial fluid is present in the form of a gel that is almost like a semisolid substance. It is because the interstitial fluid is not present as fluid but is bound in a proteoglycan meshwork. However, in abnormal conditions, when the fluid increases most of the fluid becomes free fluid that is not bound to proteoglycan meshwork. It flows freely through tissue spaces, producing a swelling called edema. This type of edema is known as pitting edema because, when this area is pressed with the finger, displacement of fluid occurs producing a depression or pit. When the finger is removed, the pit remains for few seconds, sometimes as long as one minute, till the fluid flows back into that area.

Edema also develops due to swelling of the cells or clotting of interstitial fluid in the presence of fibrinogen. This is called non-pitting edema because, it is hard and a pit is not formed by pressing.

4 Dehydration

Definition:

Dehydration is defined as excessive loss of water from the body. Body requires certain amount of fluid intake daily for normal functions. Minimum daily requirement of water intake is about 1 L. This varies with the age and activity of the individual. The most active individuals need 2 to 3 L of water intake daily. Dehydration occurs when fluid loss is more than what is consumed.

2 Homeostasis

ILOs:

A- Knowledge and understanding:

Define homeostasis, and homeostatic control systems.

B- Intellectual skills

Explain the consequences of the failure to maintain homeostasis and give examples.

Definition:

'Homeostasis' refers to the maintenance of constant internal environment of the body (homeo = same; stasis = standing) (Fig. 1).

Internal environment in the body is the extracellular fluid (ECF) in which the cells live. It is the fluid outside the cell and it constantly moves throughout the body. ECF contains nutrients, ions and all other substances necessary for the survival of the cells.

Normal healthy living of large organisms including human beings depends upon the constant maintenance of internal environment within the physiological limits. If the internal environment deviates beyond the set limits, body suffers from malfunction or dysfunction.

For the functioning of homeostatic mechanism, the body must recognize the deviation of any physiological activity from the normal limits. Fortunately, body is provided with appropriate detectors or sensors, which recognize the deviation. These

detectors sense the deviation and alert the integrating center. The integrating center immediately sends information to the concerned effectors to either accelerate or inhibit the activity so that the normalcy is restored.

Factors homeostatically regulated

- Concentration of nutrients.
- Concentration of O2 and CO2.
- Concentration of waste products.
- pH.
- Concentration of water, salt, and other electrolytes.
- Volume and pressure.
- Temperature

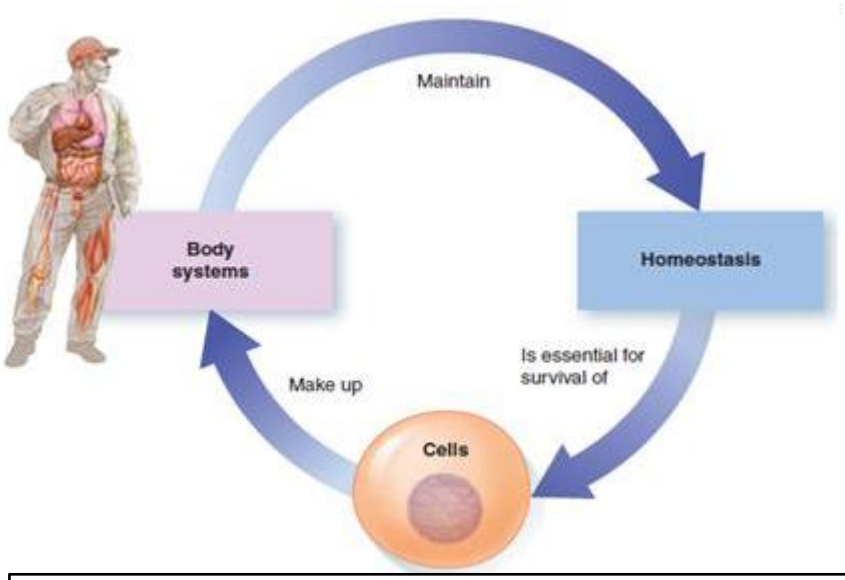


Fig. 1 Interdependent relationship of cells, body systems, and homeostasis.

Role of various systems of the body in homeostasis

One or more systems are involved in homeostatic mechanism of each function. Some of the functions in which the homeostatic mechanism is well established are given below:

1. **The pH of the ECF** has to be maintained at the critical value of 7.4. The tissues cannot survive if it is altered. Thus, the decrease in pH (acidosis) or increase in pH

(alkalosis) affects the tissues markedly. The respiratory system, blood and kidney help in the regulation of pH.

2. **Body temperature** must be maintained at 37.5°C. Increase or decrease in temperature alters the metabolic activities of the cells. The skin, respiratory system, digestive system, excretory system, skeletal muscles and nervous system are involved in maintaining the temperature within normal limits.

3. **Adequate amount of nutrients must be supplied to the cells**. Nutrients are essential for various activities of the cell and growth of the tissues. These substances also form the source of energy required for various activities of the cells. Nutrients must be digested, absorbed into the blood and supplied to the cells. Digestive system and circulatory system play major roles in the supply of nutrients.

4. **Adequate amount of oxygen should be made available to the cells** for the metabolism of the nutrients. Simultaneously, the carbon dioxide and other metabolic end products must be removed. Respiratory system is concerned with the supply of oxygen and removal of carbon dioxide. Kidneys and other excretory organs are involved in the excretion of waste products.

5. **Many hormones are essential for the metabolism of nutrients and other substances necessary for the cells**. Hormones are to be synthesized and released from the endocrine glands in appropriate quantities and these hormones must act on the body cells appropriately. Otherwise, it leads to abnormal signs and symptoms.

6. **Water and electrolyte balance should be maintained optimally**. Otherwise it leads to dehydration or water toxicity and alteration in the osmolality of the body fluids. Kidneys, skin, salivary glands and gastrointestinal tract take care of this.

7. **For all these functions**, the blood, which forms the major part of internal environment, must be normal. It should contain required number of normal red blood cells and adequate amount of plasma with normal composition. Only then, it can transport the nutritive substances, respiratory gases, metabolic and other waste products.

8. **Skeletal muscles** are also involved in homeostasis. This system helps the organism to move around in search of food. It also helps to protect the organism from adverse surroundings, thus preventing damage or destruction.

9. **Central nervous system**, which includes brain and spinal cord also, plays an important role in homeostasis. Sensory system detects the state of the body or surroundings. Brain integrates and interprets these informations and commands the body to act accordingly through motor system so that, the body can avoid the damage.

10. **Autonomic nervous system** regulates all the vegetative functions of the body essential for homeostasis.

Components of homeostatic system

Homeostatic system in the body acts through self- regulating devices, which operate in a cyclic manner (Fig.2). This cycle includes four components:

- ***Sensors*** or ***detectors***, which recognize the deviation.
- Transmission of this message to a ***control center***.
- Transmission of information from the control center to the effectors for correcting the deviation.

Transmission of the message or information may be an **electrical** process in the form of impulses through nerves or a **chemical** process mainly in the form of hormones through blood and body fluids

- ***Effectors***, which correct the deviation.

Mechanism of action of homeostatic system

Homeostatic mechanism in the body is responsible for maintaining the normalcy of various body systems. Whenever there is any change in behavioral pattern of any system, the effectors bring back the normalcy either by inhibiting and reversing the

change or by supporting and accelerating the change depending upon requirement of the situation. This is achieved by means of feedback signals

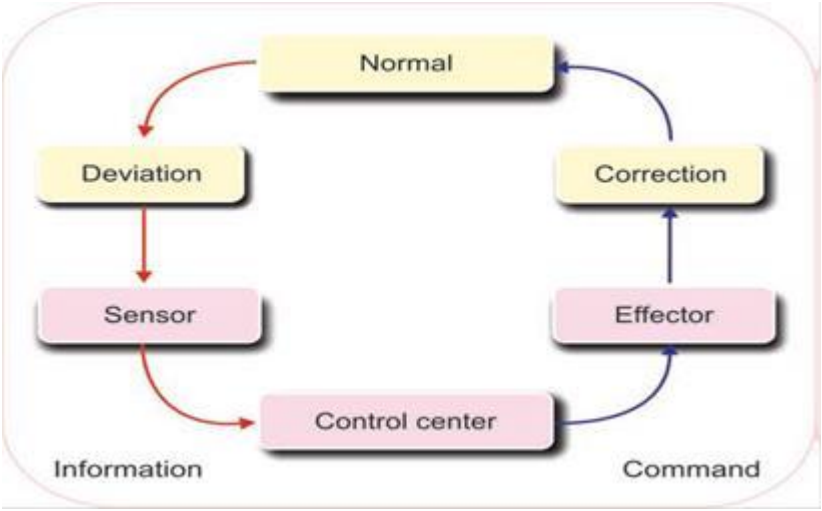


FIGURE 2: Components of homeostatic system

Whenever any change occurs, system receives and reacts to two types of feedback:

1. Negative feedback.
2. Positive feedback

1. Negative feedback.

- ⇨ Homeostatic control mechanisms operate primarily on the principle of negative feedback.

- ⇨ In negative feedback, the system reacts to arrest the change or reverse the direction of change.

- ⇨ For example, when temperature monitoring nerve cells detect a decrease in body temperature below the desired level, these sensors signal the temperature control center in the brain, which begins a sequence of events that ends in responses, such as shivering, that generate heat and raise the temperature to the proper level (Fig.3). When the body temperature reaches the set point, the temperature

monitoring nerve cells turn off the stimulatory signal to the skeletal muscles. As a result, the body temperature does not continue to increase above the set point. Conversely, when the temperature-monitoring nerve cells detect a rise in body temperature above normal, cooling mechanisms such as sweating are called into play to lower the temperature to normal. When the temperature reaches the set point, the cooling mechanisms are shut off.

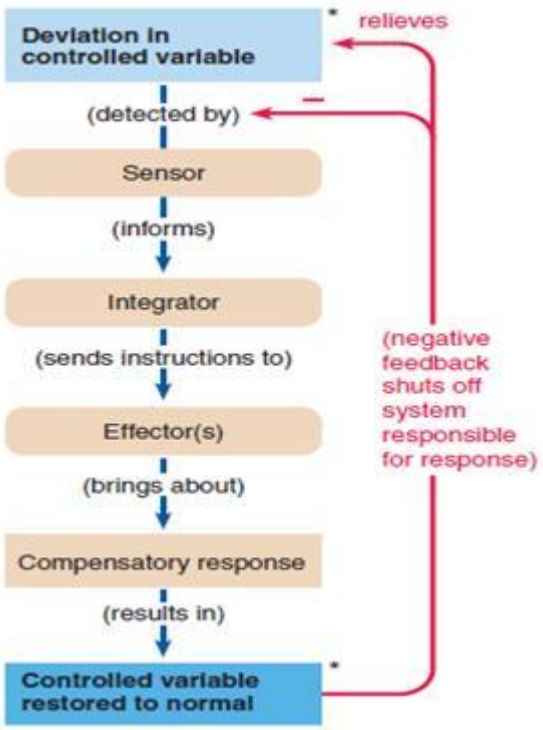


Fig. Components of negative feedback control system

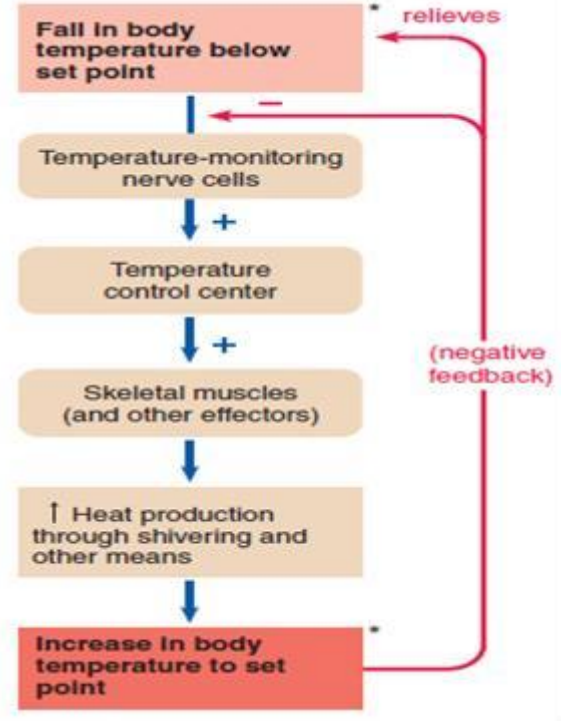


Fig. 3: Negative feedback control of body temperature

2. Positive feedback.

⇨ In positive feedback the system reacts to increase the intensity of the change in the same direction. Positive feedback is less common than the negative feedback. However, it has its own significance particularly during emergency conditions.

⇨ One of the positive feedbacks occurs during the blood clotting. Blood clotting is necessary to arrest bleeding during injury.

⇨ Other process where positive feedback occurs is as in birth of a baby.

- The hormone oxytocin causes powerful contractions of the uterus. As the contractions push the baby against the cervix (the exit from the uterus), the resultant stretching of the cervix triggers a sequence of events that brings about the release of even more oxytocin, which causes even stronger uterine contractions

NOTE

Disruption in homeostasis can lead to illness and death:

When one or more of the body systems malfunction, homeostasis is disrupted and all cells suffer because they no longer have optimal environment in which to live and function.

3 Functions of various cell organelles and cell membrane

ILOs:

Describe the functions of various cell organelles and characteristics and functions of cell membrane.

1 The Cell

All the living things are composed of cells. A single cell is the smallest unit that has all the characteristics of life. Cell is defined as the structural and functional unit of the living body.

General Characteristics of Cell:

Each cell in the body:

1. Needs nutrition and oxygen.
2. Produces its own energy necessary for its growth, repair and other activities.
3. Eliminates carbon dioxide and other metabolic wastes.
4. Maintains the medium, i.e. the environment for its survival.
5. Shows immediate response to the entry of invaders like bacteria or toxic

substances into the body.

6. Reproduces by division. There are some exceptions like neuron, which do not reproduce.

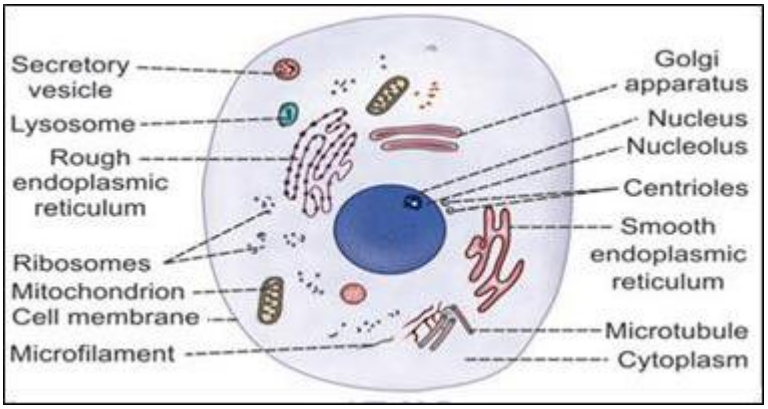


Fig. 1: structure of the cell

Structure of the cell (Fig. 1)

1. Cell membrane.
2. Cytoplasm.
3. Nucleus.

Cell membrane

Cell membrane is a protective sheath, enveloping the cell body. It is also known as plasma membrane or plasmalemma. This membrane separates the fluid outside the cell called extracellular fluid (ECF) and the fluid inside the cell called intracellular fluid (ICF). The cell membrane is a semipermeable membrane. So, there is free exchange of certain substances between ECF and ICF. Thickness of the cell membrane varies from 75 to 111Å.(figure 2)

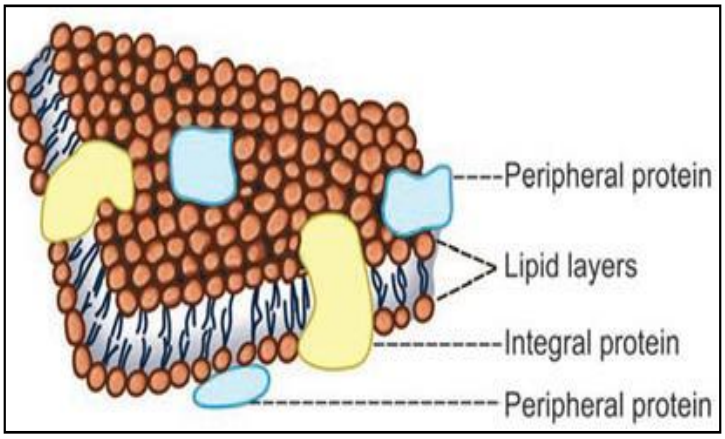


Fig. 2: Diagram of the cell membrane

Composition of cell membrane (sandwich of lipids' covered by proteins on both sides).
Cell membrane is composed of three types of substances:

1. Proteins (55%).
2. Lipids (40%).
3. Carbohydrates (5%).

Lipid Layers of the Cell Membrane (Fig. 3)

The central lipid layer is a bilayered structure. It is fluid in nature and not a solid structure. So, the portions of the membrane move from one point to another point along the surface of the cell.

Major lipids in cell membrane are:

1. Phospholipids.
2. Cholesterol

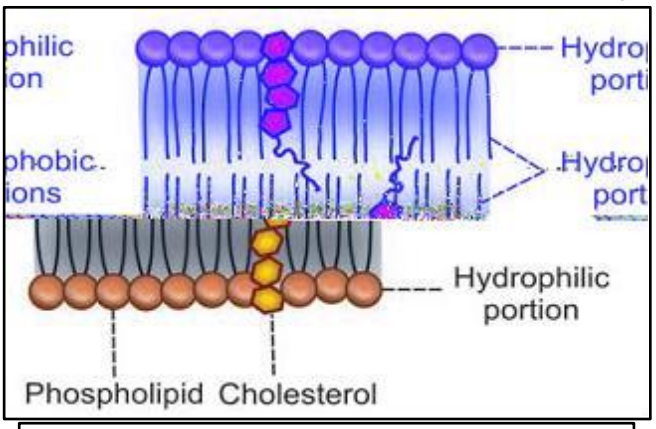


Fig. 3: Lipids of the cell membrane

1. Phospholipids:

Phospholipid molecules are arranged in two layers (Fig. 3). Each phospholipid molecule resembles the headed pin in shape. The outer part of the phospholipid molecule is called the head portion and the inner portion is called the tail portion.

Head portion is the polar end and it is soluble in water and has strong affinity for water (hydrophilic). Tail portion is the non-polar end. It is insoluble in water and repelled by water (hydrophobic).

Two layers of phospholipids are arranged in such a way that the hydrophobic tail portions meet in the center of the membrane. Hydrophilic head portions of outer layer face the ECF and those of the inner layer face ICF (cytoplasm).

2. Cholesterol:

Cholesterol molecules are arranged in between the phospholipid molecules. Phospholipids are soft and oily structures and cholesterol helps to 'pack' the phospholipids in the membrane. So, cholesterol is responsible for the structural integrity of lipid layer of the cell membrane.

Functions of Lipid Layer in Cell Membrane:

Lipid layer of the cell membrane is a semipermeable membrane and allows only the fat-soluble substances to pass through it. Thus, the fat-soluble substances like oxygen, carbon dioxide and alcohol can pass through this lipid layer. The water-soluble substances such as glucose, urea and electrolytes cannot pass through this layer.

Protein Layers of the Cell Membrane:

These layers cover the two surfaces of the central lipid layer. Protein layers give protection to the central lipid layer. The protein substances present in these layers are mostly glycoproteins.

Protein molecules are classified into two categories:

1. Integral proteins or transmembrane proteins.
2. Peripheral proteins or peripheral membrane proteins.

1. Integral proteins:

Integral or transmembrane proteins are the proteins that pass through entire thickness of cell membrane from one side to the other side. These proteins are tightly bound with the cell membrane.

Examples of integral protein:

a) Cell adhesion proteins.
b) Cell junction proteins.
c) Some carrier (transport) proteins.
d) Channel proteins.
e) Some hormone receptors.
f) Antigens.
g) Some enzymes.

2. Peripheral proteins:

Peripheral proteins or peripheral membrane proteins are the proteins which are partially embedded in the outer and inner surfaces of the cell membrane and do not penetrate the cell membrane. Peripheral proteins are loosely bound with integral proteins or lipid layer of cell membrane. So, these protein molecules dissociate readily from the cell membrane.

Examples of peripheral proteins:

a) Proteins of cytoskeleton.
b) Some carrier (transport) proteins.
c) Some enzymes.

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❒ **Examples:-**

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- Some enzymes.

Functions of proteins in cell membrane

- Integral proteins provide the structural integrity of the cell membrane.
- Channel proteins help in the diffusion of water- soluble substances like glucose and electrolytes.
- Carrier or transport proteins help in the transport of substances across the cell membrane by means of active or passive transport.
- Pump: Some carrier proteins act as pumps, by which ions are transported actively across the cell membrane.
- Receptor proteins serve as the receptor sites for hormones and neurotransmitters.
- Enzymes: Some of the protein molecules form enzymes and control chemical (metabolic) reactions within the cell membrane.
- Antigens: Some proteins act as antigens and induce the process of antibody formation.
- Cell adhesion molecules or the integral proteins are responsible for attachment of cells to their neighbors or to basal lamina.

Carbohydrates of the cell membrane

Some of the carbohydrate molecules present in cell membrane are attached to proteins and form glycoproteins (proteoglycans). Some carbohydrate molecules are attached to lipids and form glycolipids.
Carbohydrate molecules form a thin and loose covering over the entire surface of the cell membrane called glycocalyx.

Functions of carbohydrates in cell membrane

1. Carbohydrate molecules are negatively charged and do not permit the negatively charged substances to move in and out of the cell.
2. Glycocalyx from the neighboring cells helps in the tight fixation of cells with one another.
3. Some carbohydrate molecules function as the receptors for some hormones.

Functions of cell membrane

1. Protective function: Cell membrane protects the cytoplasm, and the organelles present in the cytoplasm.
2. Selective permeability: Cell membrane acts as a semi-permeable membrane, which allows only some substances to pass through it and acts as a barrier for other substances.
3. Absorptive function: Nutrients are absorbed into the cell through the cell membrane.
4. Excretory function: Metabolites and other waste products from the cell are excreted out through the cell membrane.
5. Exchange of gases: Oxygen enters the cell from the blood and carbon dioxide leaves the cell and enters the blood through the cell membrane.
6. Maintenance of shape and size of the cell: Cell membrane is responsible for the maintenance of shape and size of the cell.

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2) Cytoplasm

Cytoplasm of the cell is the jellylike material formed by 80% of water. It contains a clear liquid portion called cytosol and many organelles with distinct structure and function.

Organelles in cytoplasm:

Cytoplasmic organelles are the cellular structures embedded in the cytoplasm. Each organelle has a definite structure and specific functions

❒ Cell organells:-

- Endoplasmic reticulum.
- Golgi apparatus.
- Lysosome.
- Peroxisome.
- Centrosome and centrioles.
- Secretory vesicles.
- Mitochondria.
- Nucleus.
- Ribosomes.
- Cytoskeleton.

1. Rough endoplasmic reticulum

1. Synthesis of proteins
2. Degradation of worn-out organelles

2. Golgi apparatus

1. Processing, packaging, labeling and delivery of proteins and lipids

3. Smooth endoplasmic reticulum

1. Synthesis of lipids and steroids
2. Role in cellular metabolism
3. Storage and metabolism of calcium
4. Catabolism and detoxification of toxic substances

4. Lysosomes

1. Degradation of macromolecules
2. Degradation of worn-out organelles
3. Removal of excess of secretory products
4. Secretion of perforin, granzymes, melanin and serotonin

5. Centrosome

1. Movement of chromosomes during cell division

6. Peroxisomes

1. Breakdown of excess fatty acids
2. Detoxification of hydrogen peroxide and other metabolic products
3. Oxygen utilization
4. Acceleration of gluconeogenesis
5. Degradation of purine to uric acid
6. Role in the formation of myelin
7. Role in the formation of bile acids

7. Mitochondria

1. Production of energy
2. Synthesis of ATP
3. Initiation of apoptosis

8. Ribosomes

1. Synthesis of proteins

9. Cytoskeleton

1. Determination of shape of the cell
2. Stability of cell shape
3. Cellular movements

10. Nucleus

1- Control of all activities of the cell
2- Synthesis of RNA
3- Sending genetic instruction to cytoplasm for protein synthesis
4- Formation of subunits of ribosomes
5- Control of cell division
6- Storage of hereditary information in genes (DNA)

4 Transport across cell membrane

Materials (Fluid, electrolytes, nutrients or Gases etc) pass through the cell membrane from inside of the cell to outside of the cell and vise verse i.e. the opposite is also true.

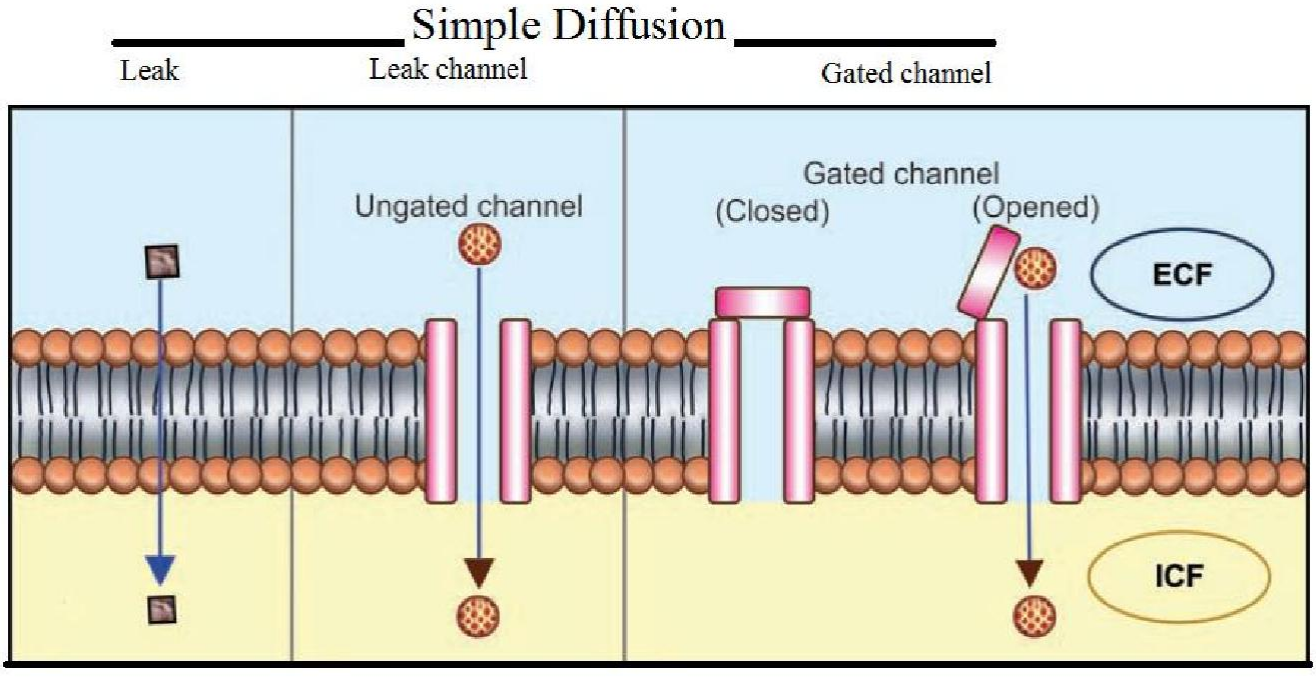
❒ Transport process classification:-

According to method of transport

- Passage through lipid bilayer of cell membrane.
- Passage through proteins of cell membrane.
- Vesicular transport (see later),

According to ATP usage

- Passive transport = No, ATP was used
- Active transport = ATP was used



Passage through lipid bilayer (Leak).

- This method is suitable for;
  - Lipid soluble substances, such as gases and fatty acids.
  - Many small-uncharged molecules.

Passage through (channels or carrier) proteins.

- This method is suitable for;
  - Water.
  - water soluble substances e.g. glucose and amino acids
  - Charged particles e.g. electrolytes Na, K etc.
  - Molecules with large molecular weight.

This method is divided to;

A- Substances pass through **protein channel** which penetrate lipid belayed from side to side.

B- Substances combine with **carrier protein** located in the cell membrane.

Protein channels

- They are tubular protein structures that connect the exterior with the interior of cells.
- They are integral proteins
- The permeability of channels depends on molecular size, shape and charge of the substance.

Water channels	Ion channels
• they are designed for Water passage across cell membrane. They are called **aquaporins.**	• They are designed for ion passage across cell membrane.

Ion channels

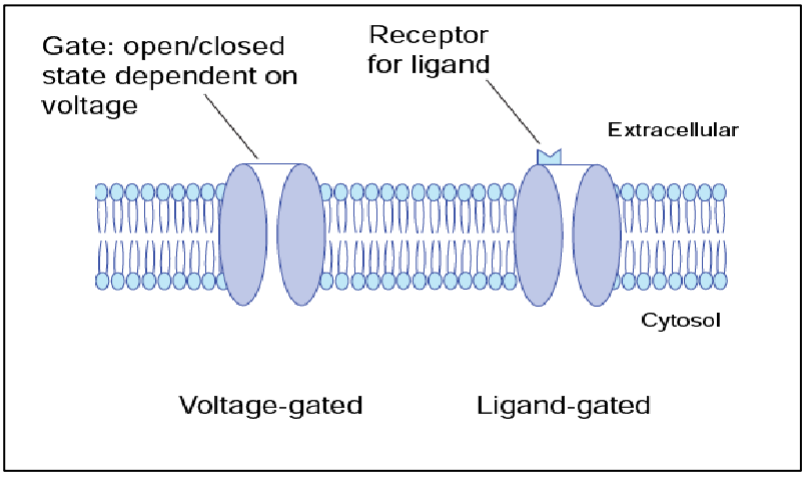
Mechanism of action of ion channels: along the length of the integral protein, a pore (opening) and a lumen are present, through them the Ions pass from one side of the membrane to the other. Thus, Ions cross the membrane without entering the lipid bilayer of the membrane.

❒ Ion channels classification:-

According to selectivity	According to gating mechanism
• **Selective:** allow passage of specific type of ions e.g. Na only or K only. • **Non selective:** allow passage of multiple types of ions.	• **Non gated:** (leak channel), always • **Gated channel:** (it is closed by a gate that open by specific method). • **Mechanisms of gating are:** 1. Voltage-gating. 2. Ligand-gating. 3. Mechanical-gating.

Voltage-gating (electrical gating):

Change in the membrane potential (voltage difference across cell membrane) beyond a certain value opens or closes the gate of the ion channels e.g. ion channels in nerve , muscle and heart are mainly voltage-gated channels like Na+ voltage gated channel and , K^+ voltage gated channel .

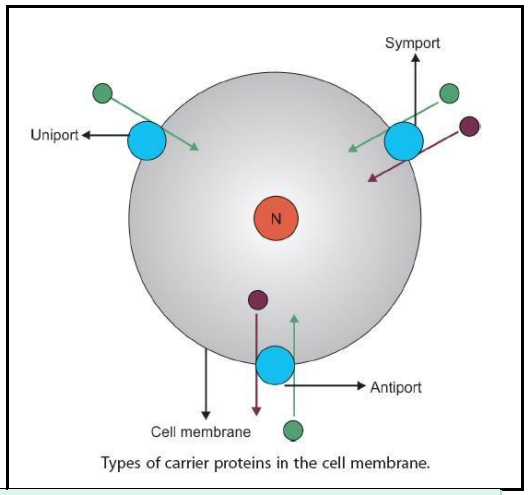


Ligand Gating (Chemical gating):

Binding of the channel with a ligand (Chemical substance) causes opening or closure of the channel. Please, take care that the chemical substance (the key) is different from the passenger (the ion).

Carrier protein

They are protein structures inserted in the cell membrane, has no lumen but can change their orientation so that they can move a substance across cell membrane. They are **integral** proteins.



Types of carrier proteins in the cell membrane.

❒ Ion channels classification:-

Uniport: A carrier protein that transports only one substance

Symport (co-transport)	Antiport (counter-transport)
• A carrier protein that transport two or more substances from one side of the membrane to the other in the same direction.	• A carrier protein that transport substances in opposite directions in which one substance is transported to the inside of the cell and other substance from inside the cell to the outside.

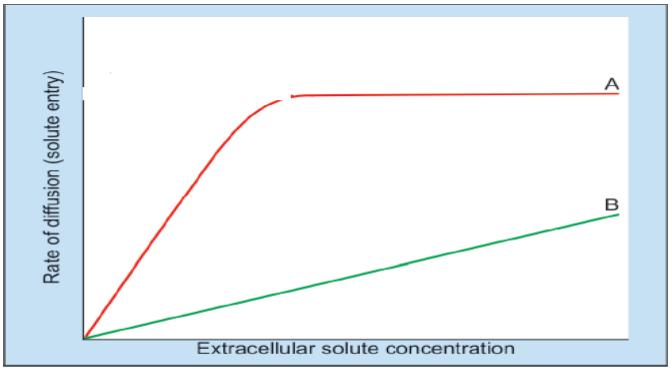
Characteristics of carrier protein

- **Competitive inhibition:** Presence of different molecules (x and y) with **similar structure** competes for the same carrier protein **binding site** so one material (x) win the **competition** and get transported and other material (y) failed to be transported **(inhibited)**.

- **Saturation (Fig.4):** The number of carrier proteins in a membrane is limited. so once all carriers are occupied i.e. **saturated** by the material (Z) , no, increase in transport can occur by simply increasing the concentration of (Z).

N.B.

In Co-transport (symport) and counter-transport (antiport), competitive inhibition does not occur as substance x and y are **dissimilar** but using the same carrier protein that has two different binding sites.



Rate of diffusion (solute entry into the cell) by carrier mediated or facilitated diffusion (A) and simple diffusion (B). In simple diffusion, though a linear relationship exists between the rate of diffusion and concentration of solute throughout the process,

Types of transport across the cell membrane

A. Passive transports (Diffusion)

1. **Simple** diffusion
2. **Facilitated** diffusion
3. **Osmosis** (diffusion of water)

B. Active transports

1. **Primary** active transport
2. **Secondary** active transport

C. Vesicular transports

1. **Exocytosis**
2. **Endocytosis**

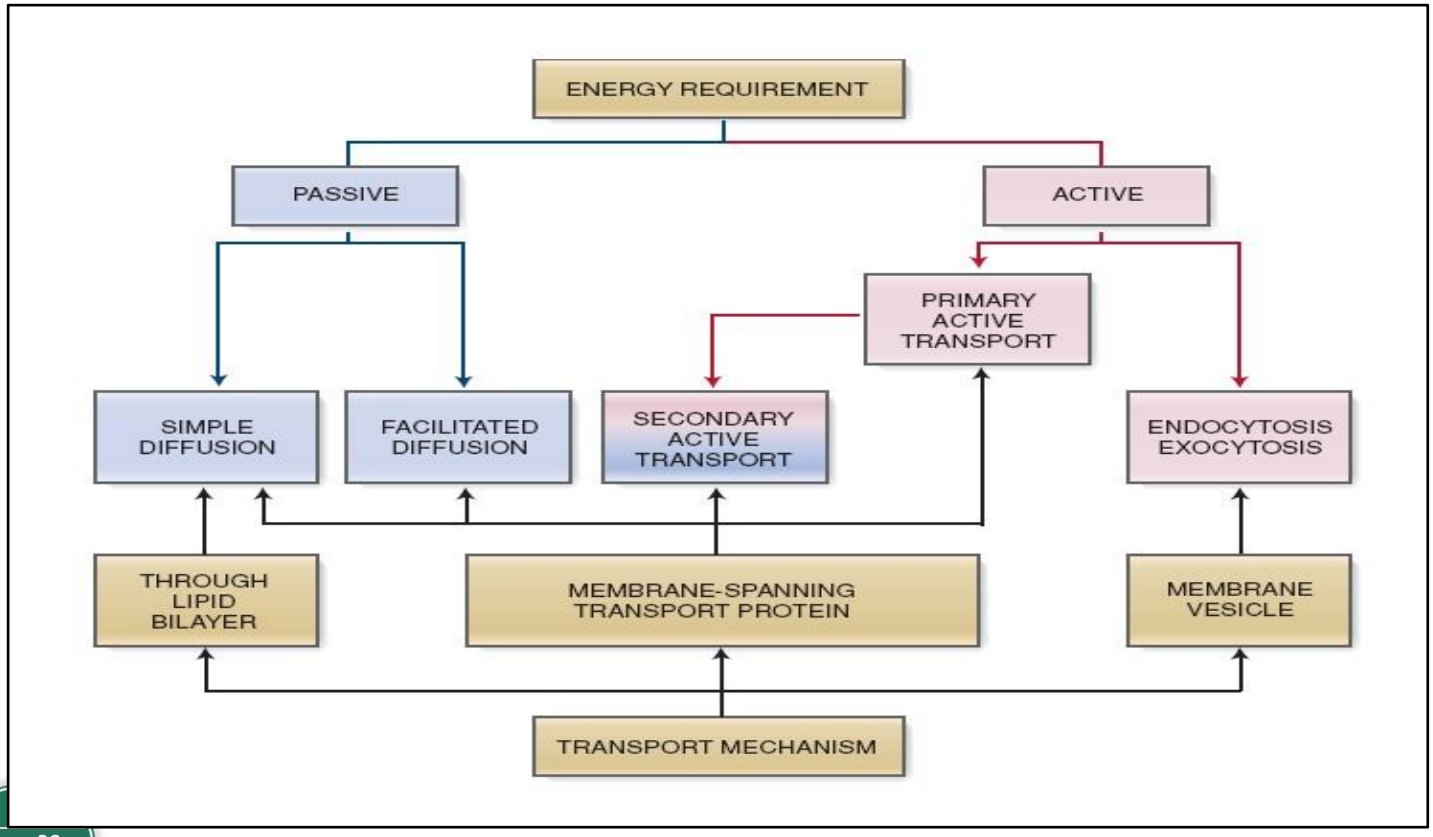
1 Passive transports (diffusion)

Definition:

Movement of **substance** according to its **electrochemical gradients.**

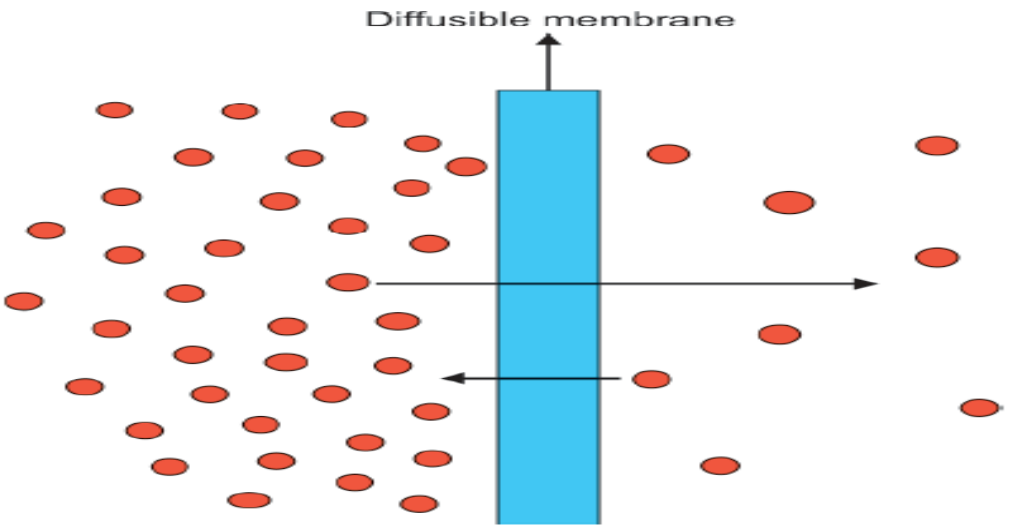
Characters of Passive transport;

1. Passive Transport is **downhill**; the movement of substance according to **Electrochemical gradient.**
2. Passive Transport does **not utilize ATP.**



Chemical gradient:

- Substance move from **high conc**. to **low conc.**
- It occurs due to **random** movement of the substance caused by their **thermal energy**. The **motion** varies **directly** with **temperature**.
- Chemical gradient occur for **both charged** and **non-charged** particles



Diffusion across the cell membrane. The molecules move in both directions, but the movement is more from higher to lower concentration of solutes. The length of arrow indicates the magnitude of movement of molecules.

Electrical gradient:

- A difference in **charges** between **two adjacent** areas produces an electrical gradient that promotes movement of ions **toward** the area of **opposite charge:**
  1. **Positively** charged ions (**cations**) tend to move **toward** the **more negatively** charged area.
  2. **Negatively** charged ions (**anions**) tend to move **toward** the **more positively** charged area
- It occurs due to the **fact** that **Ions** with like charges (those with the **same kind of charge**) **repel** each other, and **ions** with **opposite** charges **attract each other.**
- Electrical gradient occur **only** for **charged particles**

Factors that Determine Rate of Diffusion;

I - **Properties** of the substance. II - **properties** of the membrane.

I – Properties of the Substance:

- **Concentration** and **electrical** gradients of the substance (see before).
- **Permeability** of the substance through the membrane i.e. the **ease** at which substances pass through the membrane.

The factors that determine the permeability are:

1. Lipid-solubility: As the membrane is a **lipid bilayer**, the permeability of a substance is **directly** proportional to their **lipid solubility** of the substance.
2. Molecular size: permeability is **inversely** proportional to the substance molecular size.
3. Charge on the molecule: it will **affect** electrical gradient
4. Temperature: Rate of diffusion is **high** at **higher body temperature** as temperature **facilitates** motion of molecules in the solution.
5. Number of channels and carriers in the membrane for a particular substance: It **directly** proportional with permeability
6. Pressure gradient: Diffusion is **more** from **high fluid** pressure to **low** pressure, as the pressure causes **greater** number of molecule to hit the membrane.

II – Properties of the membrane:

1. Membrane **cross-sectional** area, Diffusion is **directly** proportional to **cross-sectional area.**
2. **Thickness** of the membrane, Diffusion is **inversely** proportional to **thickness**.

Types:

1. **Simple =** substance **move** through **cell membrane without** using **protein Carrier** i.e. either through **lipid layer** or **protein channels.**
2. **Facilitated =** substance **movement** through cell membrane is **facilitated** by using a **protein Carrier.**
3. **Osmosis (Diffusion of water),** Discussed later.

Facilitated diffusion:

Facilitated diffusion differs from simple diffusion in that:
1. **Faster** rate of transport.
2. **Specificity**, if the substance has its **own specific carrier** which has **one binding** site that **binds** it and **cannot** be used by **dis-similar** substance.
3. **Competitive inhibition** (discussed before).
4. **Saturation** (discussed before).

Osmosis (Diffusion of water):

Background:

Any salutation is composed of **solutes** dissolved in a **solvent** i.e.

Solution = Solutes + solvent

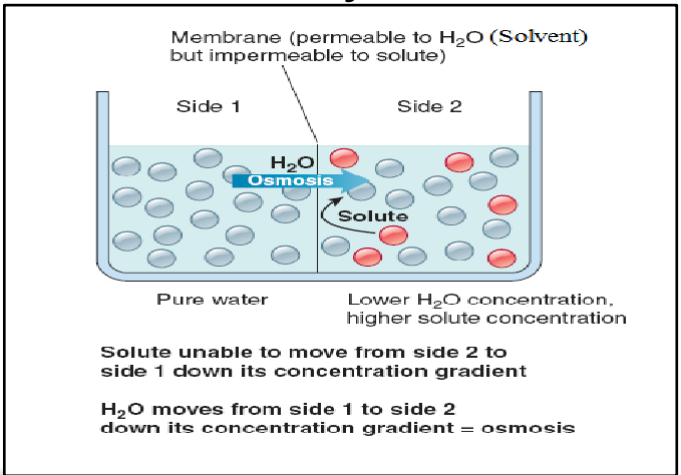
- **Inside** human body, the **universal solvent** is **water**.
- When you said that solution **(x)** is **more concentrated** than solution **(y)**, you point out **two facts:**
  1. **1st fact:** Solution **(x)** has **more solutes** in **comparison** to solution **(y)**
  2. **2nd fact:** Solution **(y)** has **more solvent** (**water**) molecules in **comparison** to solution **(x)** i.e. solution **(y)** has **more water concentration** than solution**(x).**
- If you have Solution **(x)** (**more solute** concentrated **less water** concentrated) **separated** by a **membrane** from Solution **(y)** (**less solute** concentrated **more water** concentrated), the following can occur;
  1. If **membrane** is **permeable** to **solutes**, solutes will **diffuse passively** from **high conc.** of solutes to **low conc**., of solutes according to **conc. gradient** of solutes
  2. If **membrane** is **Non permeable** to **solutes** but **permeable** to **water**, **solutes** will **not diffuse** but **water** will **move passively** i.e. **diffuse** (**Osmosis**) from (**low conc.** of **solutes high concentration** of **water**) to (**high conc.**, of solutes **low concentration** of water) **according** to **conc. gradient of water.**

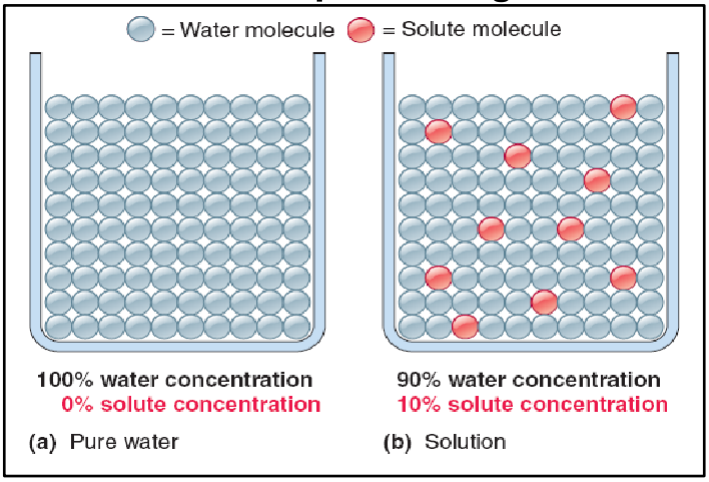
Definition1: Osmosis is the process of **movement of solvent** from the **solution** with the **lower concentration** of **solutes** to the solution with **higher concentration** of **solute**, when **both** the **solutions** are **separated** by a **membrane** which allows the **passage** of **solvent** and **not the solute particles.**

Definition2: Osmosis is **diffusion** of **water** according to **concentration gradient** of **water**.

Factors affecting the ability of solute to cause osmosis;

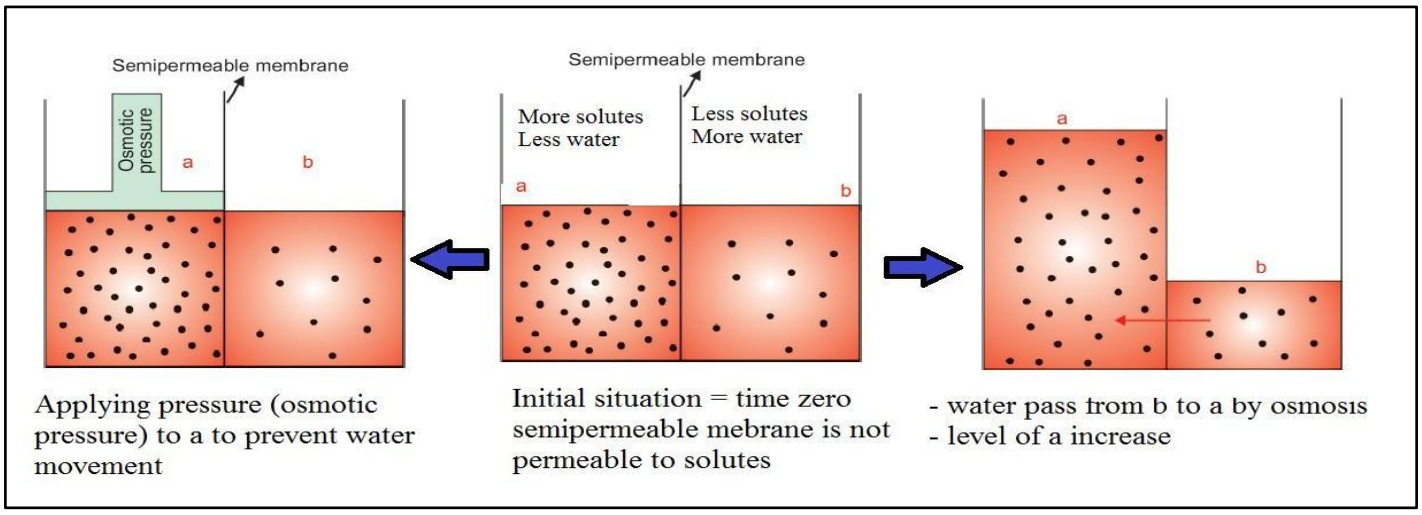
- The **solutes** is **not permeable** i.e. **cannot** diffuse from **side to side**
- **Concentration** of that **solutes** in the **solution**
- **Osmotic ability** of solute is **not related** to the **size**, **shape** or **charge of solute.**





Osmotic Pressure:

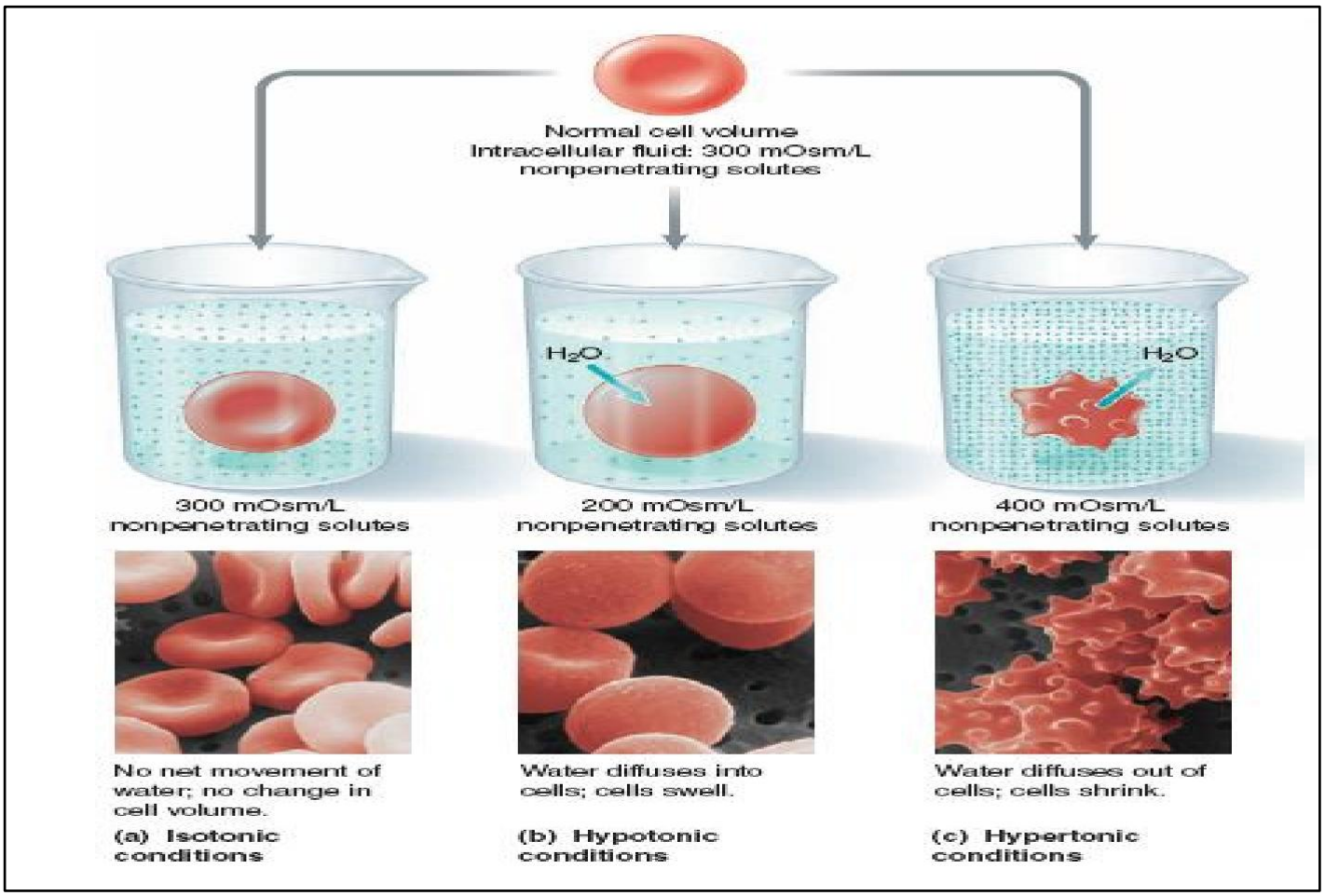
- Osmotic pressure is the **pressure** applied to the solution with **higher solute** concentration to **prevent osmotic flow** of **water** to that solution.
- **Experimentally**, osmotic pressure can be **measured** by measuring the **hydrostatic pressure** applied to **prevent water** from entering the solution with **higher solute concentration.**



Osmolarity and tonicity:

Osmolarity refers to **total number** of **solutes** in a **solution**.

- **Total number of solutes** = **solutes** which can **penetrate cell membrane** + **solute** which **cannot penetrate cell membrane.**
  1. **Solutes** which **can penetrate** cell membrane are **osmotically useless** i.e. they **cannot** cause **osmosis**.
  2. **Solutes** which **cannot penetrate** cell membrane are **osmotically useful** i.e. can **cause osmosis**.
- If you have **multiple** solution (**A, B, C, and D**) and you **compare** their **similarity** to each other , you can describe them as follow:
  1. If solution **A and B** are **equal** in **similarity** then **A** is **isoosmotic** to **B**.
  2. If solution **C** is **lower** than **B** in **osmolarity** then **C** is **hypoosmotic** to **B**.
  3. If solution **D** is **higher** than **B** in **osmolarity** then **D** is **hyperosmotic** to **B**.
- **Tonicity** refers to effect of **solutes** which **cannot penetrate** cell membrane i.e. **osmotically active,** on the **cell volume.**
  1. **Isotonic** Solutions**,** which have **tonicity same** as **intracellular fluid,** so these solution cause **no water osmosis** so **cell volume** do **not change.**
  2. **Hypotonic**, Solutions with **lower tonicity** than **intracellular** fluid, so these solution cause **water osmosis** into cell so **cell volume increase.**
  3. **Hypertonic**, Solutions with **higher tonicity** than **intracellular** fluid, so these solution cause **water osmosis** out of the cell so **cell volume decreases.**


Normal cell volume
Intracellular fluid: 300 mOsm/L
nonpenetrating solutes
H_2O
H_2O
300 mOsm/L
nonpenetrating solutes
200 mOsm/L
nonpenetrating solutes
400 mOsm/L
nonpenetrating solutes
No net movement of water; no change in cell volume.
(a) Isotonic conditions
Water diffuses into cells; cells swell.
(b) Hypotonic conditions
Water diffuses out of cells; cells shrink.
(c) Hypertonic conditions

Active transports

Definition:

- Movement of substance against to its Electrochemical gradients and utilize ATP

Characters;

- Up hill transport: Transport occurs against the electrochemical gradient of the substance, so it needs a source of energy.
- Utilize ATP. It is the source of energy so active transport is susceptible to metabolic poisons which prevent ATP from being produced or utilized.

Types;

- ❑ Primary active transport.
- ❑ Secondary active transport.

Primary Active Transport (Ion Pumps)

Definition

I

- s the transport mechanism that directly utilizes metabolic energy for the transport process. The ion pumps hydrolyze ATP to ADP hence they are called ATPases and use energy in the third phosphate bond for the transport process.

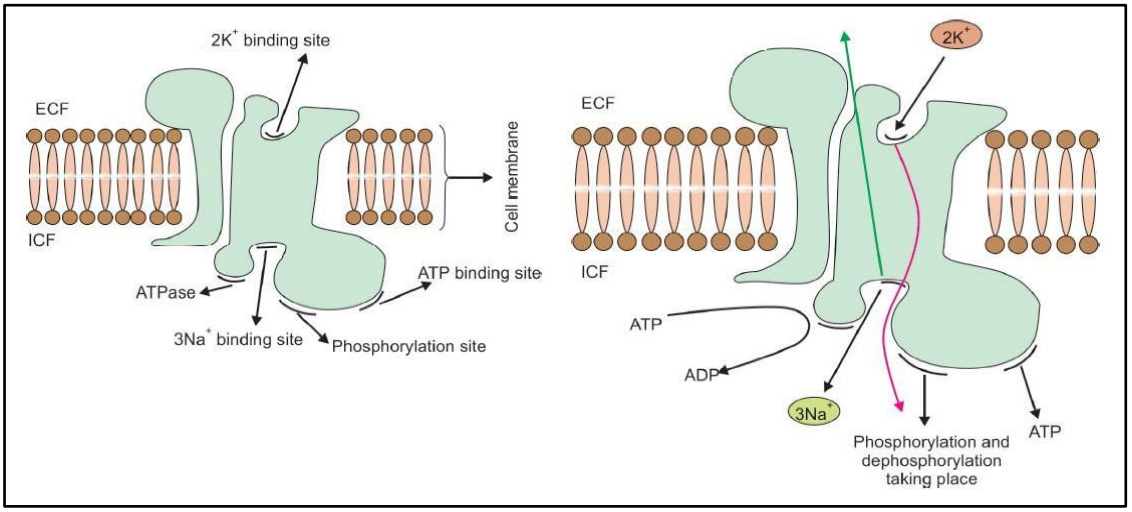
❑ The commonly occurring ATPase is Na^+–K^+ ATPase or Na^+–K^+ pump.

- Na^+/K^+ ATPase (Fig.11)
- Na^+-K^+ ATPase is present in all eukaryotic cells.

- It is antiport that pumps K^+ into the cell and Na^+ out of the cell against their concentration gradients. It is primarily responsible for maintaining the high K^+ and low Na^+ concentration inside the cells.

Structure and its Functional Aspects

- At the cytoplasmic side, the pump has ATPase activity and binding sites for 3 Na^+ and ATP and Phosphate binding site.
- At the extracellular side there are binding sites for 2 K^+.
- Binding of three ions of Na^+ and an ATP molecule with the carrier on its cytoplasmic surface causes hydrolysis of ATP giving rise to ADP and Pi.
- The phosphate group is bound to phosphate binding site.
- On activation, Na^+-K^+ ATPase pumps three Na^+ ions out of the cell and two K^+ into the cell per each ATP Na^+-K^+ pump.
- Na^+-K^+ pump is an electrogenic pump as it forces net positive charge to out of the cell.



Secondary Active Transport

Definition:

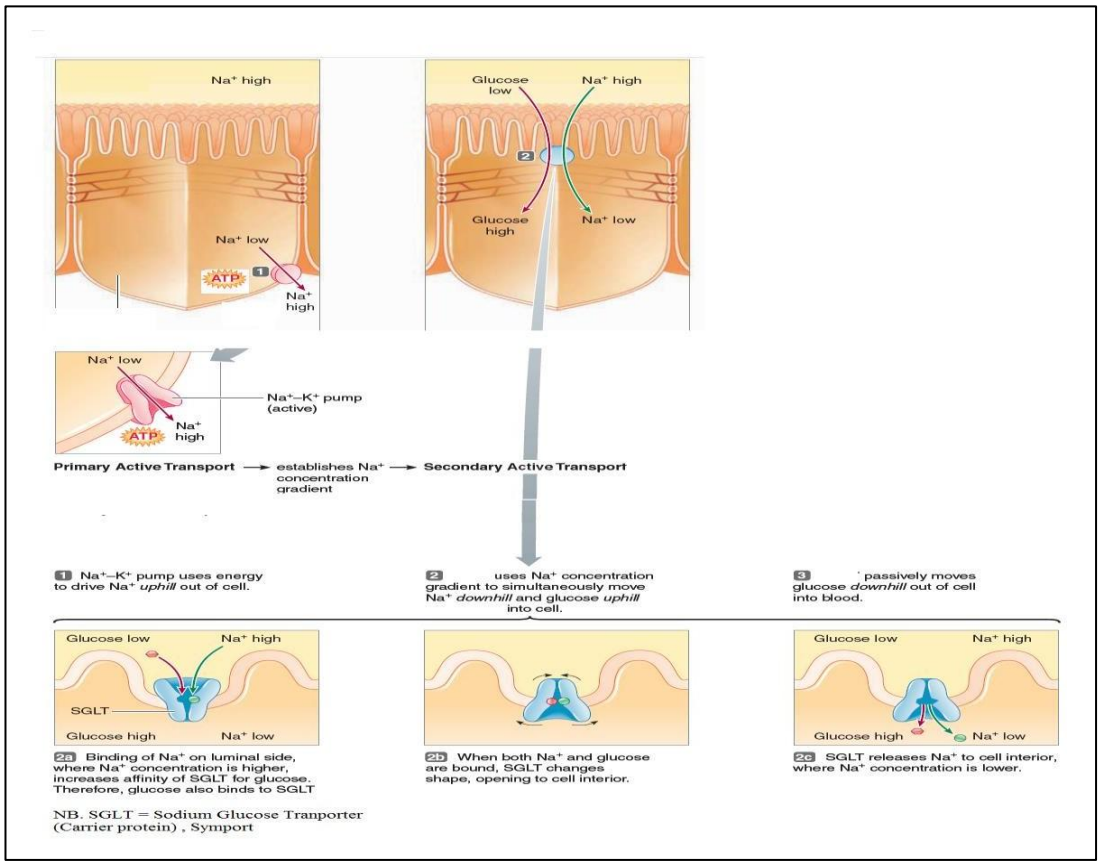
- Transfer one solute against its concentration gradient (from low to high) by using the energy produced by gradient of another solute (driver solute).

Background:

- Energy created by Na^+ gradient – Na^+ is the driver solute - is utilized for transport of other solutes. The Na^+ gradient is generated and maintained by Na^+-K^+ pump (Active transport). Though the transport system of the other solute by itself does not directly utilize energy, it depends on the function of Na^+-K^+ pump.

Steps; (Fig.12)

- The primary active transport of Na^+ out of the cell by Na^+-K^+ pump leads to decreased concentration of Na^+ in the cytoplasm of the cells.
- This causes facilitated diffusion (carrier mediated) of Na^+ from the out of the cell (high Na conc.) into the cells (Low Na. conc.).
- The carrier protein that transfers Na^+ into the cell also transports another substance or ion against their conc. gradient.
- The carrier protein transports the other substance simultaneously with sodium making usage of Na^+ conc. gradient created by the Na^+- K^+ pump so it is secondary active transport.
- The carrier protein may be
  - Symport i.e. carry Na and other substance in the same direction.
  - Antiport i.e. carry Na and other substance in opposite directions.



Vesicular transports

Definition,

- The transport process that occurs by either fusion of vesicle or formation of vesicle is called as vesicular transport.

Types (Fig. 13):

❑ Exocytosis

- Fusion of vesicle with the cell membrane and release of the cell vesicle content – particles /substances - to exterior.

❑ Endocytosis

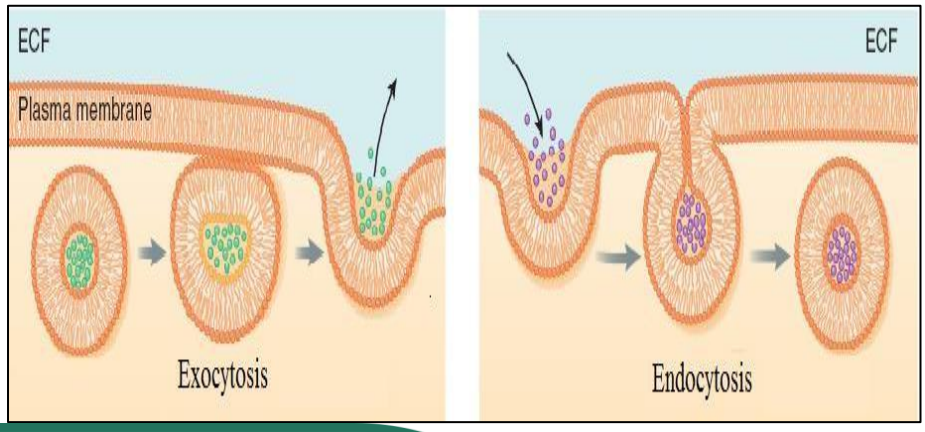
- Formation of vesicle from the cell membrane and uptake of an outside of the cell particle/substance into the cell interior.

N.B

Cell use vesicular transport method when the particle/substance <u>can not be</u> transported by diffusion or active transport process

Characters;

- It has No, relation to conc. gradient i.e. it may go with or against conc. gradient.
- It may Utilize ATP.
- It affects cell membrane surface area.
  - Removing areas of cell membrane in process of vesicle formation in endocytosis.
  - Adding areas to cell membrane in process of exocytosis



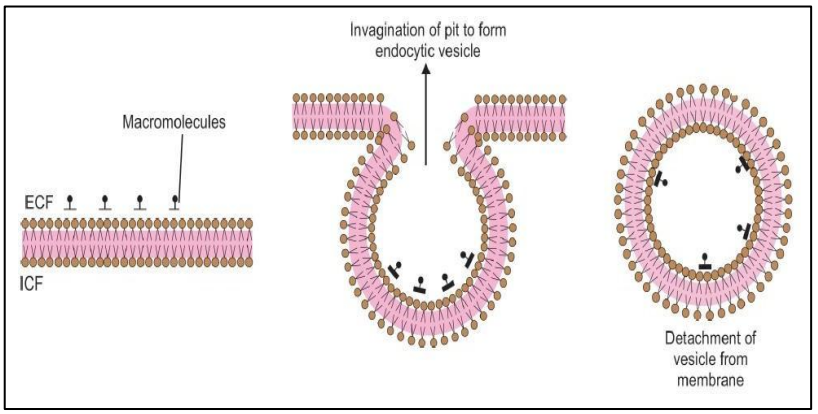
Mechanism of Vesicular transport

❑ Endocytosis mechanism (Fig.14):

- Vesicle formation, the substance makes contact with the cell membrane that invaginates to form an endocytotic vesicle.
- Vesicle detachment, the non-cytoplasmic side of the membrane then fuses and the vesicle is pinched-off into the cytosol.

❑ Exocytosis mechanism:

- Vesicle formation, the substance is packed within the vesicle inside the cell.
- The vesicle moves toward cell membrane and contact it.
- The vesicle fuses with the cell membrane.
- The vesicle releases its content to cell exterior.



Membrane Potentials
(Resting membrane potential and Action Potential)

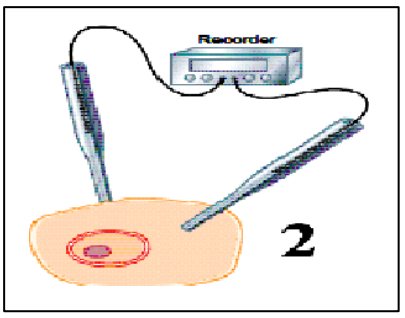
Introduction:

- Ions Distribution inside and outside of cell:
- Extracellular fluid contains many Na^+, Cl^- & small amount of K^+.
- Intracelluar Fluid contains,
  - Many potassium ion (K^+), small amount of sodium (Na^+) and chloride (Cl^-) ions.
  - Many -ve charged organic particles e.g. proteins, phosphates in DNA and ATP

Ion	ICF mEq/L	ECF mEq/L
Na	14	140
K	140	4
Cl	4	120

- **mEq/L = Milliequvilant per liter i.e. a unit for measuring ion conc. in body fluids**

- **Methods of Measurement of Membrane potentials**
- **By a cathodal ray oscilloscope.**



Resting Membrane Potential (RMP)

Definition:

- It is a potential difference between outer and inner surface of membrane during rest (i.e. unstimulated membrane).

N.B

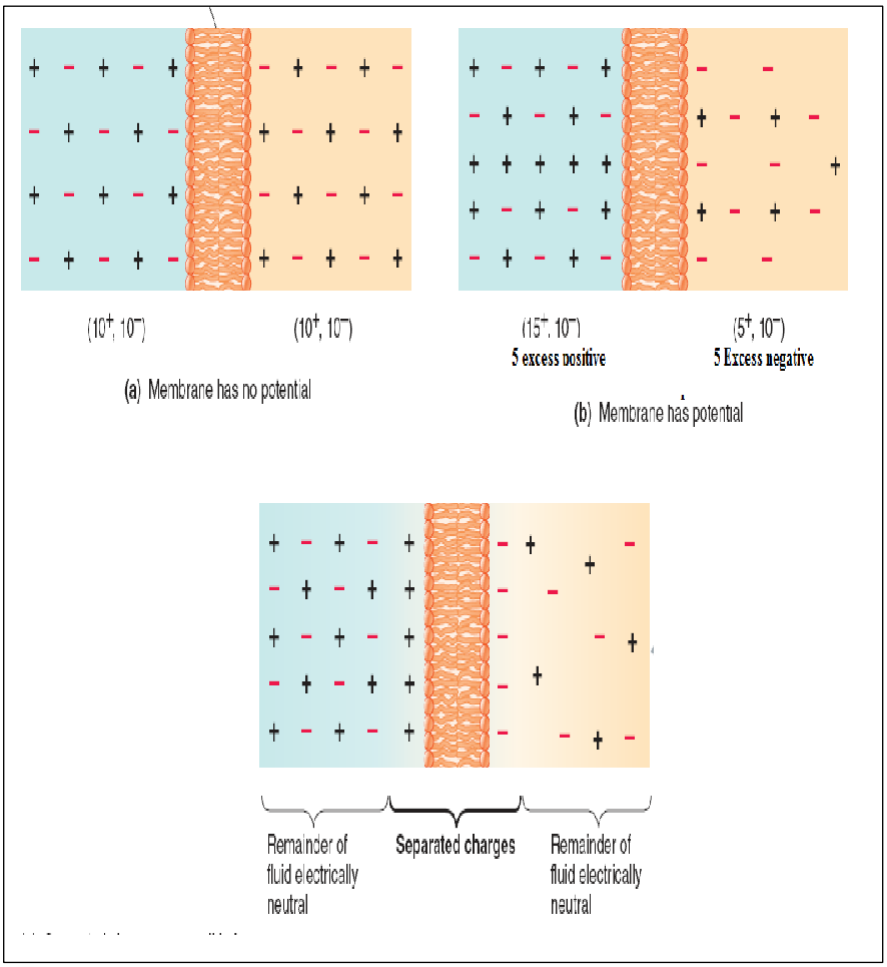
RMP is also called transmembrane potential.

Recording:

- If 2 microelectrodes connected to galvanometer (cathodal ray oscilloscope) are place on cell membrane of living cell at rest as follow;
  - Both on Outer surface of an axon.
  - Both on Inner surface of an axon.
  - One on the outer surface and the other on the inner surface.

Results of recording,

- In case of A or B, There will be no potential (voltage) difference between 2 microelectrodes.
- In case of C, there is a potential difference (RMP) between 2 microelectrodes, so the cell membrane is in polarized state.



NB1

The whole interior and exterior of the cell membrane in neutral, only different charges surround cell membrane form both sides.

N.B2

It is not specific; it is recorded from almost all cell types though each single cell type has a particular RMP.

N.B3

As a convention, it described inner of membrane relative to outer of the membrane e.g. If inside of the cell is 80 mv less than outside of the cell then we describe this as RMP = - 80 mv.

❑ Normal Value: Variable according to type of cell e.g.

- -70 mv in medium sized axon.
- -90 mv in large sized axon, skeletal muscles & cardiac muscle.

Generation of Resting Membrane Potential (Fig.3):

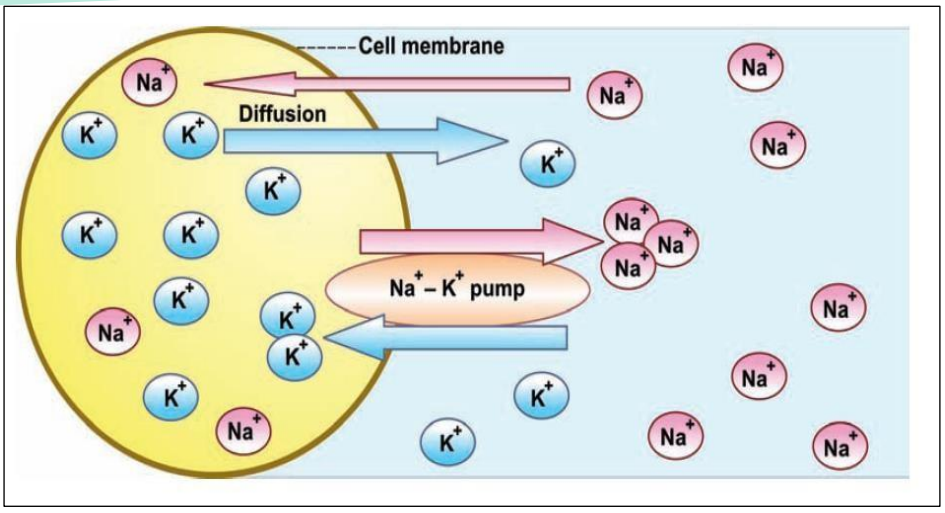
❑ Passive Factors (do not consume ATP) ,

- -65.1 mv out of -70mv (About 93%) of resting membrane potential is due to passive factors:
  - Mainly Diffusion of Na+ and K+ by selective permeability for Na+ and K+ through ion specific leak channels.
  - Non diffusable particles (DNA and protein etc.).

N.B

The role of K^+ diffusion (K^+ efflux) is more important than role of Na^+ diffusion (Na^+ influx) as the membrane is 50-100 times more permeable to K^+ than Na^+ (Fig.4) because:

- High K^+ concentration gradient than Na^+ conc. gradient.
- Number of leak K^+ channels is more than Na^+ leak channels.
- Na^+ channels pores are surrounded by calcium ions (positive charge) which repel Na^+ (positive charge).
- Hydrated K^+ ion are smaller than hydrated Na^+ ion. Ions inside body are surrounded by a jacket of water molecules (hydrated ions).



❑ Active Factor (consume ATP): Na^+/K^+ pump

- -4.9 mv of -70 mv (About 7%) of resting membrane potential is due to Na^+/ K^+ pump. Na/K pump is an electrogenic pump pumping of 3 Na^+ ions out of the cell and 2 K^+ ions into the cell.

Maintenance of RMP (role of Na/K pump)

K^+ diffusing out of the cell and Na^+ diffusing into the cell will come to an end once the concentration gradient is lost. However, that does not happen, as Na^+-K^+ ATPase helps in rebuilding the concentration gradient. It serves to pump back the Na^+ that diffuses into the cell and K^+ that diffuses out of the cell. Thus, Na^+/K^+ pump plays an important role in maintaining RMP.

NB. Na^+/K^+ pump has a small role (only 7%) in **generating RMP** and a great role in **Maintaining RMP**

:Equilibrium Potential (EP) (Nernst Potential)

Definition: is the membrane potential at which there is no diffusion of a particular type of ion.

NB

It is calculated by an equation (Nernst equation).

Principle

Passive movement of any ion is under the control of two forces (electrochemical gradient):

1- The concentration gradient. **2- The electrical gradient.**

- When the two forces are equal in effect but opposite in direction, there is an (equilibrium), there is no net movement of the ion.
- The greater the conc. gradient of a particular ion the greater the EP that will prevent its diffusion across cell membrane.
- There is a specific EP for each ion type in the same single cell.

NB1. As a convention, EP describes inner of membrane relative to outer of the membrane e.g.

- If EP for Na^+ is + 61 mv, this mean that potential of inside of the cell should be larger than potential outside cell by 61 mv.
- If EP for K^+ is - 91 mv, this mean that potential of inside of the cell should be less than potential outside cell by 91 mv.

NB2, A different equation (Goldman equation) is used if we want to calculate EP of different ions at the same time in a single cell

Relation between Equilibrium Potential (EP) and RMP

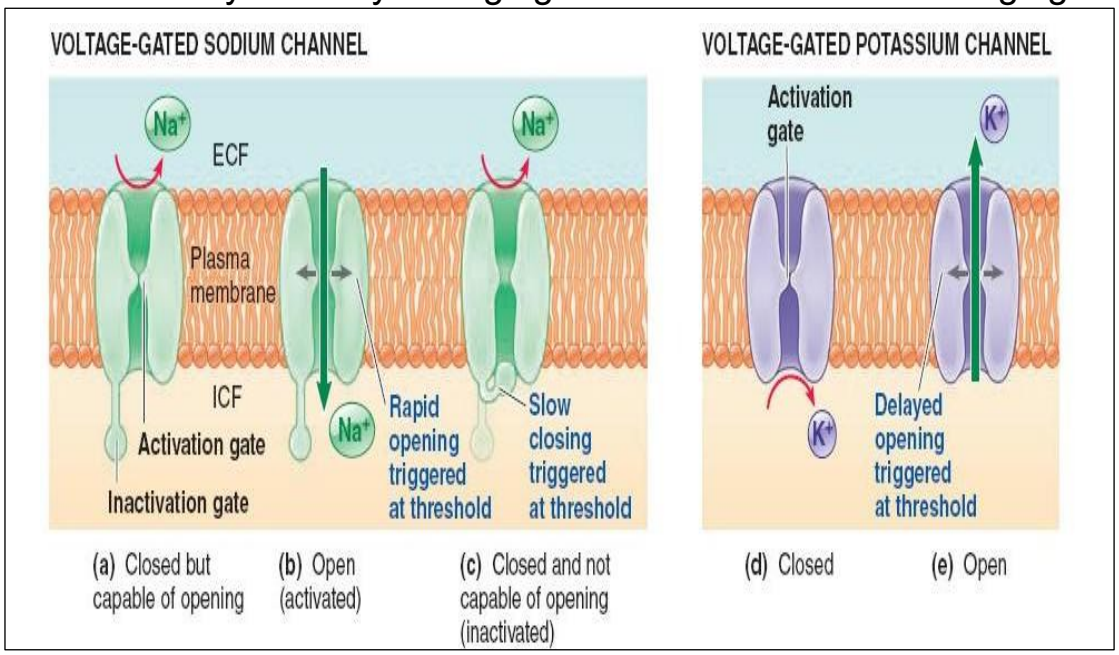
Historically, EP was used as an explanation for RMP; actually this was a failure as RMP value is different from any EP for any single ion type.

- If passive movement for any single ion type alone is the only factor responsible generation of RMP then EP for that ion should equal RMP.
- Actually, comparing measured RMP with EP of individual different ions (calculated by Nernst equation) clearly show that K^+ is most important ion in RMP generation as EP for K^+ is nearer to the value of RMP.

Action Potential (AP)

Introduction

We must study carefully Voltage gated Na^+ channel and Voltage gated K^+ channel (Fig.6)



The voltage-gated Na^+ channel has two gates: an activation gate **(outer - Fast)** and an inactivation gate (inner - slow). Both gates must be open to permit passage of Na^+ through the channel, and closure of either gate prevents passage.

Voltage-gated Na^+ channel can exist in three conformations (status);

1. **Closed state**, closed but capable of opening (**activation gate closed, inactivation gate open**).
2. **Opened state** (**activated state**), both gates open.
3. **Inactivated state**, Closed and not capable of opening, (activation gate open, inactivation gate closed.

The voltage-gated K^+ channel has only an activation gate, which can be either closed or open.

Action Potential Definition: Rapid brief (transient) changes in membrane potential following stimulation of excitable cell (nerves and muscles) by adequate stimulus.

Stages (Phases) and Ionic Bases (causes) of Action Potential:

1- **Stimulus artifact:** When a stimulus is applied, there is a slight irregular deflection of baseline for a very short period. This is called stimulus artifact. The artifact occurs because leakage of current (electrons) from stimulating electrode to detecting electrode in the fluid outside axon.

Significance: It indicates the time of application of the stimulus.

2- **Latent period:** It is the time taken between application of the stimulus (stimulus artifact) and the beginning of the action potential curve. It is time that action potential takes to travel from stimulator to recording electrodes. Duration of latent period depends on the distance between stimulating and recording electrodes and conduction speed of the axon.

Significance: used to measure axon conduction speed.

Speed of conduction = distance between stimulating and recording electrodes/latent period.

3- **Depolarization (Ascending Limb):** the cell membrane loss its polarized state. During Depolarization the membrane potential approaches zero and then reach a positive value. It is divided into:

a- Slow depolarization from – 70 mv to – 55 mv:

- From -70 mV to -63 mV: This is due to addition of -ve charge to the outer surface of the cell membrane from stimulating electrode.
- From -63 mV to -55 mV: This is due to opening of a few voltage gated Na^+ channels. The opening of the voltage gated Na^+ channels depolarize the membrane further, leading to the opening of greater number of Na^+ channels (Hodgkin cycle).

b- Rapid Depolarization at -55 mv:

- The depolarization become rapid and irreversible (all or none rule) – See later. This level is called the threshold for stimulation (firing level). This is due to rapid opening of the all voltage gated Na^+ channels. This leads to massive influx of sodium ions.

c- **The membrane is completely depolarized** and the potential difference between the outer and inner surface of the membrane is zero **(isopotential = depolarization = Total depolarization).**

d- **The reversal of polarity** = overshoot from zero level to +35 mV. The inside becomes positive relative to outside

⇨ At +35 mv (Peak) Na^+ channels are **inactivated, no Na^+ can enter the cell, though still Na^+ is more concentrated outside.**

4- **Repolarization (Descending Limb):** during which the membrane potential returns to the resting level. It is divided into 2 parts:

A- **Rapid repolarization :**

⇨ It is the first 70 % of the repolarization process.

⇨ Rapid depolarization and rapid repolarization is called spike potential.

Causes of Rapid Repolarization: Closure of Na^+ channels and increasing of K^+ outflow. The maximal opening of K^+ channels just at the same time that the Na^+ channels are inactivated i.e. at +35 mv. Rapid efflux of K^+ occur by electrochemical gradient (K is more inside cell and cell is +35 mv).

B- **After-Potential**: After the 70% of repolarization process, the membrane potential change slowly , It is divided into:

a- **Negative after potential** (**after-depolarization**):

- Outer surface becomes more negative than in resting condition. Membrane is hypopolarized and more excitable (near to firing level).
- **Cause**: The outward diffusion of potassium ion is decreased due to loss of some of electrical driving force (electric gradient).

b- **Positive after potential** (**after-hyperpolarization**):

- Outer surface becomes more positive than in resting condition. Membrane is hyperpolarized (-90 mv) and less excitable (away from firing level).
- **Cause**: Slow closure of K^+ channels and K^+ keep effluxing from the cell.

:Ionic Activity after Action Potential - Role of Na+/K+ pump

During a single action potential, a very small amount of Na^+ and K^+ move through cell membrane during depolarization and repolarization respectively, so after a single Action potential: conc. gradient for both Na^+ and K^+ decrease.

- Conc. of Na^+ is still lager in outside in comparison to inside.
- Conc. of K^+ is still lager inside the cell in comparison to outside.

After any action potential,

- ⇨ It is a must to regain Na^+ original conc. gradient by pumping the very small amount of Na^+ that influxed during depolarization. Na^+ must be pumped into outside (from low conc. to high conc.).
- ⇨ It is a must to regain K^+ original conc. gradient by pumping the very small amount of K^+ that effluxed during repolarization. K^+ must be pumped into the cell (from low conc. to high conc.).

All or None Law:

Definition:

Action potential occurs with a constant amplitude and shape irrespective of magnitude of the stimulus.

Explanation of the law:

A sub-threshold stimulus (non-adequate) fails to excite the tissue. Only a stimulus of threshold magnitude elicits an action potential. If supra-threshold stimuli are applied, the action potentials resulting from them have the same amplitude, duration and form as those produced by threshold stimuli, provided the experimental conditions remain the same.

Mechanism:

Maximum activation of voltage-gated Na^+ channels occurs at certain membrane potential (depolarization) e.g. -55 mv (firing level), Following that, any extra degree of stimulus intensity is of no further use i.e. after the threshold level is achieved, the amount of sodium influx becomes independent of the stimulus intensity.

Autonomic Nervous system

ILOs:

At the end of this chapter student can:

1- Compare and contrast the anatomical features of the sympathetic and parasympathetic systems.
2- For each neurotransmitter in the autonomic nervous system, list the neurons that release them and the type and location of receptors that bind with them.
3- Describe the mechanism by which neurotransmitters are removed.
4- Distinguish between cholinergic and adrenergic receptors.
5- Describe the overall and specific functions of the sympathetic system.
6- Describe the overall and specific functions of the parasympathetic system.

Introduction

Any activity in our life is under control of one of the main control systems:

1- Endocrine system: This controls the slow action by *hormones*.
2- Nervous system: this controls the rapid action by *neurotransmitters*.

Division of nervous system

A. Central nervous system:

a) **Brain**:

- Brain stem (mid brain, pons and medulla oblongata).
- Cerebellum.
- Cerebral cortex.

b) **Spinal cord**:

- 31 segments (8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccegeal).

B. Peripheral nervous system:

a) **Cranial nerves**:

- Autonomic (involuntary).
- Somatic (voluntary).

b) **Spinal nerves**:

- Autonomic
- Somatic

Q: Compare between somatic and autonomic nervous system?

Autonomic nervous system (ANS):

- Also known as the visceral or involuntary nervous system, the ANS functions without conscious, voluntary control.
- It innervates **cardiac muscle**, **smooth muscle**, and various endocrine and exocrine **glands**, this nervous system influences the activity of most tissues and organ systems in the body.
- Therefore, the ANS makes a significant contribution to homeostasis. The regulation of blood pressure, gastrointestinal responses to food, contraction of the urinary bladder, focusing of the eyes, and thermoregulation are just a few of the many homeostatic functions regulated by the ANS.

Divisions of the A.N.S:

The A.N.S. is divided into the following 2 systems:

I. **The sympathetic nervous system:**
   - This arises from the lateral horn cells (LHCs) of all thoracic and upper 3 lumbar segments of the spinal cord (***thoraco-lumbar outflow***).

II. **The parasympathetic nervous system:**
   - This arises from the 2nd, 3rd and 4th sacral segments of the spinal cord as well as from the nuclei of some cranial nerves III, VII, IX and X cranial nerves (***Cranio-sacral outflow***).

Autonomic ganglia

Definition:

It is the site of physiological contact between pre and post ganglionic fibers.

Types:

a) Lateral (sympathetic).
b) Collateral (mixed) or pure sympathetic.
c) Terminal (parasympathetic).

Lateral (paravertebral):

- These are 23 ganglia that form the sympathetic chain.
- Divided into: 3 cervical, 12 thoracic, 4 lumber, 4 sacral.

Collateral:

- Around the large blood vessels as:
  - Caeliac ganglion.
  - Superior mesentric ganglion.
  - Inferior mesentric ganglion.

Terminal:

(in the wall of organ, no post-ganglionic or very short post-ganglionic fiber may be present).

Functions:

1) Distributing center.
2) Relay station between pre and postganglionic fiber.

Sympathetic nervous system

- Sympathetic NS works during **"Fight or flight"**.
- It is also called the "E" division. It concerned with body activities during: **E**xercise, **E**xcitement, **E**mergency, and **E**mbarrassment.
- The sympathetic N.S. can be divided into 4 divisions which are: the **cervical**, **cardio-pulmonary**, **splanchnic** and **somatic** divisions.



(1) The cervical sympathetic division:

⇨ This division supplies various structures in the region of the head and neck.

⇨ It arises from the LHCs of the upper 2 thoracic segment of the spinal cord. Preganglionic fibers reach the sympathetic chain then they proceed upwards and relay in the superior cervical region. Postganglionic neurons arise from this ganglion then they pass along branches of the carotid arteries and supply the following structures:

(I) **The eyes**, in which sympathetic stimulation causes:

(a) Contraction of the radial muscle of the iris (dilator pupillae muscle) leading to **mydriasis** (Pupil dilation).

(b) Contraction of muscles in the eyelids (called the superior and inferior tarsal muscles) leading to **widening of the palpebral fissure**.

(c) Contraction of Muller's muscle behind the eyeballs leading to exophthalmos specially in animals.

(d) Vaso Constriction (V.C.) of the blood vessels of the eye.

(e) Relaxation of the ciliary muscle during far vision.

(II) **The salivary glands** in which sympathetic stimulation causes V.C. leading to (trophic secretion) little in amount, viscid rich in enzymes.

(III) **The lacrimal glands**, in which sympathetic stimulation produces V.C. and a little trophic secretion.

(IV) **The skin of the head and neck**, in which sympathetic stimulation causes:

a) V.C.
b) Sweat secretion.
c) Hair erection (as a result of contraction of the erector pilae (or pilomotor) muscles in the skin).

(V) ***The cerebral vessels***, in which sympathetic stimulation causes weak V.C.

Q: WHAT IS HORNER SYNDROME?

(2) The cardio-pulmonary sympathetic division

- ⇨ This division supplies the heart and lungs.
- ⇨ It arises from the LHCs of **the upper 4 thoracic** segments of the spinal cord. Preganglionic neurons reach the sympathetic chain, where they relay in the **lower cervical and upper 4 thoracic ganglia** from which postganglionic neurons arise and supply the following structures:

(1) ***The heart***: Sympathetic stimulation activates the heart through increasing all properties of cardiac muscle. *Contraction* becomes stronger; *excitability* and speed of c*onductivity* are much increased.

(2) ***The coronary vessels*** (= vessels that supply the cardiac muscle): Sympathetic stimulation leads to vasodilatation (V.D.) of these vessels.

(3) ***The bronchioles***: Sympathetic stimulation inhibits secretion of mucus from the bronchial glands and produces *bronchodilatation.*

(4) ***Pulmonary vessels*** in which sympathetic stimulation leads to V. C.

(3)The splanchnic (visceral) sympathetic division

⇨ This division supplies various abdominal and pelvic viscera.

⇨ It arises from the LHCs of the lower 6-7 thoracic and upper 3 lumbar segments of the spinal cord. Preganglionic neurons reach the sympathetic chain where they pass without relay, then they divide into the following 2 splanchnic nerves:

A. *The greater splanchnic nerve*:

- Rely mostly in the *collateral ganglia* in the abdomen except those supplying the adrenal medulla, which relay inside the gland. From these ganglia postganglionic neurons arise and supply the abdominal viscera producing the following effects (or functions):

(1) **V.C**. of the majority of the arterioles in the stomach, small intestine, proximal part of the large intestine, kidneys, pancreas and liver.

(2) **Relaxation** of the walls of the stomach, small intestine and proximal part of the large intestine and **contraction** of their corresponding sphincters (leading to food retention). Gastrointestinal secretions are also reduced.

(3) **Contraction** of the smooth muscle in the capsule of the spleen, so pouring its stored concentrated blood (about 50 ml) into the blood stream.

(4) Stimulation of **glycogenolysis** in the liver (= breakdown of glycogen into glucose molecules) thus increasing the blood glucose level.

(5) Sympathetic stimulation also increases the metabolic rate (athergenic effect) and stimulates **lipolysis** in adipose tissue (thus increasing both the blood glucose and free fatty acid levels).

(6) Stimulation of the adrenal medulla to secrete hormones known as **catecholamine** (adrenaline and noradrenaline).

B. *The lesser splanchnic nerve*:

- Arise From L1 - L_3, relays in inferior mesentric ganglion. It supplies the pelvis. It has the following functions:

a) Rectum → retention of stool by relaxation of its wall and contraction of internal anal sphincter.

b) Urinary bladder → retention of urine by relaxation of its wall and contraction of internal uretheral sphincter.

c) Sex organs → ejaculation.

(4) Functions of the sympathetic nerves in the skeletal muscles

- The skeletal muscles receive both noradrenergic sympathetic nerve fibers (mainly cause V.C.) as well cholinergic sympathetic nerve fibers (cause strong V.D.).

- Accordingly, sympathetic stimulation may cause V.C. or V.D. in the skeletal muscles, depending on the underlying condition e.g. in case of hemorrhage, V.C. predominates (which helps elevation of the arterial blood pressure) while during muscular exercise, V.D. predominates (which increases the blood supply to active muscles).

- Sympathetic stimulation also accelerates the chemical reactions in the active muscles, and this together with V.D. lead to a stronger contraction, delayed fatigue and early recovery after exereise. Such effect was called the **orbelli phenomemnon**.

Parasympathetic nervous system

- Parasympathetic NS works during **"rest or digest"**.



- It is also called the "D" division. It concerned with body activities as: **D**igestion, **D**efecation, and **D**iuresis.

Distribution of parasympathetic

1- Cranial division

A) Oculomotor nerve (III N):

- It arises from oculomotor nucleus, relays in ciliary ganglion, post ganglionic (short ciliary nerve) supplies two muscles:

1- Constrictor pupille muscle leads to miosis.
2- cilliary muscle for near vision.

Q: What is accommodation?

B) Fascial Nerve (VII N):

- It arises from superior salivary nucleus, relays in *sphenopalatine* ganglion and submandibular ganglion. Post ganglionic fibers supplies *Sublingual and submandibular* salivary glands.

C) Glossopharyngeal Nerve (IX N):

- It arises from inferior salivary nucleus, relays in ***otic*** ganglion. Post-ganglionic fibers supply the *parotid salivary* gland.
  The net result is ***true*** salivary secretion, which is profuse, watery little in enzymes.

D) The vagus (10th cranial) nerve:

- This constitutes 75 % of the parasympathetic fibres in the body.
- Its preganglionic fibres arise from the dorsal vagus nucleus in the medulla oblongata, relay in **terminal ganglia** from which short postganglionic fibers arise and supply the following organs in the chest and abdomen:

1- **The heart:** Vagus stimulation of the heart inhibits all properties of the cardiac muscle (leading to slowing of the heart rate and a decrease in its contractility, excitability and conductivity). The vagi do not supply the ventricles, and this phenomenon (or property) is called vagus escape.

2- **The lungs:** Vagus stimulation causes bronchoconstriction, mucus secretion from the bronchial glands and V.D. of pulmonary vessels.

3- **The gastrointestinal tract (GIT):** Vagus stimulation causes contraction of the walls of the esophagus, stomach, small intestine and proximal part of the large intestine, and relaxation of their corresponding sphincters. Therefore, parasympathetic stimulation promotes deglutition (= swallowing) as well as evacuation of food from the GIT.

4- The **gall bladder**: Vagus stimulation helps gallbladder evacuation by causing contraction of' its wall and relaxation of the sphincter of Oddi.

5- Vagus stimulation causes secretion from the stomach (the gastric juice), the pancreas (the pancreatic juice and the hormones insulin and glucagon), the duodenum (mucus from Brunner's glands) and the liver (bile).

6- In the liver it also produces V.D. (thus increasing the hepatic blood Flow).

2- The sacral parasympathetic outflow (sacral autonomic nerves):

This part of the parasympathetic N.S. originates from the 2nd, 3rd and 4th sacral segments, from which preganglionic fibers arise and pass through the sympathetic chain without relay, then unite together forming the ***pelvic nerve***. This nerve relays in ***terminal*** (hypogastric) ganglia from which postganglionic fibers arise and supply the following organs:

1- **The *urinary bladder*:** Parasympathetic stimulation leads to contraction of the muscle in the wall of the urinary bladder (= detrusor muscle) and relaxation of the internal urethral sphincter (resulting in ***micturition***).

2- **The *rectum*** and distal part of the descending colon: Parasympathetic stimulation leads to contraction of the walls of these organs and relaxation of the internal anal sphincter (resulting in ***defecation***).

3- **The seminal vesicles and *prostate*:** Parasympathetic stimulation leads to ***secretion*** from these glands.

4- **The *erectile tissues*** (the penis in males and the clitoris in females): Parasympathetic stimulation produces V.D. in these tissues (in addition to other pelvic viscera) leading to their ***erection.***

Chemical transmitters in ANA

Definition:

These are the substances which transmit the nerve impulse from presynaptic to postsynaptic membrane.

The main chemical transmitters in ANS are:

a) Acetyl choline (Cholinergic neurotransmission).
b) Noradrenaline (noradrenergic neurotransmission).

A. Cholinergic transmission:

Synthesis of acetyl choline:

- ⇨ Occurs In cytoplasm of synaptic knobs as follows:

Acetyl CoA + Choline → Acetyl choline + CoA

- ⇨ This reaction catalyzed by choline acetyl transferase enzyme

Storage: after synthesis of acetyl choline, it stored inside vesicles.

Release of Acetyl choline:

Ca^{++} channels in the presynaptic membrane open → Ac.ch. Release by exocytosis.

Binding to receptors: after release of acetyl choline, it crosses the synaptic cleft and binds to cholinergic receptors on the postsynaptic membrane.

Fate of Acetyl choline:

- ➢ Degradation of Ac.ch. by acetyl choline esterase:

Ac.ch. → choline + acetate

- ➢ Then choline is reuptaken by presynaptic neurons and Recycled resynthesis of Ac.ch.
- ➢ Diffusion into the blood then degraded by psudocholine estrase??

Sites of release of ac.ch. (cholinergic fibers):

1) Central cholinergic fibers:

- All autonomic ganglia, at the endings of all preganglionic neurons (sympathetic and parasympathetic) including those supplying the adrenal medulla.
- The neuromuscular junctions (= motor end plates) i.e. at the endings of the somatic motor nerves that supply skeletal muscles.

2) Peripheral cholinergic fibers:

- All parasympathetic postganglionic nerve endings.
- The sympathetic postganglionic nerve endings that supply the eccrine sweat glands and skeletal muscles blood vessels.

Types of cholinergic receptors:

1- Muscarinic receptors:

⇨ This name from muscarine, a substance which has the same action as ac.choline in peripheral cholinergic sites.

2- Nicotinic receptors:

⇨ Name from nicotine which in small dose has the same action of ac.choline in central cholinergic sites.

B. Adrenergic transmission:

- Most sympathetic postganglionic neurons liberate noradrenaline (so these fibres are called noradrenergic fibers), but some of these neurons are cholinergic?

Steps of synthesis of noradrenaline:

1- The amino acid phenylalanine is hydroxylated into the amino acid tyrosine by activity of the enzyme phenylalanine hydroxylase.
2- Tyrosine is then hydroxylated to DOPA (=dihydroxy phenyl alanine) by activity of the enzyme tyrosine hydroxylase.
3- DOPA is converted to dopamine by activity of the enzyme **DOPA decarboxylase**.
4- Dopamine is oxidized to noradrenaline (= norepinephrine) by activity of the enzyme dopamine beta hydroxylase.
5- Adrenaline (= epinephrine) can then be formed in the adrenal medulla (and also by certain neurons in C.N.S.) by methylation of noradrenaline under effect of the enzyme (= Phenylethanolaminc-N-Methyl-Transferase).

Both adrenaline and nor adrenaline as well as dopamine are known as **catecholamines**.

Inactivation (or fate) of catecholamines:

This occurs by the following mechanisms:

a) Active re-uptake into the vesicles at the nerve endings which have released them.
b) Oxidation of both adrenaline and noradrenaline by activity of the enzyme mono-amine oxidase (**MAO**) to dihydroxy-mandelic acid.
c) Methylation of both adrenaline (= epinephrine) and noradrenaline (= norepinephrinc) into metanephrine nor-metanephrine respectively by activity of the Catechol-O-Methyl-Tramferase enzyme (**COMT**).

Q: 1- What is pheochromocytoma?
2- Antidepressant?

Types of adrenergic receptors:

The various effects of catccholamines arc produced through acting on 2 types of receptors in the tissues called the ***alpha*** and **beta** receptors.

1- **The alpha receptors:**
- ⇨ There are 2 main subtypes of these receptors called alpha 1 and alpha 2 receptors.
- ⇨ Both generally produce excitatory effects e.g. V.C. of blood vessels (arterioles and systemic veins) and contraction of muscles (e.g. the dilator pupillae muscle, erector pilae muscles, various sphincters, spleen capsule muscles and those which cause ejaculation of semen).
- ⇨ However, in certain sites they produce inhibitory effects (particularly the alpha 2 receptors) e.g. in the stomach and intestinal walls and the islets of Langerhans of the pancreas.

2- **The beta receptors:**
- ⇨ There are 2 main subtypes of these receptors called **beta 1** and **beta 2** receptors.
- ⇨ However, there are also beta 3 receptors in some areas (specially in the adipose tissue).
- ⇨ ***Beta 1 receptors:*** These produce mainly excitatory effects and are located mainly in the heart (in addition to some other tissues e.g. the adipose tissue and the juxtaglomerular cells in the kidneys). However, in certain areas they produce inhibitory effects in the intestinal wall.
- ⇨ ***Beta 2 receptors***: These generally produce inhibitory effects e.g. V.D. of vessels (arterioles and systemic veins) and relaxation of the bronchial muscle, ciliary muscle, the spleen capsule muscles and the walls of the stomach, intestine and urinary bladder. However, in certain site they produce excitatory' effects e.g. in the liver, heart and islets of Langerhans of the pancreas.

Body temperature regulation

INTRODUCTION

Living organisms are classified into two groups, depending upon the maintenance (regulation) of body temperature:

1. Homeothermic animals.
2. Poikilothermic animals.

Homeothermic (warm blooded) animals:

Homeothermic animals are the animals in which the body temperature is maintained at a constant level irrespective of the environmental temperature. Birds and mammals including man belong to this category. They are also called **warm blooded animals.**

Poikilothermic (cold blooded) animals:

Poikilothermic animals are the animals in which the body temperature is not constant. It varies according to the environmental temperature. Amphibians and reptiles are the poikilothermic animals. These animals are also called **cold blooded animals.**

MEASUREMENT:

Body temperature can be measured by placing the **clinical thermometer** in different parts of the body such as:

1) Mouth (oral temperature).
2) Axilla (axillary temperature).
3) Rectum (rectal temperature).

Normal body temperature

Normal body temperature in human is 37°C (98.6°F), when measured by placing the clinical thermometer in the mouth (oral temperature). It varies between 35.8°C and 37.3°C (96.4°F and 99.1°F).

Temperature at different parts of the body:

Axillary temperature is 0.3°C to 0.6°C (0.5°F to 1°F) lower than the **oral temperature.** The **rectal temperature** is 0.3°C to 0.6°C (0.5°F to 1°F) higher than oral temperature.

Variations of body temperature:

I. **Physiological Variations**:

1. ***Age***:
   - In infants, the body temperature varies in accordance to **environmental temperature** for the first few days after birth. It is because the temperature regulating system does not function properly during infancy.
   - In children, the temperature is slightly **(0.5°C**) more than in adults because **of more physical activities**.
   - In old age, since the heat production is less, the body **temperature decreases** slightly.

2. ***Sex***:
   - In females, the body temperature is **less** because of **low basal metabolic** rate, when compared to that of males.

3. ***Diurnal variation***
   - In early morning, the temperature is 1°C less.
   - In the afternoon, it reaches the maximum (about 1°C more than normal).

4. ***After meals:*** The body temperature rises slightly (0.5°C) after meals.

5. ***Exercise***: During exercise, the temperature raises due to production of heat in muscles.

6. ***Sleep***: During sleep, the body temperature decreases by 0.5°C.

7. ***Emotion:*** During emotional conditions, the body temperature increases.

8. ***Menstrual cycle:*** In females, immediately after ovulation, the temperature rises (0.5°C to 1°C) sharply.

II. **Pathological Variations:**

⇨ Abnormal increase in body temperature is called ***hyperthermia*** or **fever** and decreased body temperature is called ***hypothermia.***

Heat balance

Regulation of body temperature depends upon the balance between heat produced in the body and the heat lost from the body.

I. **Heat gain or heat production in the body**:
Various mechanisms involved in heat production in the body are:

1. ***Metabolic Activities***:
   - ⇨ Major portion of heat produced in the body is due to the metabolism of food stuffs. It is called **heat of metabolism.**
   - ⇨ Heat production is more during metabolism of fat. About 9 calories of heat is produced during metabolism of fats. Carbohydrate metabolism produces 4.7 calories of heat. Protein metabolism produces 4.5 calories.

2. ***Muscular Activity***:
   - ⇨ Heat is produced in the muscle both at rest and during activities. During rest, heat is produced by muscle tone.
   - ⇨ Heat produced during muscular activity is called **heat of activity.** About 80% of heat of activity is produced by skeletal muscles.

3. ***Role of Hormones***:
   - ⇨ Thyroxine and adrenaline increase the heat production by accelerating the metabolic activities.

4. ***Radiation of Heat from the Environment***:
   - ⇨ Body gains heat by radiation. It occurs when the environmental temperature is higher than the body temperature.

5. ***Shivering***
   - ⇨ Shivering refers to shaking of the body caused by rapid involuntary contraction or twitching of the muscles as during exposure to cold. Shivering is a compensatory physiological mechanism in the body, during which enormous heat is produced.

6. ***Brown Fat Tissue***
   - ⇨ Brown adipose tissue is one of the two types of adipose tissues, the other being white adipose tissue.

II. Heat loss occurs by the following methods:

Maximum heat is lost from the body through skin and small amount of heat is lost through respiratory system, kidney and GI tract. When environmental temperature is less than body temperature, heat is lost from the body.

1. ***Conduction***:
   - ⇨ Three percent of heat is lost from the surface of the body to other objects such as chair or bed, by means of conduction.

2. ***Radiation***:
   - ⇨ Sixty percent of heat is lost by means of radiation, i.e. transfer of heat by infrared electromagnetic radiation from body to other objects through the surrounding air.

3. ***Convection***:
   - ⇨ Fifteen percent of heat is lost from body to the air by convection. First the heat is conducted to the air surrounding the body and then carried away by air currents, i.e. convection.

4. ***Evaporation – Insensible Perspiration***:
   - ⇨ When water evaporates, heat is lost. Twenty two percent of heat is lost through evaporation of water. Normally, a small quantity of water is continuously evaporated from skin and lungs. We are not aware of it. So, it is called the insensible perspiration or insensible.

5. ***Panting***:
   - ⇨ Panting is the rapid shallow breathing, associated with dribbling of more saliva. In some animals like dogs which do not have sweat glands, heat is lost by evaporation of water from lungs and saliva by means of panting.

Regulation of body temperature:

Body temperature is regulated by ***hypothalamus***, which sets the normal range of body temperature. The set point under normal physiological conditions is 37°C.

Hypothalamus has two centers which regulate the body temperature:

1. Heat loss center.
2. Heat gain center.

1) **Heat loss center**:
   - ⇨ Heat loss center is situated in **preoptic nucleus** of anterior hypothalamus.
   - ⇨ Neurons in preoptic nucleus are heat sensitive nerve cells, which are called **thermoreceptors.**
   - ⇨ Stimulation of preoptic nucleus results in cutaneous vasodilatation and sweating.
   - ⇨ Removal or lesion of this nucleus increases the body temperature.

2) Heat gain center
   - ⇨ Heat gain is otherwise known as heat production center. It is situated in **posterior hypothalamic nucleus.**
   - ⇨ Stimulation of posterior hypothalamic nucleus cause shivering.
   - ⇨ The removal or lesion of this nucleus leads fall in body temperature.

Mechanisms of temperature regulation:

1. When Body Temperature Increases

- When body temperature increases, blood temperature also increases. When blood with increased temperature passes through hypothalamus, it stimulates the thermoreceptors present in the heat loss center in preoptic nucleus. Now, the heat loss center brings the temperature back to normal by two mechanisms:
  A. **Promotion of heat loss.**
  B. **Prevention of heat production**.

***A) Promotion of heat loss*:**

- When body temperature increases, heat loss center promotes heat loss from the body by two ways:
  i. By increasing the secretion of sweat: When sweat secretion increases, more water is lost from skin along with heat.
  ii. By inhibiting sympathetic centers in posterior hypothalamus: This causes cutaneous vasodilatation. Now, the blood flow through skin increases causing excess sweating.

B) Prevention of heat production

- Heat loss center prevents heat production in the body by inhibiting mechanisms involved in heat production, such as shivering and chemical (metabolic) reactions.

***2.* When Body Temperature Decreases**

- When the body temperature decreases, it is brought back to normal by two mechanisms:
  1. **Prevention of heat loss.**
  2. **Promotion of heat production.**

A) *Prevention of heat loss*

- When body temperature decreases, sympathetic centers in posterior hypothalamus cause cutaneous vasoconstriction. This leads to decrease in blood flow to skin and so the heat loss is prevented.

B) *Promotion of heat production*

- Heat production is promoted by two ways:
  i. *Shivering:* When body temperature is low, the heat gain center stimulates the primary motor center for shivering, situated in posterior hypothalamus near the wall of the 3rd ventricle and shivering occurs. During shivering, enormous heat is produced because of severe muscular activities.

  ii. *Increased metabolic reactions:* Sympathetic centers, which are activated by heat gain center, stimulate secretion of adrenaline and nor adrenaline?

Metabolism

- There is a balance between energy intake (Anabolism) and energy output (Catabolism).
- This results in constant body weight.
  - More anabolism → ↑ body weight e.g. in children.
  - More catabolism → ↓ body weight e.g in hyperthyroidism

- Anabolism

- It is the conversion of small molecules **with energy consumption** to large molecules.

e.g. Amino acids → Proteins.
Glucose → Carbohydrates.
Fatty acids → Fats

- Catabolism

- It is the conversion of large molecules **with energy production** into small molecules.

$$\text{Glucose } (C_6H_{12}O_6) \rightarrow 6CO_2 + 6H_2O + \text{Energy}$$

Metabolism :

- All chemical reactions occurring in living cells that make it possible to continue living.
- These chemical reactions lead to release of large amount of energy.

Fate of Energy liberated by catabolism:

1) Formation of energy stores as ATP, Creatine phosphate & Glycogen (27%).
2) Liberated as heat.
3) Used directly by cells for certain work as:
   - Membrane transport (active transport).
   - Muscle contraction.
   - Glandular secretion (HCl secretion by gastric glands).
   - Synthesis of proteins.
   - Synthesis of hormones.
   - Conduction of nerve impulses.
   - Cell division and growth.

- The amount of energy is expressed in **calories**.

Calorie (C)

- It is the Amount of heat necessary to raise the temperature of 1Kg of water one °C from 15°C to 16°C.
- Kilocalorie (Kc or C) = 1000 calorie (c).

Physical and physiological heat

- Any substance (carbohydrate, fat or protein) can be oxidized inside the body or outside the body with liberation of heat.
- **Physical heat :** is the amount of **heat** in calories which are produced by complete oxidation of food outside the body (by burning with pure O_2).

  This can be measured using **Bomb calorimeter** in which the food is ignited by an electric spark.
- **Physiological heat*:*** is the amount of **energy** in calories which are produced by complete oxidation of food inside the body.

Substance	Physical heat value in Cal. / gram	Physiological heat value in Cal. /gram
Carbohydrate	4.1	4.1
Fat	9.3	9.3
Protein	5.3	4.1 only

We notice that :

1. In the body, calorie values obtained for carbohydrate & fat are similar to that outside the body.
2. The Physiological heat value of proteins is less than their Physical heat value **why?**
   - Because the nitrogen of protein (which represents 16% of the protein molecule) are not oxidized because its oxidation would produce large amounts of nitric acid → ↓ Blood pH to fatal levels. So it is converted into Nitrogenous substances mainly urea which is excreted in urine.

Respiratory quotient (RQ)

- Definition:

- It is the ratio between the volume of CO_2 produced to the volume of O_2 consumed in the same time of metabolism.

$$RQ = \frac{\text{the volume of CO2 produced}}{\text{the volume of O2 consumed}} \text{ in same time.}$$

Significance of RQ determination:

a) The RQ is an index of the nature of the substances oxidized in different tissues and conditions.
b) Calculation of metabolic rate.
c) RQ estimation has been used as an evidence for the transformation of one food substance into another.

1. The RQ is an index of the nature of the substances oxidized in different tissues and conditions :

Substance oxidized	RQ
carbohydrates	*1.0*
protein	*0.8*
fat	*0.7*

- **For carbohydrates:**

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$$

$$RQ = \frac{6\ CO2}{6\ O2} = 1$$

- So, for each molecule of O_2 consumed, there is a molecule of CO_2 is produced.
- This is because carbohydrates contain enough O_2 in their molecules to oxidize all the H^+ atoms present & form H_2O. Extra O_2 is needed from the atmosphere, is used to oxidize carbon atom and appears as CO_2.

- **For fats :**

$$C_{18}H_{36}O_2 + 26\ O_2 \rightarrow 18\ CO_2 + 18\ H_2O + \text{energy}$$

$$RQ = \frac{18\ CO2}{26\ O2} = 0.7$$

- This is because oxygen atoms present in fat & protein molecules are less than that for carbohydrates. So, excess O_2 is needed from atmosphere to oxidize the hydrogen and carbon atoms.
- For proteins RQ was found to be 0.8.
- Mixed food → RQ = 0.85.

❖ *RQ of the brain is 1.0 (0.97- 0.99) indicating that glucose is its principal fuel.*
❖ *RQ in diabetes mellitus is 0.7 indicating that fat is the chief source of energy.*

2. Calculation of metabolic rate :

- By determination of the O_2 Heat value from RQ (***by metabolic tables***).

Metabolic rate = O_2 Heat value X O_2 consumed.

Table (2)

R.Q.	O_2 Heat value (Calories)	Fat %	Carbohydrate %
0.70	4.70	100	Zero
0.75	4.75	84	16
0.80	4.80	67	33
0.82	4.82	60	40
0.85	4.85	50	50
0.90	4.90	33	67
0.95	4.95	16	84
1.00	5.00	Zero	100

O_2 Heat value: is the amount of heat in calories resulting from oxidation of a food substance using one liter of O_2.

3. RQ estimation has been used as an evidence for the transformation of one food substance into another:

- Transformation of carbohydrates into fats (lipogenesis) → RQ above unity. (A *substance rich in O_2 is converted into a substance poor in O_2 so, O_2 consumption is ↓ and CO_2 is produced without atmospheric O_2 consumption*).
- Transformation of fats or proteins into carbohydrates (gluconeogenesis) → RQ below unity. (*due to ↑ O_2 consumption & ↓ and CO_2 production)*

Factors affecting the RQ:

1. Factors increasing the RQ above unity:

a) **Immediately after a meal:** RQ approaches unity because the main food that is metabolized is carbohydrates.

b) **Conversion of carbohydrates into fats**.

c) **Early stages of severe muscular exercise**: the RQ as high as 2 due to:

- Accumulation of lactic acid which when buffered by $NaHCO_3$ gives CO_2. (So, there is CO_2 production without O_2 consumption).

Lactic acid + $NaHCO_3$ → Na lactate + CO_2 + H_2O

- Hyperventilation.

d) Metabolic acidosis: because the acids will be buffered by $NaHCO_3$ & also due to hyperventilation.

e) **Hyperventilation**: due to CO_2 wash more than O_2 uptake.

2. Factors decreasing the RQ below unity:

a) ***8-10 hours after a meal:*** the body already used up most of its carbohydrates & the main food that is metabolized is fat (RQ approached 0.7).

b) ***Conversion of fats into carbohydrates***.

c) ***Recovery from muscular exercise***:

- Due to oxidative removal of lactic acid.
- CO_2 produced is used for reformation of bicarbonate.

So, O_2 consumed ↑ while CO_2 produced ↓.

d) Metabolic alkalosis: due to hypoventilation.

e) ***Hypoventilation***: due to CO_2 retention.

f) ***Starvation***.

g) ***Diabetes mellitus***: due to:

- ↓ Carbohydrate metabolism.
- ↑ Fat metabolism.
- Ketone bodies formation (rich in O_2) from fat (poor in O_2).

Excess RQ

- Definition :

- It is the RQ of the excess metabolism due to muscular exercise.

- It is determined as follows:

$$\text{Excess RQ} = \frac{A-a}{B-b}$$

Where :

A= total CO_2 produced during the whole period of exercise & recovery.
a = CO_2 produced during a similar period of rest.
B= total O_2 consumed during the whole period of exercise & recovery.
b = O_2 consumed during a similar period of rest.

- Significance of excess RQ:

- It indicates the **duration of muscular exercise** & **type of food oxidized during it**.
  - **Excess RQ of unity** → means that the exercise is of short duration & the energy is mainly derived from oxidation of carbohydrates.
  - **Excess RQ of unity then drops gradually** → means that the exercise is of long duration & the energy is derived from oxidation of carbohydrates & fats.
  - **Excess RQ below unity from the start** → means that the only available fuel is fat as in diabetes and starvation.

- Factors affecting the excess RQ:

1. Duration of muscular exercise.
2. Type of food oxidized during it.

Basal metabolic rate (BMR)

- Definition :

It is the amount of energy production in unit time (one hour) under basal conditions.

- It represents the amount of energy required to perform all the essential activities of the body systems as CNS, heart, respiration, kidneys & skeletal muscle tone i.e. it represents the unavoidable cost of life.
- It isn't the minimal energy expenditure because it can be reduced by many factors as sleep.
- It represents 60% of the daily energy expenditure in sedentary individuals.

- Metabolic rate increased by:

- Exercise.
- Eating.
- Exposure to cold.

So, BMR is measured under the following standard basal conditions:

1. **Complete physical and mental rest:**
   - The person lies comfortably in bed for at least 30 minutes before measurement.
   - He must be awake.
   - Stress should be avoided.
2. **Postabsorptive state:**
   - **12:14** hours after the last meal to avoid the stimulating effect of food on metabolic rate (specific dynamic action).
3. **Comfortable external temperature:**
   - Comfortable zone of temperature 20 - 25 ° C for dressed person & 28 – 31 ° C when naked.

- Measurement :

- Measured by direct or indirect calorimeter and divide on body surface area.
- BMR is expressed in …… Calories / m^2 / hr.
- Normally in adults = 40 Calories / m^2 / hr ± 15 %.
- Clinically, BMR is expressed as a % deviation from the normal standard thus, BMR of 60 Calories / m^2 / hr is expressed as + 50% above the normal standard.
- Average adult must take about 2000 C/day to meet his basal needs.

-Factors affecting BMR:

a) Physiological factors:

1) Age:

- In A newly born infant ➔ 25 Calories / m^2 / hr (so infants are greatly susceptible to cold).
- **2-5 years ➔ 60 Calories / m^2 / hr**. (the highest BMR)
- At puberty ➔ 55 Calories / m^2 / hr.
- **20 years ➔ 40 Calories / m^2 / hr.**
- After 20 years ➔ BMR is decreased by 1 calorie for every 10 years (due to ↓ in muscle mass and its replacement by fat).

2) Sex:

- BMR in females is lower than in males by 7 %.
- It is not due to hormonal difference because it is present in children.
- It may be due to greater stores of fat in females which have a little metabolic activity.

3) Physical habits: Athletics have higher BMR (by 10 %) than ordinary persons due to ↑in muscle mass.

4) Dietetic habits: The habit of eating protein rich diet for long period ↑ BMR by 10% than eating carbohydrates.

5) Climate: BMR is higher in cold and temperate zones (e.g. Eskimo) than in hot zones by 10%.

6) Race:

- Pure races e.g. Chinese & Indians have lower BMR than mixed races as Americans & Egyptians.
- Dark races have lower BMR than white races.

7) Pregnancy:

- BMR ↑ gradually during pregnancy until it is doubled at the end of pregnancy due to:
  - ✓ Increased thyroxine, cortisol & sex hormones.
  - ✓ Added metabolism of fetus.

8) Sleep: ↓ BMR by about 10%.

b) Pathological factors:

Factors that ↑ BMR	*Factors that ↓ BMR*
1. *hyperthyroidism:* **↑ BMR up to + 100% or higher.** **Because thyroxin is the most powerful stimulant for cell metabolism.**	***1. hypothyroidism:***
2. hyperpituitarism: **due to ↑ growth hormone & TSH (both ↑ body metabolism)**	***2. hypopituitarism:***
3. hyperadrenalism: **Hypersecretion of suprarenal medulla as in Pheochromocytoma ➔ excessive secretion of adrenaline.**	***3. hypocorticism:*** ↓ Secretion of cortisol which stimulates metabolism.
4. Fever: **↑ Body temperature by 1 ° C ➔ ↑ BMR by 13%.**	***4. hypothermia:*** ↓ Body temperature by 1 ° C ➔ ↓BMR by 13%.
5. Blood disease: **Overactivity of the bone marrow ➔ marked ↑ BMR as in anaemia, polycythaemia & leukaemia.**	***5. malnutrition***
6. Heart failure: **↑ BMRdue to dyspnea ➔ ↑ activity of respiratory muscles.**	
7. Diabetes insipidus: **↑ BMR to maintain body temperature constant by raising the temperature of the large amounts of water taken orally which are usually of temperature lower than body temperature.**	

c) Chemical factors :

1. Thyroxin.
2. Adrenaline.
3. Amphetamine.

Q: **what are the possibilities of an adult person with a metabolic rate of 28 Cal. / m^2 /hr?**

❖ Specific dynamic action

- Definition :

- It is the power of the food to stimulate the metabolic rate (thermic effect of food).
  - The metabolic rate starts to ↑ 1 hour after taking food.
  - Reaches the maximum after 3:4 hours.
  - ↓ to basal levels 12:18 hours after food intake.
- It results from the different chemical reactions associated with digestion, absorption and storage of food in the body.

- It depends on:

1. Type of food ingested:

- It is 30% for protein.
- 6% for carbohydrates.
- 4% for fat.

food	Physiological heat	SDA
fat	9.3	4%
carbohydrates	4.1	6%
protein	4.1	30%

2. Quantity of food:

- SDA is proportional to the quantity.

3. External temperature:

- If external temperature is 33° C ➔ SDA is maximal.
- If external temperature is 5° C ➔ SDA disappears because it is retained in the body to maintain body temperature.

-N.B:

Energy released due to SDA is a **waste heat** but it is useful on exposure to cold. It **helps in maintaining body temperature constant** so; high protein diet results in discomfort in hot weather but not in cold one.

NOTES

1. **Liver** is essential for **SDA.**
2. GIT glands or endocrine glands aren't a cause of **SDA.**
3. **SDA** isn't related to the biological value of proteins i.e. A low biological value protein produces the same **SDA** as a high biological value protein.

Food intake

- Control of food intake is achieved by a specific center in hypothalamus and limbic system. It is formed of:
  a) **Feeding center**:
     - In lateral hypothalamus.
     - Its stimulation → ↑ appetite and food intake.
     - Its destruction → anorexia.
  b) **Satiety center**:
     - In ventromedial hypothalamus.
     - Its stimulation → ↓ appetite and food intake.
     - Its destruction → hyperphagia.

- Normally the feeding center is continuously active and inactivated only when the **satiety center is stimulated.**

- In addition, the following hypothalamic nuclei play a major role in regulating food intake:
  - **The paraventricular** : its lesions often cause excessive eating.
  - **Dorsomedial**: its lesions usually depress eating behavior.
  - **Arcuate nuclei**: are the sites where multiple GIT hormones and adipose tissue converge to regulate food intake, as well as energy expenditure.

- These hypothalamic nuclei also influence the secretion of several hormones that are important in regulating energy balance and metabolism, including thyroid and adrenal & pancreatic hormones.

- Two types of neurons concerned with regulation of appetite and energy expenditure in the arcuate nuclei :

A. pro-opiomelanocortin (POMC) neurons: produce

- ✓ α – melanocyte-stimulating hormone (α-MSH).
- ✓ Cocaine- and amphetamine-related transcript (CART).

Their Activation decreases food intake and increases energy expenditure.

B. NPY-AGRP Neurons that produce the orexigenic substances:

- ✓ Neuropeptide Y (NPY).
- ✓ Agouti-related protein (AGRP).

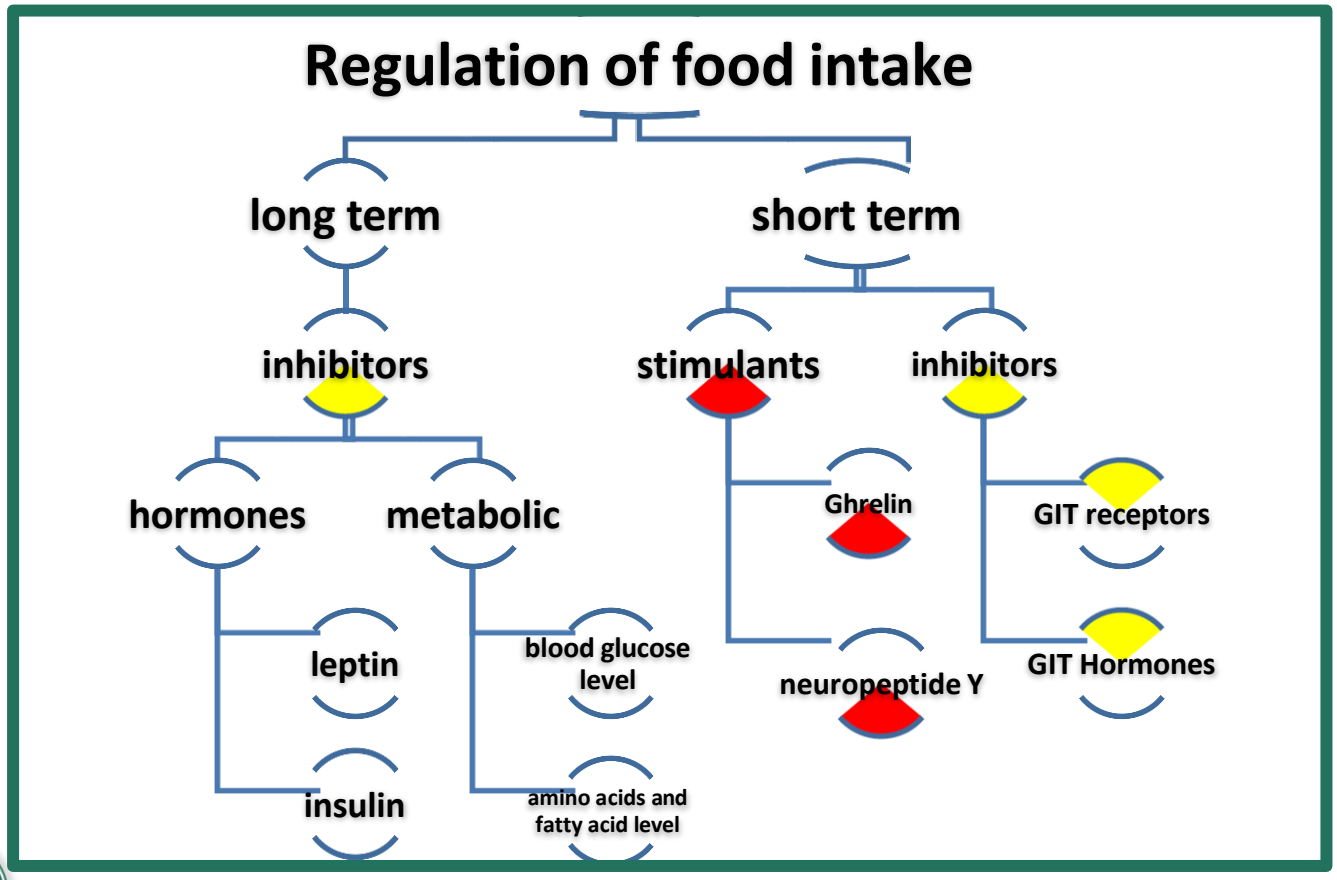
Their activation increases food intake and reducing energy expenditure.

The POMC neurons in arcuate nuclei release α-MSH → acts on melanocortin receptors (MCR-3 and MCR-4) on the paraventricular nuclei → reduces food intake & increase energy expenditure.

Regulation of food intake

A. Short term regulation: concerned primarily with preventing overeating at each meal.

B. Long-term regulation: concerned primarily with maintenance of normal quantities of energy stores in the body.



A. Short term regulation:

1. **GIT receptors**:
   - ✓ Present in buccal cavity (metering mechanism), oesophagus, stomach and intestine.
   - ✓ Stimulation of these receptors e.g. by distension of the stomach → vagal afferent→ stimulates the satiety center and decrease food intake.
2. **GIT hormones**: as
   - ✓ **Cholecystokinin** (CCK) from I cells in duodenal mucosa.
   - ✓ **Peptide YY** (longer duration than CCK): from the entire gastrointestinal tract, but especially from the ileum and colon.
   - ✓ **Ghrelin** hormone: released from gastric mucosa and pancreas during fasting and stimulates feeding.

CCK is short acting hence can't be used in treatment of obesity. It acts by exciting sensory ending in duodenum that travel in vagus nerve → decrease appetite and food intake

B. Long term regulation:

1. **Blood glucose**:
   - ★ **Hyperglycaemia** → stimulates the satiety center but in diabetics there is hyperphagia inspite of high blood glucose level because the satiety center needs insulin for glucose transport.
   - ★ **Hypoglycamia** stimulates the feeding center.

2. **Amino acids and fatty acid levels**: inhibit feeding by negative feedback. The level of fatty acids is the most important long term inhibitor.

3. **Thermoreceptors**:
   - ★ Exposure to cold → stimulate feeding center.
   - ★ Exposure to hot → stimulate satiety center.

   This phenomenon is caused by interaction within the hypothalamus between the temperature-regulating system and the food intake–regulating system.

4. **Hormones**:
   - ★ **Insulin**, calcitonin → decrease appetite and food intake by acting on their central receptors in the brain.
   - ★ **Glucagon** → increase appetite and food intake.

★ **Leptin**: from adipose tissue and cross BBB by facilitated diffusion and act on POMC and AGRP/NPY neurons of the arcuate nuclei and paraventricular nuclei → inhibit feeding center to control body weight and increase energy expenditure through inhibition of neuropeptide Y and AGRP & activation of POMC neurons.

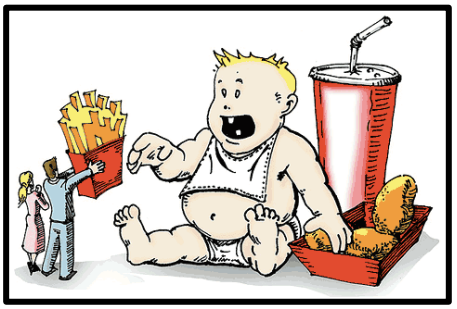
5. **Drugs**:
   - ★ Amphetamine stimulates the satiety center.
6. **Psychological factors.**

Obesity

- Definition:

- ➢ Deposition of excessive amount of fat in storage areas of the body.

- Functions of Fat tissue:



- Gives the body its shape.
- Protects the viscera from external trauma.
- Insulator.
- Energy reserve.
- It is no longer considered only an energy storage tissue but a major endocrine organ, secretes leptin, several cytokines, angiotensinogen, plasminogen activator inhibitor-1 (PAI-1), adiponectin, resistin and others.

Obesity:

- Occurs When energy input **>** energy output.
- It is said to be present if :
  - ➢ Fat = 25% in males.
  - ➢ Or = 35% in females.

∗ This is known by estimating the ***body mass index*** (*BMI*).

$$BMI = \frac{weight\ (in\ Kg)}{Height\ (in\ m2)}$$

- BMI of 25 – 29.9 is considered overweight.
- BMI of 30 or more is obesity.

- Types of obesity :

1) **Hyperplastic type : (↑ No. of adipocytes)**
   - Occurs in children as the number of adipocytes ↑ only in childhood.
   - So excessive energy intake in children leads to the hyperplastic type.
2) **Hypertrophic type :**
   - Occurs in adults.
   - There ↑ in the size of the adipocytes.

Research studies have shown that new adipocytes can differentiate from fibroblast-like preadipocytes at any period of life and that the development of obesity in adults is accompanied by increased numbers, as well as increased size, of adipocytes.

- Causes of obesity



1) **Hyperphagia:**
   - ↑ in food intake with energy expenditure remains normal.
   - It is the most common cause of obesity.
   - It may be due to psychological interference with the normal function of the hypothalamic feeding center.
2) **Sedentary life style** (relative Hyperphagia):
   - ↓ Energy expenditure with food intake remains normal.
3) **Endocrinal causes :**
   - Hypothyroidism ➔ myxoedema.
   - Hyperinsulinism.
   - Cushing's syndrome.
   - Frohlich's syndrome (adipose-genital syndrome): that affects the satiety center in hypothalamus → hyperphagia.
4) **Psychogenic obesity.**
5) **Genetic obesity:** due to an abnormality in feeding center or its neurotransmitters.

- Complications of obesity

a) Accelerated atherosclerosis.
b) Type II diabetes mellitus.
c) Cancer.
d) Shortness of breath.
e) Osteoarthritis.

- Treatment

- ↓ Energy intake.
- ↑ Energy expenditure.
- Amphetamine (↓ appetite and ↓ hunger pains).
- **Orlistat**: altering lipid absorption by the gut.
- Treatment of the psychological disturbance.
- **Surgical**: reduce the size of the stomach reservoir and/or bypass it altogether

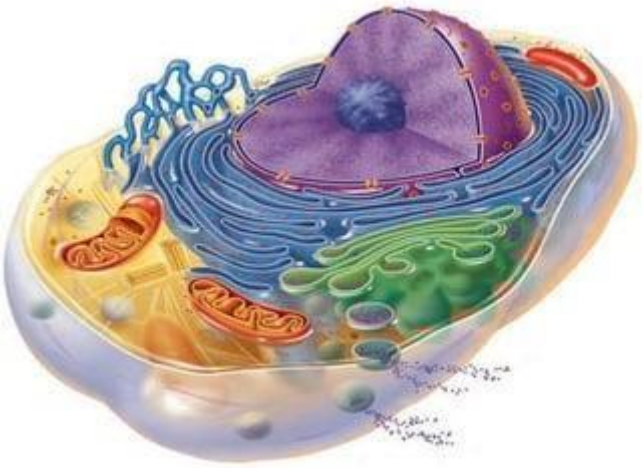
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Sohag University
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Histology department

First Level

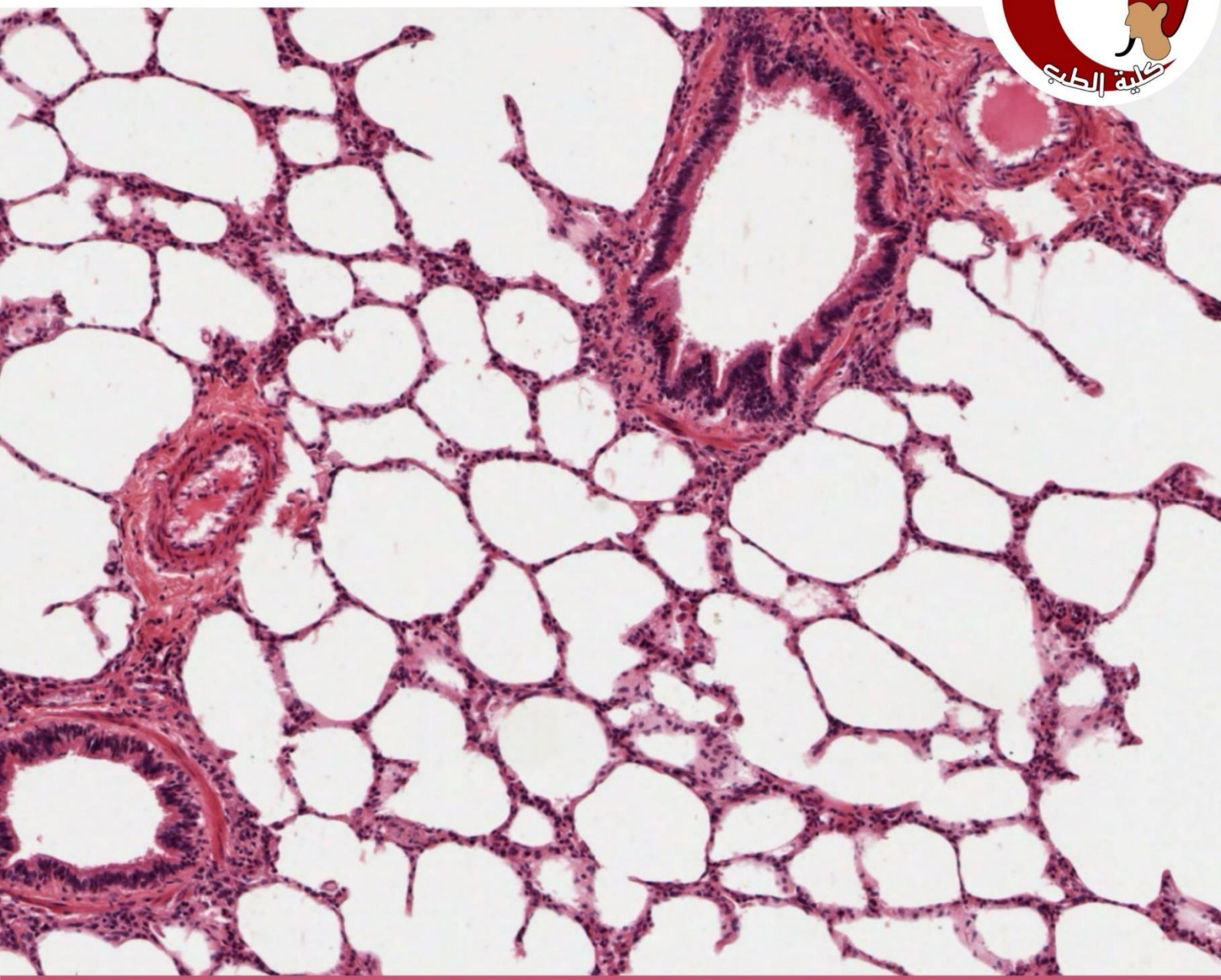
Block CSF-103 Cell Structure and Function



BY
Histology and Cell Biology Department
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2023-2024

Sohag university
Faculty of medicine

General Histology



CSF-103

2024/2025

INTRODUCTION

Biology is the science of studying different aspects of life. A living object is one which is made up of living cells and is characterized by the ability to interact with its environment. The cell is the building and functional unit in the human body. So, we start from the beginning to study the cells.

The human body is formed of human systems (e.g. digestive system). The latter is formed of a number of organs (e.g. stomach) which is formed of a number of tissues (e.g. muscular tissue). The latter is formed mainly of cells (e.g. muscle fibers) which are formed of a number of organelles (e.g. myofibrils). The latter is composed of macromolecules (e.g. actin and myosin) and these are formed of atoms. So, the human body represents the highest of these different levels of organization.

The name "Histology" is derived from the Greek word for a tissue "*Histos*", and "*-logos*" = the study of. Today the concept of Histology as a subject includes far more than just the study of tissues, but includes understanding of the structure and function of cells, tissues, organs and organ systems, which can better be described as "Microscopic Anatomy". It is evident from the title that the study of Histology or Microscopic Anatomy involves the use of microscopes (light and electron) as basic tools. In addition to understanding the histology and ultrastructure of cells, tissues and organs, it is also necessary to complement the morphological observations with an understanding of the biochemistry, physiology and biophysics of these structures.

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1 MICROSCOPY AND MICROTECHNIQUE

Histology: is the science of studying tissues and cells

Microscopy

- The use of any microscope is to serve two main functions; magnification and resolution.

Magnification:

- ✓ is the increase in linear dimensions of an object without optical defects.
- ✓ This is done through 2-stage magnifying system.
- ✓ The specimen is magnified first by many objective lenses system, then by a second lens, ocular lens or eye-piece lens.
- ✓ The total magnification of the instrument is the product of magnification resulting from the objective and the ocular lenses.
- ✓ Magnification should be hand in hand with clarity (resolution) otherwise the image is blurred.

Resolution:

- ✓ is the power of the microscope to distinguish fine details.
- ✓ The limit of resolution (L.r) is defined as the shortest distance between two details below which they appear as one or cannot be seen at all with the microscope.
- ✓ The resolution power of the microscope increases as the limit of resolution decreases.
- ✓ The L.r. of the human eye is 200 um.
- ✓ While that of the ordinary light microscope is 0.24 um and it is 0.24 nm in electron microscopes. 1mm =1000 um 1um=1000 nm

Types Of Microscopes:

1- The ordinary light microscope:

- in which the day or artificial lamp light is used for illumination.
- The optical components consist of 3 systems of lenses: condenser, objective and eye piece. The details of its components are shown in Fig. (1).

2- Phase contrast microscopy:

- It permits direct observation of living cells.
- It is based on the principle that light changes its speed when passing through cellular and extracellular structures with different refractive indices.
- These changes are used by the phase contrast system to cause the structures to appear lighter or darker relative to each other.

3- Fluorescent microscopy:

- Certain substances when irradiated with invisible ultraviolet light, they emit light in the visible portion of the spectrum.
- These substances are called fluorescent and the phenomenon is called fluorescence.
- Some components are auto fluorescent such as vitamin A .
- Fluorescent stains that have affinity to certain macromolecules may be used e.g. acridine orange which can combine with DNA and RNA.

4- Electron microscopy:

- **There are three types of electron microscopes:**

A- Transmission electron microscopes (TEM):

- It is used to examine the fine details of the cells and tissues.
- The illumination source is electron beam emitted by heating a cathode (tungsten filament).It is of very short wave length (λ).
- The beam follows a straight bath through vacuum.
- It is deflected and refracted when it passes through electro-magnetic lenses.
- The L.r is 0.24 nm.

- The magnification power is several thousand times.
- The resulting magnified image is projected on a fluorescent screen and could be photographed.

B- High voltage electron microscope (H.V.E.M.):

- It is supplied with high voltage electric current resulting in emitting electron beam of higher penetrating power.
- It is used to examine relatively thick sections (5 um.) and in studying living cells.

C- Scanning electron microscope (S.E.M.):

- It is used to study the details of surfaces and contours of the cells.
- It produces a very narrow electron beam that is moved or scanned from point to point across the specimen.

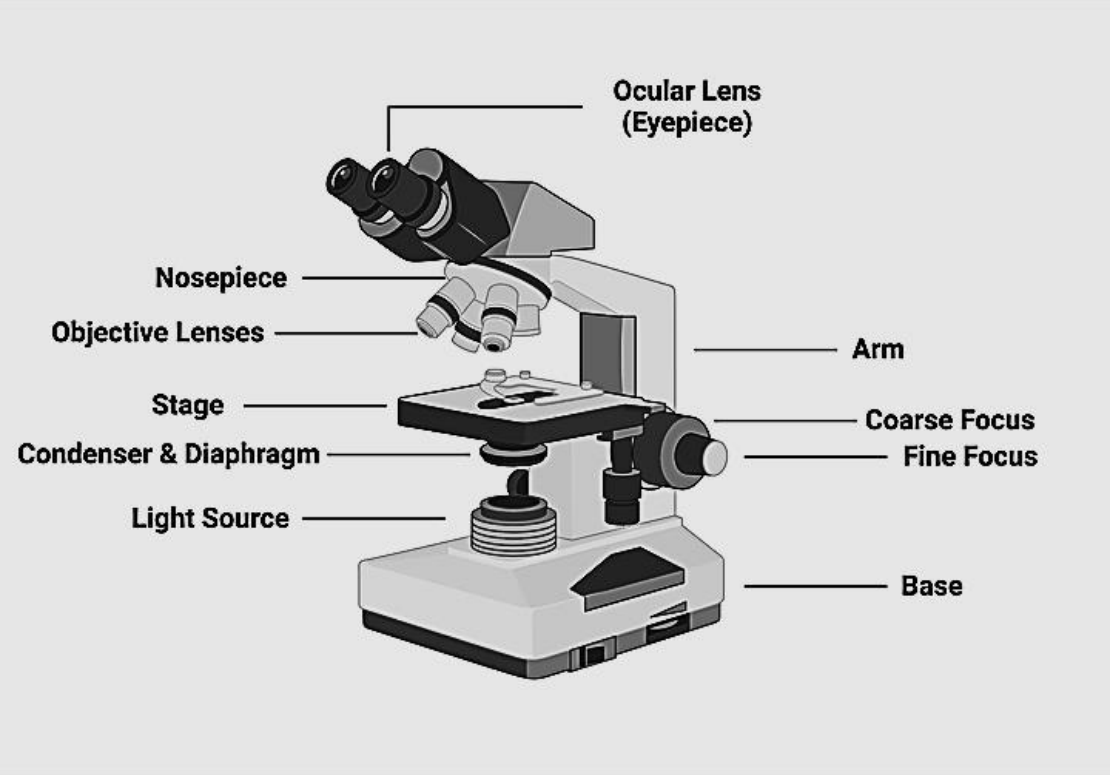


Fig. (1): Schematic drawing of the different parts of the ordinary light microscope (A) and the optical bath through it.

Microtechnique

Methods of studying cells:

- The microscopic study of cells and tissues includes 2 major techniques:

A: Examination of living cells: (tissue culture)

- It is known as in vitro study.
- The piece of tissue is treated with trypsin to dissociate its components and the cells are put in sterile dish with the appropriate culture medium.
- It may be primary, secondary or established cell line.
- **It is used to:**
  - ✓ Study the effect of a toxic substance on certain cells.
  - ✓ To make karyotypes for chromosomal study.
  - ✓ To form clones of cells from one cell.
  - ✓ In established cell line, the cells are kept culture after culture

B: Examination of fixed tissues and cells:

- The piece of tissue (0.5 – 1 cm) is taken by biopsy, surgical excision or postmortem undergoes several steps to be prepared in the form of a histological slide.
- There are several methods used to prepare sections:

[A] Paraffin techniques for examination by light microscope: will be sicussed in setails in practical sessions:

1- **tissue sampling**: as previously mentioned.

2- **fixation of tissue**. : The tissue sample must be put immediately in a fixative e.g. formalin.

Function:

- Prevents postmortem changes (autolysis and putrefaction).
- Fixation hardens the tissue and makes it stainable.
- It decreases the shrinkage and distortion of cells.

- It preserves cell constitutes insoluble and prevents their removal by the subsequent steps.

3- **Washing:** of excess fixative.

4- **Dehydration**: gradual removal of water from the tissue by using ascending grades of alcohol.

5- **Clearing**: to replace a paraffin solvent instead of alcohol e.g. xylol.

6- **Impregnation**: infiltration of the tissue with a hardening material, soft paraffin (melted at 50 – 55 c) in the oven.

7- **Embedding**: in fresh melted hard paraffin using special metal hold then left to cool.

8- **Sectioning**: sections (5 – 10 um) are cut by using the microtome with a very sharp knife. Sections of the accumulated ribbon can be separated and floated on warm water.

9- **Mounting**: sections are mounted on glass slides.

10- **Staining:**

It is essential to distinguish the different components of tissues as they absorb stains on a selective basis.

[B]Frozen sections:

- It is used to prepare fresh and fixed sections where histochemical reactions especially enzyme histochemistry are to be applied.
- **It has two drawbacks;** formation of large ice crystals inside the cell and the difficulty of obtaining permanent preparations.

[C]Freeze drying technique:

- The tissue is immediately frozen in liquid helium (-170c) and is kept under vacuum.
- Water evaporates and the tissue is directly embedded in paraffin.
- **Advantages of this method:** fixation is homogenous and soluble materials are not extracted.

[D] Preparation of tissues for electron microscopy:

1- **Fixation:** cold gluteraldehyde 4% in buffer (overnight) followed by fixation in 1% osmic acid for 1 hour.

2- **Dehydration:** in alcohol

3- **Clearing:** in propylene oxide

4- **Embedding**: in a harden medium to allow cutting very thin sections.

Epoxy resin is an example which is left in the oven to polymerize at 60 c.

5- **Sectioning:** a special cutting instrument, the ultra-microtome is used for cutting thin sections (1-2 um.) by using glass knives and ultra-thin sections (60 – 80 nm.) by using diamond one.

6- **Mounting**: ultra-thin sections are mounted on small rounded copper grids.

7- **Staining:** heavy metals are used which combine with tissue components rendering some of them more electron dense than others. The stains used are uranyl acetate and lead citrate. Osmic acid is a fixative and a satin with an affinity to lipid structures and the outer 2 layers of the cell membrane rendering them electron dense.

[E] Special tissue preparations for microscopy:

a. Filming: used in examination of tissue fluids e.g. blood film.

B. Smearing: used in examination of sticky secretions e.g. bone marrow smear.

c. Teasing: suitable for examination of fibers e.g. muscle and nerve fibers.

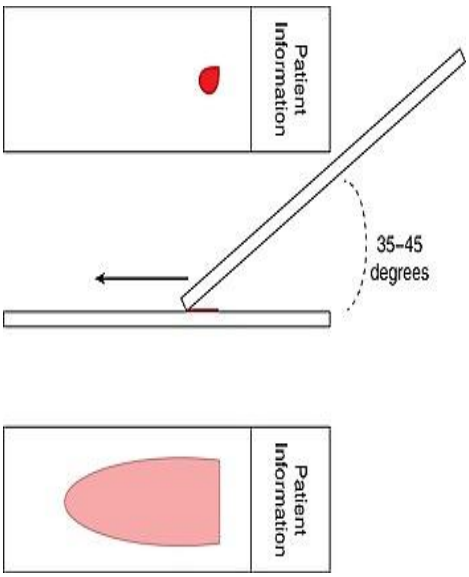
d. Spreading: suitable for examination of thin tissue e.g. loose connective tissue.

e. Grinding: used to examine bone by ground preparation.

Blood film



Teased muscles



Types of staining methods:

1- General histological methods as H&E.

2- Special histological methods: In these methods certain dyes are known to be attached to structures or a particular structure in cells and tissues respectively as silver stain.

3- Vital staining methods:

these include 2 different main techniques:

A-Vital staining technique: staining of tissues in the living state inside the body e.g. staining of macrophages using Indian ink.

B-Supra vital staining technique: staining of tissues in the living state outside the body e.g. staining of mitochondria in a fresh state using Janus green B.

4- Histochemical methods: a particular component such as an enzyme, glycogen or fat is localized inside the cells by using either physical or chemical procedures on histological sections.

The Stain:

- The stain is a coloring agent which can penetrate and be retained by tissue elements.
- For routine uses, hematoxylin and eosin (H&E) are used as general histological stain.
- Hematoxylin gives deep purple satin while eosin gives pink to red color.
- Substance is coloured when it has a specific group called a **"chromophore"** which shows absorption band in the region of visible light.
- The stain can be retained by the tissue components when it has a specific group called **"auxochrome"** of the dye which may be acidic or basic group that can be retained by the stained material by reacting with it.

Basophilic and Acidophilic staining :

- **Hematoxylin** stain is a basic dye that has the ability to stain acidic structures in the cell (e.g. nuclei) and the stained structure is known as basophilic structure.
- Eosin is an acidic dye that has the ability to stain basic structures in the cell (e.g. the cytoplasm) and the stained structure is known as eosinophilic or acidophilic structure.

2 Cytology: Plasma membrane

Introduction

The cell (Fig.1)

- The cell is the structural and functional unit of the body.
- The cells are bound together to form tissues.
- The tissues are combined to form organs.
- Several organs having interrelated functions constitute the different systems of the body as: digestive system, urinary system

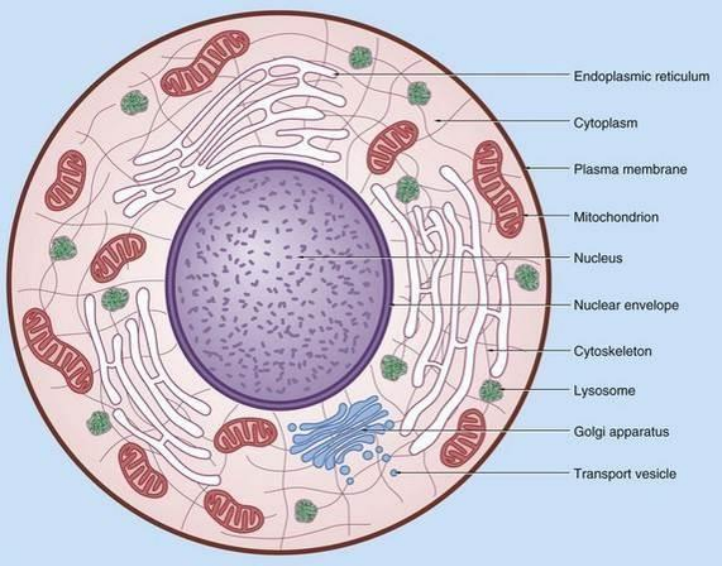


Fig 1; Diagram of the cell components

Size of Cells:

- They are variable in size.
- Some cells are very small as certain cells of the cerebellum, while others are very large as the muscle and ovarian cells.

Shape of Cells:

- The different body cells vary in shape.
- Some cells are rounded, others are oval, flat, cubical or columnar in shape.

Structure of Cells:

- Each cell is composed of two main parts:-

1. The Cytoplasm
2. The nucleus

The Cytoplasm

- The cytoplasm is a complex structure.
- **It is formed of three main components:-**

[A] Cytoplasmic Organelles (Cell Organoids):

- **They Are Of Two Types :**

1) Membranous Cytoplasmic Organelles :

- ✓ They are enclosed in membranes.
- ✓ They perform essential metabolic functions for each cell.
- ✓ **Examples:** The Cell membrane or plasma membrane, the Mitochondria, The Golgi apparatus,..... etc

2) Non-membranous Cytoplasmic Organelles.

- ✓ They are not enclosed in membranes.
- ✓ **Examples:** Ribosomes and centrioles....

[B] Cytoplasmic Matrix:

- It is a colloidal solution of proteins, lipids, carbohydrates, minerals, enzymes, small molecules and ions.

[C] Cytoplasmic Inclusions (Cell Inclusions):

- They are temporary components of certain cells.
- They are usually an accumulation of **stored food as :** glycogen and fat
- or an accumulation **of pigments as :** carbon and melanin.

The Cell Membrane

Definition:

- It is a **covering** membrane which forms the outermost cover of the cytoplasm.
- It is also named **Plasmalemma** or **Plasma Membrane.**
- It regulates the passage of materials in and out of the cell.
- **Thickness :** Its thickness ranges 7 nm to 10 nm

***By* The Light Microscope (L/M):**

- It cannot be distinguished (not visible) because it is very thin.
- its thickness is 7 – 10 nm which is below the Lr of the L.M (240 nm)
- Its appearance by L.M. is due to its oblique cut of 2 contiguous cell membranes or due to condensation of the stain on the cell coat.

***By* The Electron Microscope (*E.M.*) *(Fig 2):*;**

- It appears as if it is formed of three layers, therefore it is called **trilaminar membrane**.
- Its outer and inner layers appear as dark lines,
- While its middle layer appears as light area.

Molecular Structure Of The Cell Membrane(Fig 3 & 4):

- It is **formed of lipids, proteins and carbohydrates.**
- 60% proteins, 35% lipids and 5% carbohydrates.
- The Lipid, Protein and Carbohydrate Molecules Are Present In the Following Arrangement In

1) The Lipid Molecules

- They are formed of : **Phospholipid** and **Cholesterol** Molecules.

[A] Phospholipid molecules (Fig. 4):

- is formed of Two Parts :

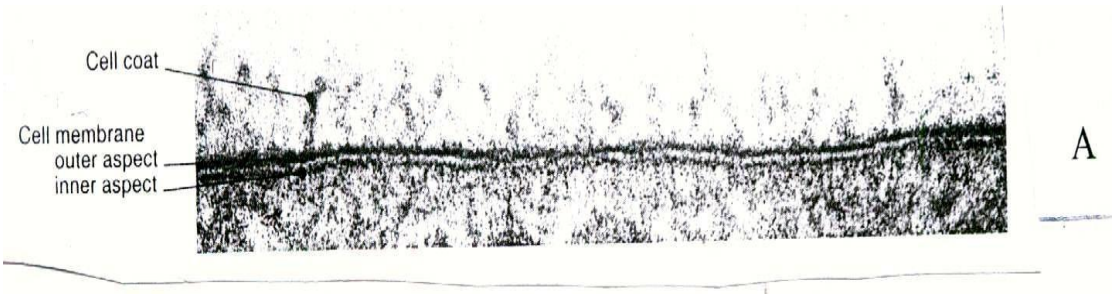


Fig.(2):The trilaminar appearance of the cell membrane by E.M.

a) **Hydrophilic Lipid Part.**

b) **Hydrophobic Lipid Part.**

a) The Hydrophilic Lipid of The Phospholipid Molecule.

- It forms the heads of the lipid molecules.
- It is called hydrophilic *because* it has a great affinity for aqueous solutions (it likes water).
- It is composed of charged lipid and is called Polar Region.
- It is directed towards the surfaces of the cell membrane (intracellular and extracellular surfaces).

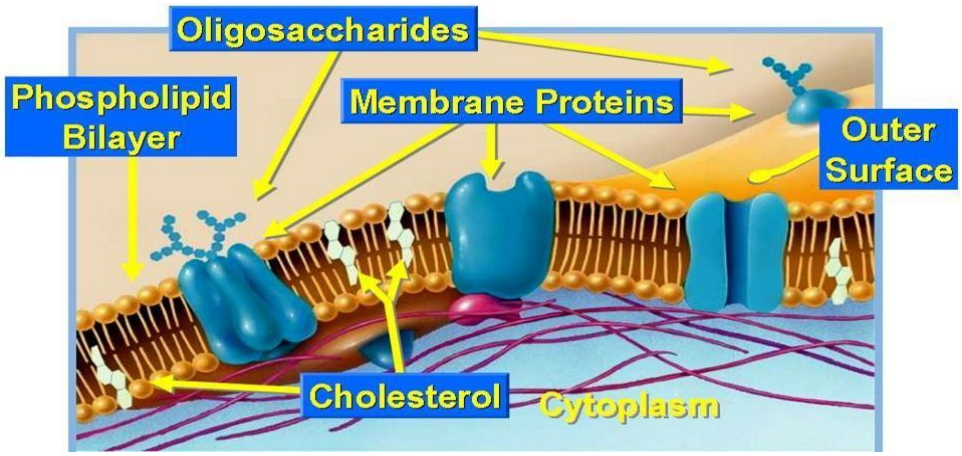


Fig. (3): The molecular structure of the cell membrane

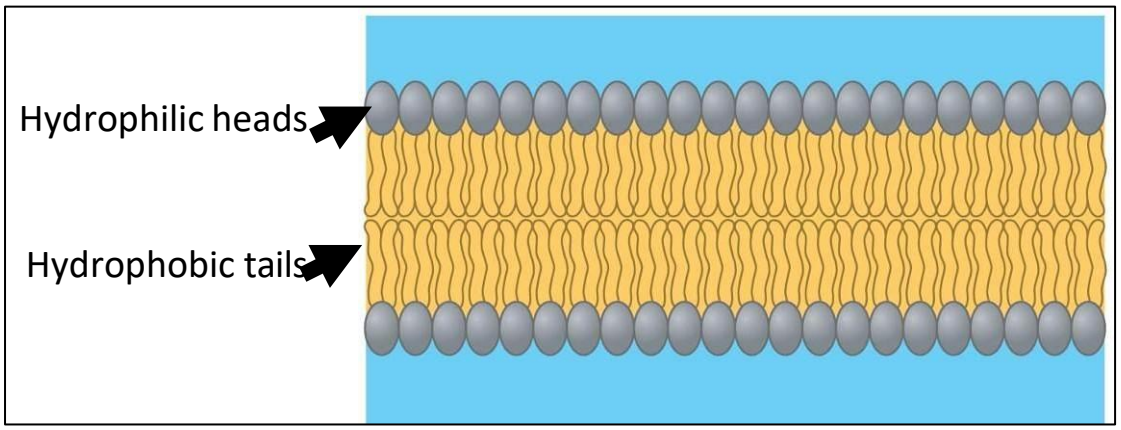


Fig 4 : Phospholipid molecule

b) The Hydrophobic Lipid Part of The Phospholipid Molecule.

- It forms the tail of the lipid molecule.
- It is called hydrophobic because it has no affinity for aqueous solutions (it dislikes water).
- It is formed of non-charged lipid and is called Non Polar Region.
- The hydrophobic ends of the lipid molecules are directed inwards, they face each other in the central part of the cell membrane.

[B] The Cholesterol Molecules:

- They are mainly present in the cytoplasmic aspect of the cell membrane and have a stabilizing effect, helping to keep the phospholipid molecules packed together and thus increase the integration of the plasma membrane.

2) The Protein Molecules (Fig.5)

- The protein molecules are present in the cell membrane in these forms:

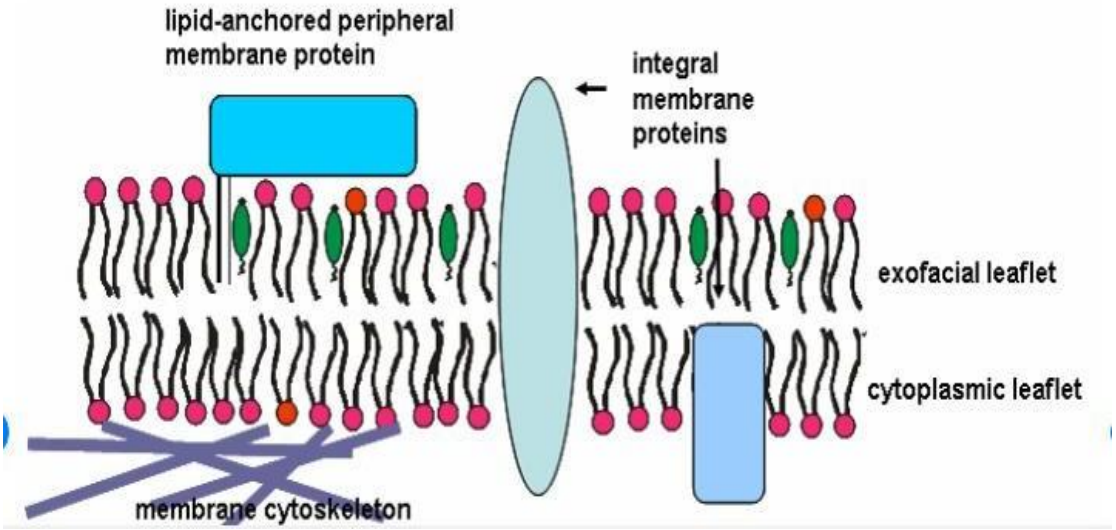


Fig.5 :plasma membrane

[A] Extrinsic or Peripheral Protein:-

- It is formed of protein molecules which are loosely bounded to both surfaces of the cell membrane.

[B] Intrinsic or Integral Protein:

- **The intrinsic protein molecules are present in the cell membrane in two forms:**

1) **Small intrinsic protein molecules** which are present as small collections of protein molecules distributed among the lipid molecules.
2) **Large intrinsic protein molecules**. They are present as large globular masses of protein molecules. They may extend through its entire thickness known as transmembrane proteins.

3) The Carbohydrate Molecules

- These are in the form of oligosaccharides attached to the lipids (**glycolipids**) or the proteins (**glycoproteins**).
- They project to the outer surface of the cell to form the cell coat or the glycocalyx.

NOTE

- Fluid mosaic nature of the PM: At body temperature, the lipid bilayer is fluid allowing free movement of the integral proteins within the membrane.
- Unit membrane theory: By E.M. all cellular membranes have the same trilaminar structure.

Functions of the Cell Membrane:

- **The Main Function of The Cell Membrane Is To Control The Exchange of materials Between the Cell And Its Surroundings;**
- **This occurs By Several Ways:-**

1. **Passive Diffusion of Substances:-**

- The cell membrane can allow substances to enter or leave the cell depending on the relative concentrations inside and outside.
- Examples; dissolved gases, essential ions and oxygen to pass into the cell. It also allows carbon dioxide and other metabolic wastes to leave the cell.

2. **Active Transport of Materials**

- Large molecules of sugars, amino acids and fatty acids in order to pass through the cell membrane, they should be combined with some catalysts.
- This combination requires energy which is provided by the enzymatic reactions of the mitochondria.

3. **Selective Transport of Materials**

- Transport by the use of receptors.
- These receptors determine which materials to enter the cell and which substances to be shut out.
- Examples; transport of drugs, hormones, bacteria and viruses which act on their target cells.

4. **Transport of Solid Materials By (Phagocytosis)**

- The cell membrane can engulf or allow solid materials to enter into the cytoplasm by the process of phagocytosis.
- This solid particle after its entrance into the cytoplasm is called phagosome.

5. **Transport Of Fluid Material By (Pinocytosis).**

- By this process, the cell membrane can engulf droplets of **fluid**. The engulfed fluid becomes surrounded by a membrane to be changed into a **Pinocytic Vesicle.**

6. **Expulsion Of Residual Bodies By (Exocytosis) :**

- After the entrance of the solid or the liquid materials into the cytoplasm, they will be digested by the lysosomal enzymes.
- After digestion, the cell can extrude these bodies outside [4] the cell by the process of exocytosis.

7. **Sodium-Potassium Pump Function :**

- The cell membrane is continuously pumping sodium ions to the outside of the cell.
- As a result of this process the concentration of sodium is higher in the tissue fluid than in the cytoplasm.
- The cell membrane can also pick up potassium ions to the inside of the cell to make the concentration of potassium ions more in the cytoplasm than in the tissue fluid.

Functions Of Cell Coat Or Glycocalyx:

8. **Cell Adhesion:**
   - The cell coat help in holding adjacent cells together.
9. **Cell Recognition:**
   - The cell coat enable cells to recognize cells of their own special kind.
   - This is of importance in transplantation of kidney, heart and skin grafts.
10. **Formation Of Cell Immunity** :
    - The cell coat plays an important role in the develpment of cell immunity against infections..
11. **The Cell Membrane** may be **modified** to form the following structures which form specific functions in certain cells as : **Micro-Villi , Cilia , Flagella.**

3 Membranous Organelles

- **Membranous organelles:** characterized by the presence of membrane.
- **They include:**
  1) The cell membrane (plasmalemma).
  2) The endoplasmic reticulum
  3) Mitochondria.
  4) Peroxisomes.
  5) Golgi apparatus
  6) Secretory vesicles.
  7) Lysosomes.

1) The Endoplasmic Reticulum:

- This is a network of interconnected membranous structures.
- **It is classified into:**
  1) Rough endoplasmic reticulum (RER)
  2) Smooth endoplasmic reticulum (SER)

[A] The Rough Endoplasmic Reticulum Fig.(1) :

- It is not visible but their site appears as localized area of basophilia.
- ***E.M.***: appears as tubules, vesicles or cisternae of SER attached on their surfaces ribosomes. They are usually connected with the outer nuclear membrane
- ***Function:*** Formation of the secretory proteins and lysosomes, packing and transfer them to Golgi.

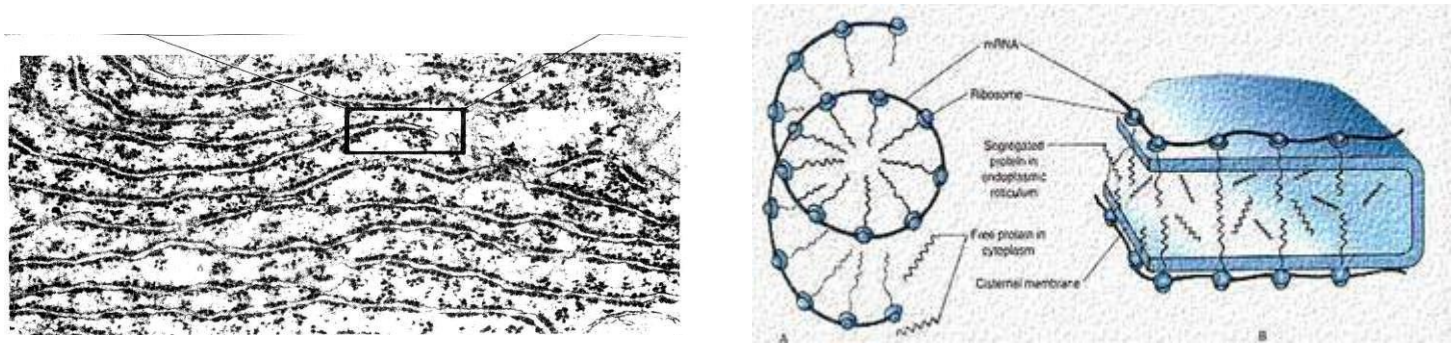


Fig.(1) Rough endoplasmic reticulum

[B] Smooth Endoplasmic Reticulum:

- ***L.M:*** it is not visible but the cells rich in it have highly acidophilic cytoplasm.
- ***E. M.:*** it consists of branching and anastomosing smooth tubules not studded with ribosomes. It may be continuous with the RER from which integral proteins are formed.
- **Function : it is involved in many functions** :
  1) Lipid synthesis (especially steroids) e.g. adrenal cortical cells.
  2) Detoxification of drugs and toxins e.g. liver cells.
  3) Mineral metabolism e.g. cells of the lacrimal gland.
  4) Glycogen synthesis e.g. liver cells.
  5) Impulse transmission within the cell e.g. the skeletal and cardiac muscle cells.
  6) Synthesis of phospholipids for all cell membranes.

2) Mitochondria:

- They are known as the power houses of the cell.
- **L.M.:**
  - ✓ Invisible by H&E.
  - ✓ Special stain for their demonstration e.g. iron Hx.
  - ✓ Because they contain all the enzymes concerned with food oxidation, e.g. cytochrome oxidase, therefore histochemical methods for detection of this enzyme are used to localize mitochondria.

- **E.M.: Fig.(2 & 3):**

✓ It is surrounded by two membranes.

✓ *The outer is straight and *the inner is thrown into folds called cristae or crests which is variable in size, shape and number from one cell to other.

✓ The outer membrane is thick permeable, contains special transmembrane proteins = porins that serve like channels for transport of substances into intermembrane space.

✓ The inner mitochondrial membrane is thin, less permeable and contains a large number of protein complexes, the respiratory chain enzymes which represents the electron transfer system.

✓ The membranes of the crests show small particles known as elementary particles each one has projected globular head and a connecting stalk. They represent ATP synthetase responsible for the generation of ATP.

✓ The inner chamber is surrounded by the inner membrane and contains the **matrix**.

✓ **It contains:**

1) RNA
2) ring-like DNA (inherited from the mother)
3) dense matrix granules (calcium particles)
4) small ribosomes
5) enzymes concerned with the kreb's cycle and β oxidation of fatty acids.

✓ Mitochondria are sometimes known as symbions as they resemble the structure of bacteria and they multiply by binary fission (Fig.4)

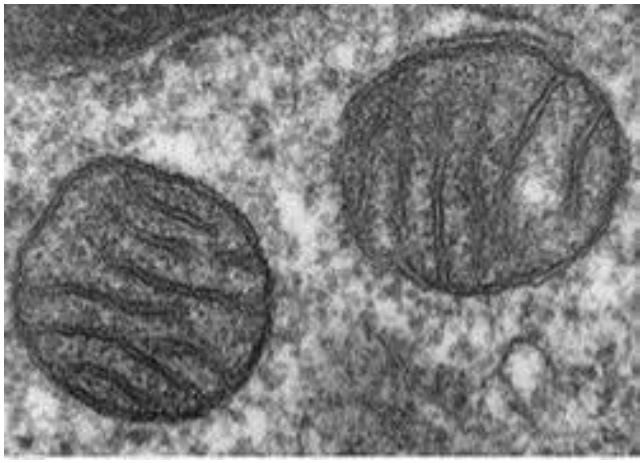


Fig. (2): Electron micrograph of the mitochondria.

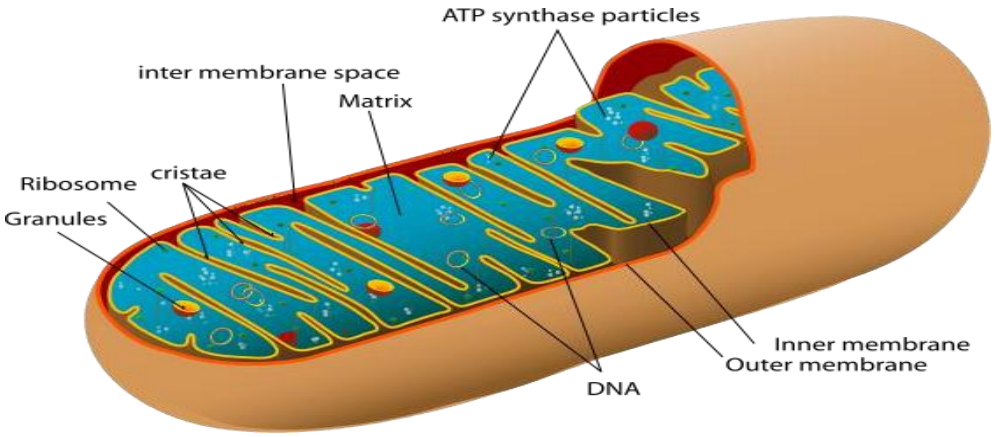


Fig. (3): schematic drawing of the electron micrograph of the mitochondria.

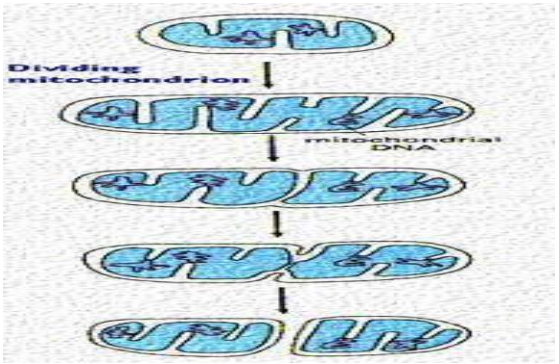


Fig. (4): Dividing mitochondria

Function:

1- Energy production.

2- Regulation of cytosolic calcium concentration.

3) Peroxisomes or microbodies

- They are small spherical membranous organelles, believed to arise from RER.
- By E.M they are morphologically similar to lysosomes and appear as finely granulated with dense core called nucleoid.
- They contain 50 different enzymes as catalase, urate oxidase, fatty acid oxidase and amino acid oxidase.
- They are numerous in liver cells.
- Peroxisomes originally were defined as organelles that carry out oxidation reactions.
- They like mitochondria, as they are the major sites of oxygen utilization as β oxidation of fatty acids with releasing of a harmful H_2O_2 and maintenance of body temperature.
- The catalase enzyme decomposes H_2O_2 by converting it to water.

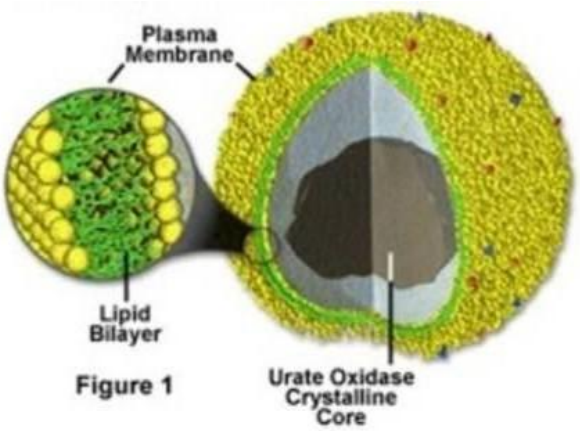


Fig. (5): peroxisomes structure

4 Membranous Organelles

4) Golgi apparatus:

- It presents in protein secreting cells located at the apical surface as epididymal cells or perinuclear as nerve cells.
- ***L.M.:***
  - ✓ In H&E stained sections it appear as unstained area in deeply basophilic cells e.g. plasma cells, osteoblast and is called negative Golgi image.
  - ✓ They need special stain for their demonstration, silver stain.

- ***E.M Fig.(1):***
  - ✓ It is formed of membrane bounded cisternae or saccules.

- They are arranged in the form of parallel stacks, the number of which depends on the protein secreting activity of the cells.
- The stacks array is curved so; it has a convex (cis or formation) surface and a concave (maturation) surface.
- Transfer or transport vesicles are present in association with the formation surface. They are budded from RER cisterna.
- Secretory vesicles are longer and present in association with the maturation surface budded from it.

Functions:

1) It is the site for condensation and packing of proteins transferred from the RER into secretory vesicles.
2) It is the site for packing hydrolytic enzymes of the primary lysosomes.
3) It is the site for addition of carbohydrates to from glycoprotion.
4) It is the site for sulphation of some products.

5) It is the main center for membrane traffic and recycling in the cell.

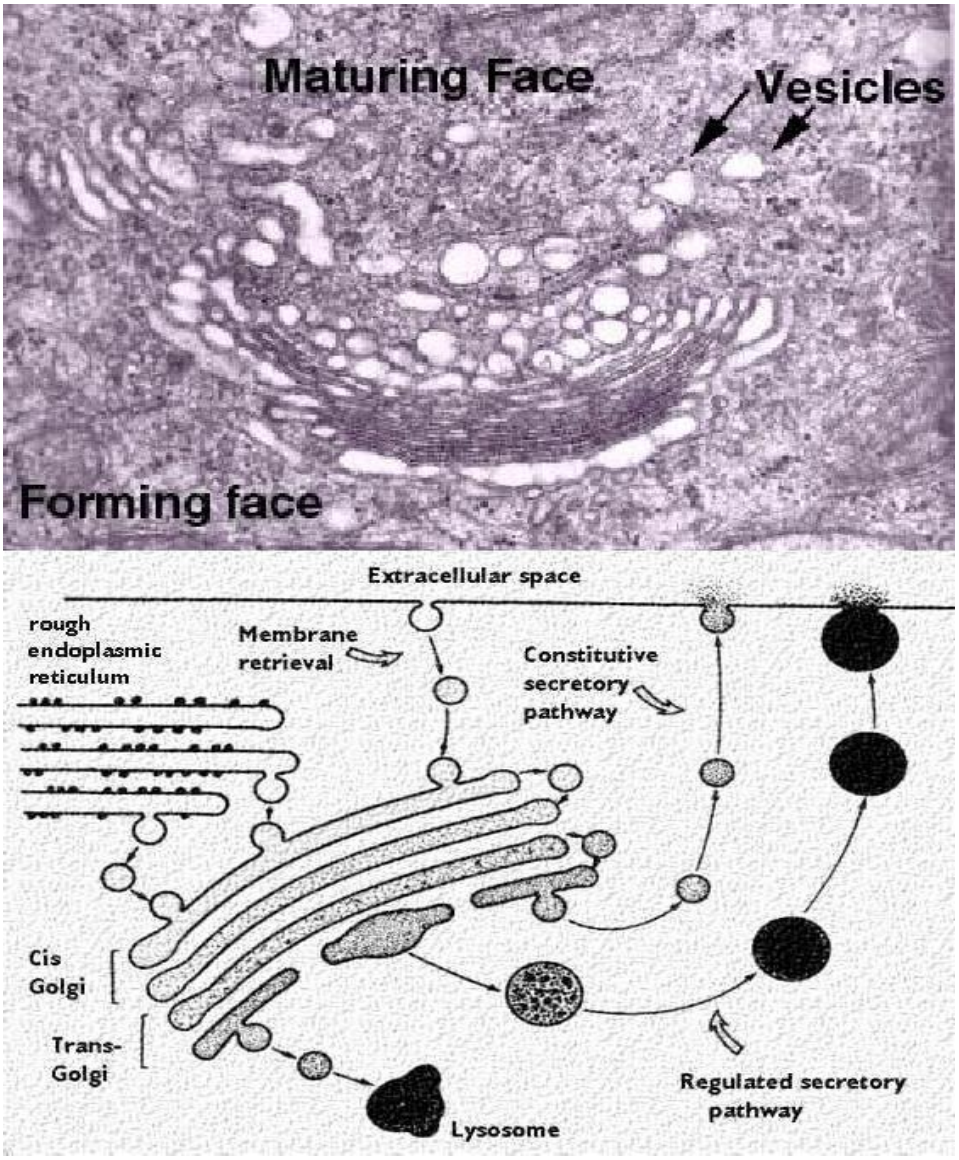


Fig. (1): Electron micrograph and Schematic drawing of the Golgi body and its function

5) Secretory vesicles.

- **Origin:** They are formed in the endoplasmic reticulum and are processed and packed in Golgi apparatus.
- **Site:** They are present throughout the cytoplasm and numerous in the cell active in secretion of stored substance.
- **Structure:** they are small membranous enclosed sacs that transport integral membrane proteins to the plasma membrane, and release the stored soluble materials within the vesicle to the outside of the cell. The concentrated secretion that excreted from the cell e.g. hormones or neurotransmitters released by exocytosis as the vesicles fuse with the plasma membrane (Fig.2). Those contain digestive enzymes are called zymogen granules.

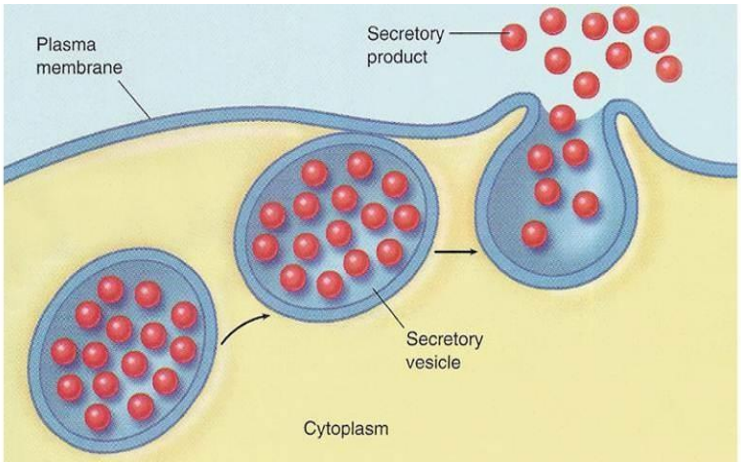


Fig. (2): secretory vesicles

6) Lysosomes:

- These are spherical membranous organelles rich in hydrolytic enzyme.
- ***L.M.***:
  - ✓ They are invisible, could be demonstrated histochemically by detection of acid phophatase enzyme.
- ***E.M.:***
  - ✓ They are membrane bounded vesicles.

- ✓ The primary lysosomes are homogeneous in electron density while secondary ones are heterogeneous.

- ***Origin:***
  - ✓ Lysosomes are budded into the cytoplasm from the Golgi apparatus with enzymes inside.
  - ✓ These enzymes are made in the rough endoplasmic reticulum, which are then delivered to the Golgi apparatus via transport vesicles.
- ***Functions of Lysosomes***
  - ✓ Lysosomes are often called 'garbage system' of the cell because of their degradation activity.
  - ✓ They contain hydrolytic enzymes work in acidic media.

- Acid phosphatase
- Proteases, which hydrolyze the proteins into amino acids
- Lipases, which hydrolyze the lipids into fatty acids and glycerides
- Amylases, which hydrolyze the polysaccharides into glucose
- Nucleases, which hydrolyze the nucleic acids into mononucleotides.

- ***Fate (Fig.3):*** **The newly formed lysosome is known as primary lysosome.**
  - ✓ It fuses with small pinocytotic vesicle→ secondary lysosome.
  - ✓ It fuses with fused multiple pinocytotic vesicles (multivesicular body) →secondary lysosome.
  - ✓ It fuses with phagosomes→ secondary lysosome.
  - ✓ It fuses with old organelle e.g. old mitochondria → secondary lysosome that called autophagic vacuole or cytosome.
  - ✓ Secondary lysosomes digest their contents and transformed into residual body that is either extruded outside or remains inside the cell.

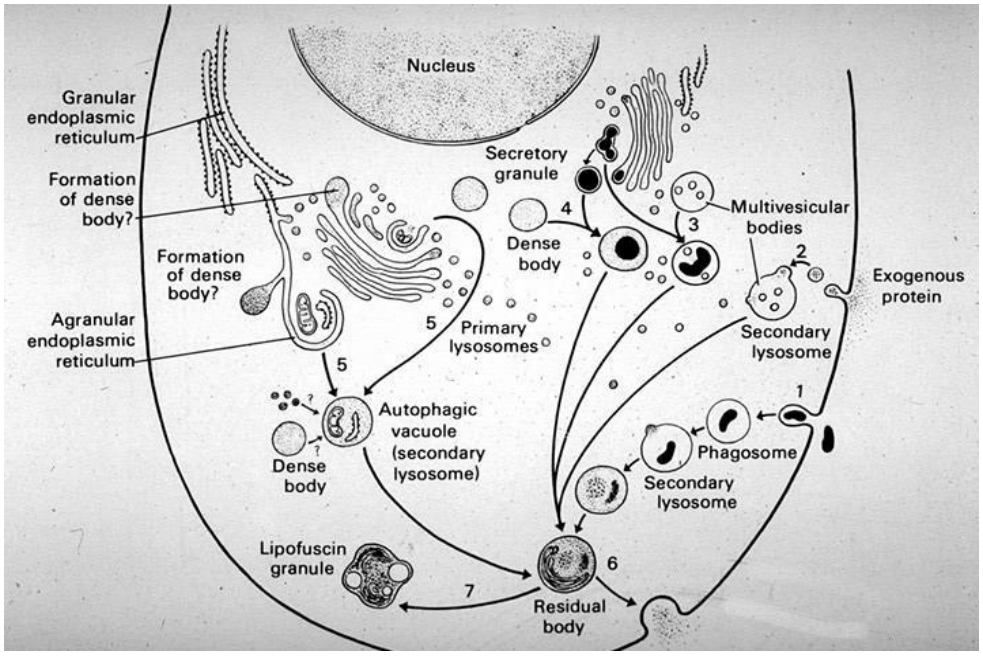


Fig. (3): Lysosomes

- **The lysosomal role in the cell can be summarized into:**

a. Heterophagy:

- ✓ The taking into the cell of exogenous material by phagocytosis or pinocytosis and the digestion of the ingested material after fusion of the newly formed vacuole with a lysosome.

b. Autophagy:

- ✓ A normal physiological process that deals with the destruction of cells in the body. It is essential for maintaining homeostasis, for normal functioning by protein degradation, turnover of destroyed cell organelles for new cell formation

c. Autolysis:

- ✓ It refers to the killing of an entire set of cells by the breakdown of the lysosomal membrane.

d. Fertilization:

- ✓ The acrosome of the sperm head is a giant lysosome that ruptures and releases enzymes on the surface of the egg. This provides the way for sperm entry into the egg by digesting the egg membrane.

5 Non membranous organelles & cell inclusions

Non-membranous organelles:

- Living, permanent, active structures present in the cytoplasm with special functions.
- They are not surrounded by membranes.
- **Include:**
  1) Ribosomes
  2) **Cytoskeleton:** Microtubules- Filaments .
  3) Centrioles

1) Ribosomes

- Small particles.
- They are formed of ribo-nucleoprotein particles (RNA) & protein both are formed in the nucleolus.
- Consist of two subunits, large and small one.
- **There are two types of ribosomes:**

1) Free ribosomes:

- ✓ Involved in synthesis of proteins utilized by the cell growth and differentiation.
- ✓ Single ribosomes are not involved in protein synthesis and have to be organized in the form of polysomes by the aid of mRNA.

2) Attached ribosomes:

- ✓ Attached to the outer surface of rER.
- ✓ They are concerned with synthesis of secretory proteins exported outside the cell.

2) CYTO-SKELETON

[A] Microtubules

- They are rigid thin tubular structure of 25 nm in diameter.
- They are formed of protein tubulin organized into thread like protofilaments.
- The wall of each microtubule is formed of 13 protofilaments.
- They are invisible by L.M. and needs special staining methods.
- Their assembly occurs in the microtulule organizing center near the centriole.
- The length can change rapidly by the process of polymerization and depolymerization of tubulin subunits

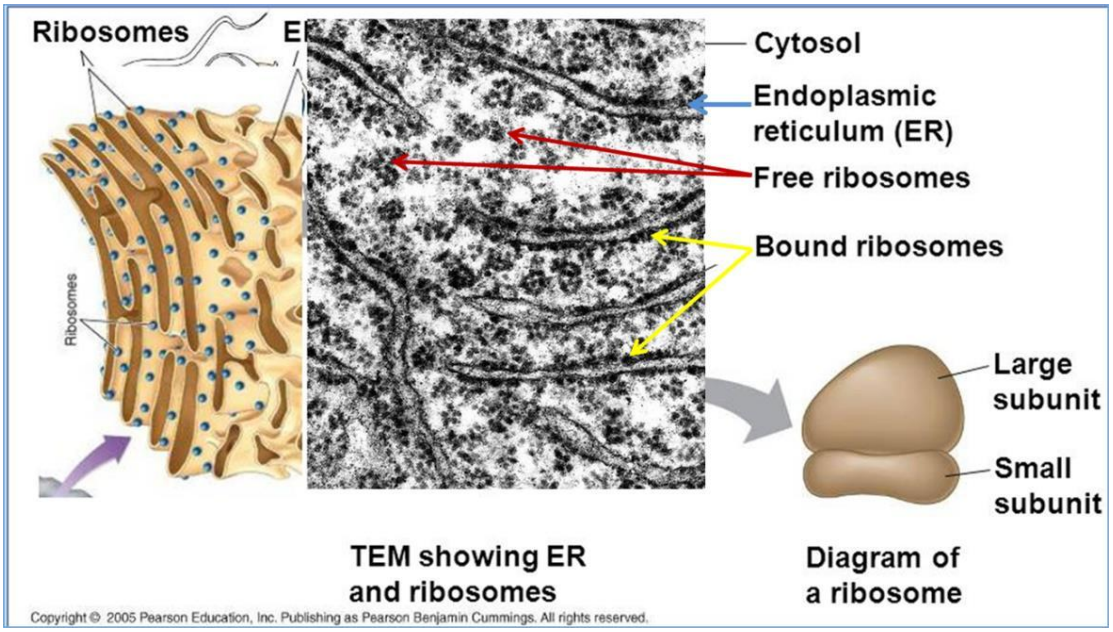


Fig 1: The types and structure of ribosomes

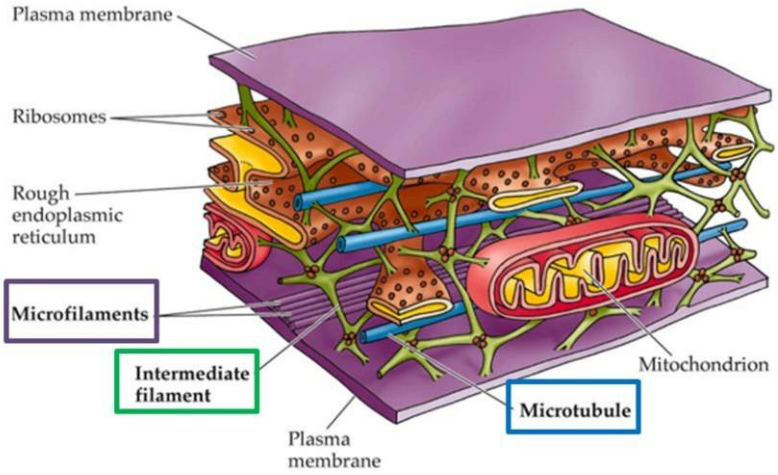


Fig 2: components of the cytoskeleton

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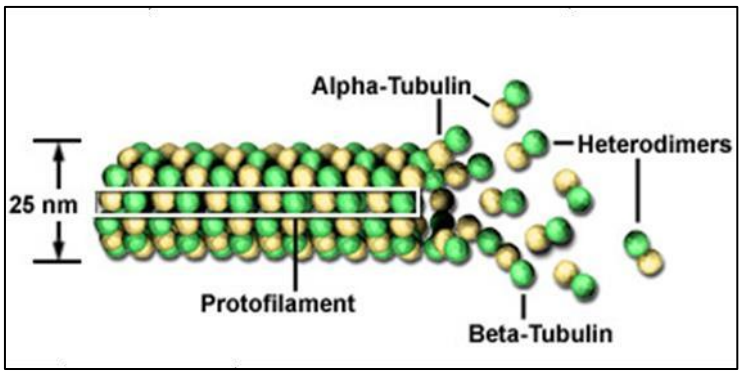


Fig 3: The structure of microtubule

Function:

1) Displacement of macromolecules and organelles through the cytoplasm.
2) Mechanical support of the cell and determine its morphology.
3) Form the mitotic spindles during cell division.
4) They constitute important structures in the cell e.g. centrioles, cilia and flagella.

[B] Filaments

- Thread like non membranous organelles.
- **They are of three types:**

1) Thin or microfilaments:

✓ They are formed of actin protein, present under the PM forming a network. In some cells e.g. muscle cells they are highly organized and contractile.

2) Intermediate filments:

✓ There are many types according to their chemical composition:

a) Keratin: forming the tonofilaments in stratified epithelia.

b) Desmin: present in smooth muscle cells.

c) Vimentin: present in fibroblasts.

d) Neurofilaments: present in nerve cells.

e) Glial filaments: present in glial cells.

3) Thick (myosin) filaments:

✓ They are filaments present in skeletal and cardiac muscle cells.

Function of filaments :

1) Mechanical support of the cell.
2) Distribute the tensile force on the cell.
3) Involved in muscle contraction.
4) Play a role in cell division.

3) CENTRIOLES

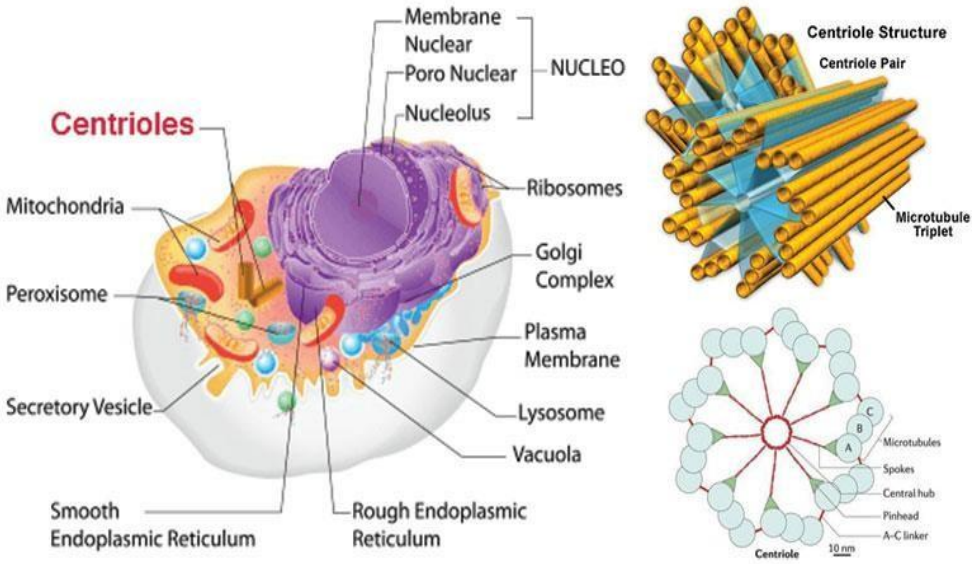


Fig 4: The structure of centrioles

- **L.M.**: appears as 2 dots close to the nucleus with special stains e.g. iron Hx.
- **E.M**: Each one appears as hollow cylinder the wall of which is formed of 9 bundles of triplet (3) microtubules.

- **Function of centrioles**
  - ✓ They are important in cell division.
  - ✓ They are duplicated during G2 to become 2 pairs.
  - ✓ Each pair migrates to one pole of the cell and form mitotic spindles in the microtubule organizing center.
  - ✓ They serve as the basal body of cilia and flagella.

2) Cell inclusions

Cell inclusions include non-living substances stored in the cytoplasm and not bound by membranes. Previously they were considered with no role in any metabolic activity. Recently it is known that many of them participate in the normal functioning of cell.

They are classified into:

1-Stored food

Glycogen:

- In liver and muscle cells.
- **L.M.**: Can be localized by PAS reaction.
- **E.M. (Figs.5,6)**: it appears as clusters of granules either as individual dense, roughly small particles, referred to as the β– particles as in liver or aggregate in two or more particles as rosette-like of larger size called α- particles as in muscle. It is located near the smooth endoplasmic reticulum.

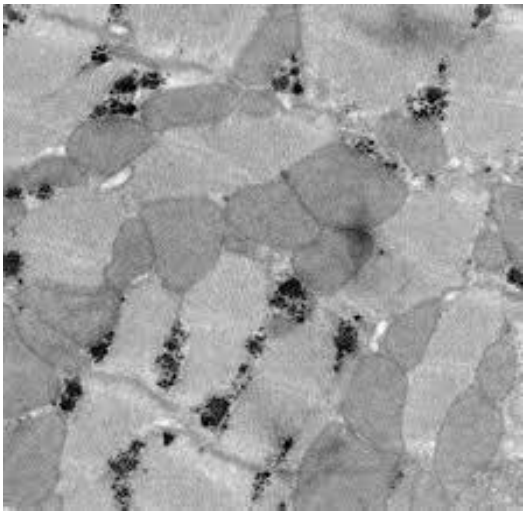


Fig. (5) E.M. picture of glycogen in muscle

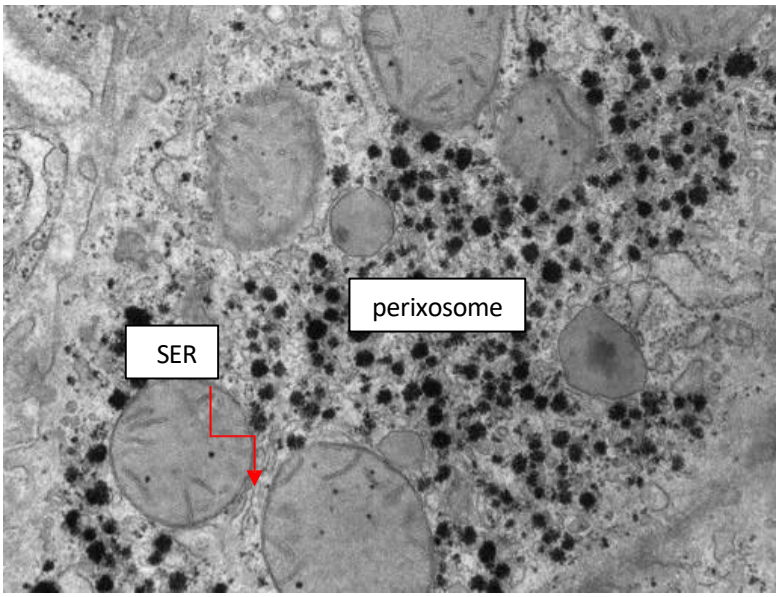


Fig. (6) E.M. picture of glycogen in liver

Fat:

- In fat cells or adipocytes.
- **L.M.:** H&E it appears as empty spaces.
- Special method for its demonstration is Sudan black stain of fresh sections. It appears as black granules
- **E.M. (Fig.7):** appears as large rounded droplets with moderate electron density.

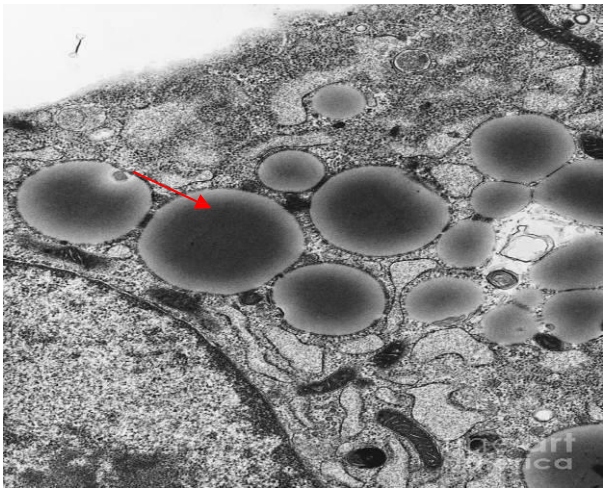


Fig. (7) E.M. of fat

2- pigments

Endogenouns pigments: synthesized inside the body e.g.:

- **Hemoglobin** in R.B.CS.
- **Myoglobin** in muscle cells.
- **Hemosiderin** is iron containing pigment that is accumulated in the cytoplasm of the phagocytes (for example in the normal spleen, liver, and bone marrow). This brown granular material has accumulated as a result of the breakdown of RBC's and release of the iron in hemoglobin. The macrophages clear up this debris, which is eventually recycled."
- **Bilirubin** in the blood and liver cells.
- **Melanin** in the epidermis, iris, hairs and substantia nigra. It is synthesized by special cells called melanocytes. on the ultrastructural level it appear as a homogenous dense granule separated from cytoplasmic matrix by a membrane unit
- **Lipofuscin** in active long lived cells e.g. nerve, liver and muscle cells. It is waste material stored inside the lysosomes. It represents aggregations of undigested material (covered by membrane). It appears yellowish-brown color in LM and electron dense granules in EM (Fig.8).

  It is easily seen at the poles of the cardiac nuclei, brown-red in color. As one ages, the amount of lipofuscin will continue to increase, especially in the heart. One can determine whether a heart is relatively young or old, as a direct reflection of the amount of lipofuscin present.

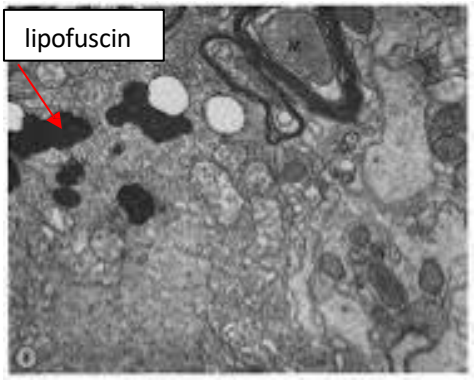


Fig. (8) E.M. picture of lipofuscin pigment

- **Exogenouns pigments:** they are taken into from outside e.g.
  - Lipochrom or carotenoids in carrots and tomatoes.
  - Dust & carbon particles in lung macrophages.
  - Minerals as lead and tattoo marks in the gum epithelia and skin cells.

3- Crystals

Most of these crystalline forms contain storage materials or cellular structure remnants which is located everywhere in the cell either in the cytoplasm or the nucleus. They have special function in cells of the inner ear (maintaining balance). Presence of crystals in cells of the testis (crystalloids of charcot- bottcher) appears to be linked with morbidity and infertility.

6 Lecture (6) Nucleus

Definition:

- **Nucleus** is control center of the cell. It contains the ***genes*** which control all the structural and functional aspects of the organism.
- Most human cells are eukaryotic cells containing true nucleus.

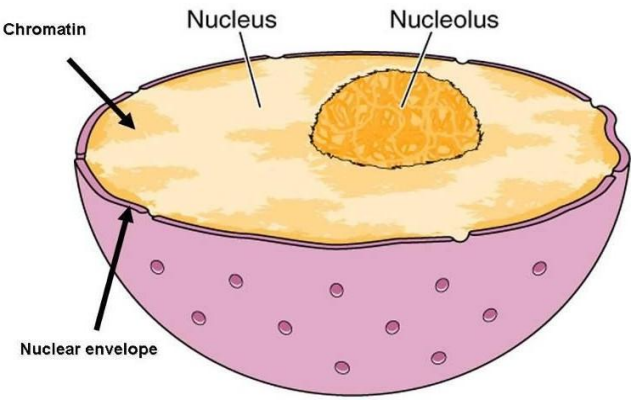


Fig.(1): General structure of the nucleus

Number:

- Usually each cell contains **one** nucleus.
- **Two** nuclei (bi nucleated) may be present in some liver cells, while many nuclei (multinucleated) are present in the skeletal muscle cells.

Size :

- The nucleus is variable it may be; small, medium-sized or large.

Shape :

- There are different shapes for the nuclei.
  It may be spherical, oval, kidney- shaped, spiral, lobulated or flattened.

Site:

- The nucleus is usually located in the widest area of the cytoplasm (central).
- It can be pushed to one side by certain structures or contents such as Fat cells& Plasma cell

Staining :

- The nucleus is a basophilic structure. It can be stained with any basic stain as hematoxylin because it is rich in nucleic acids (DNA & RNA).

Types of nuclei :

- **Condensed:** or inactive nucleus due to its high content of inactive condensed chromatin.
- **Vesicular or open-face:** or active nucleus due to its high content of active euchromatin.

- **Structure by E M: The nucleus consists of FOUR Components:**

1-Nuclear envelope.	2- Chromatin	3-Nucleolus.	4-Nuclear matrix

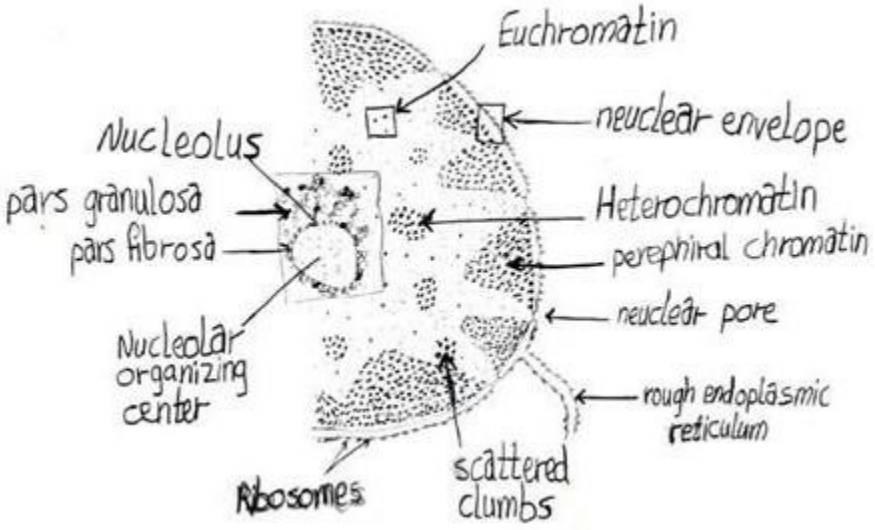


Fig. 2; Drawing E M structure of the Nucleus

1) Nuclear envelope

- **With L.M**: it appears as dark basophilic line. The cause of the staining properties is the chromatin adherent to its inner surface and the ribosomes associated to its outer surface.
- **With E.M**: it is formed of 2 membranes (8 nm thick each) with a 25 nm wide space inbetween. Each membrane is formed of a unit membrane structure.
  - The outer membrane is continuous with the rough endoplasmic reticulum and is covered with ribosomes.
  - The inner one is lined with peripheral chromatin.

Nuclear pores: circular pores are present in the nuclear envelope to allow the exchange of special substances between the nucleus and the cytoplasm (regulates the passage of protein to enter the nucleus and ribonucleoprotein **to** leave it).

The nuclear pore complex: is a cylindrical channel extending from the nuclear pore into the cytoplasm. From inside the nucleus, the peripheral chromatin encircles it. From the cytoplasmic side, an electron – dense ring encircles the short channel. The ring is partially covered by fibrillar material.
The nuclear envelope disappears during cell division

2) Chromatin

Definition:

- Chromatin materials are the basophilic particles and threads from which chromosomes are formed during cell division (they are the substances of the chromosomes).

Staining:

- They are basophilic in staining due to presence of DNA.

With The Light Microscope:

- The chromatin materials appear as coarse and fine basophilic granules.

With The Election Microscope:

- There Are Two Types of Chromatin;
  1. Euchromatin
  2. Heterochromatin

	1. Euchromatin	2. Heterochromatin
LM	Lightly stained	Dark basophilic clumps
EM	Fine organized structure	Coarse granules
Activity	More active	Less active
DNA	Less coiled	More coiled
Examples	Liver &nerve cells	Lymphocytes

Heterochromatin:

condensed clumps represent the inactive coiled parts of chromosomes and presents in the following areas:

1. **Peripheral chromatin**: associated with the inner surface of the nuclear envelope.
2. **Nucleolar associated chromatin or nucleolar organizer chromosomes**: there are 5 pairs of chromosomes that contain genes code for RNA. They are chromosomes number 13, 14, 15, 21 and 22.
3. **Sex chromatin or Barr body**: represents one of the x-chromosomes presents in the female.
4. **Scattered clumps** of chromatin in the nucleoplasm.

- **Chromatin structure:**

  - it is formed of:
    1- DNA double helix.
    2- Histones: strong basic proteins.
    3-Non- histones: less basic proteins.
    4-Small amount of RNA.
  - The DNA is organized around histones into nucleosomes that are wound into a helix to form chromatin. The latter is then wound again into a supercoiled chromosomes **Fig. (3).**

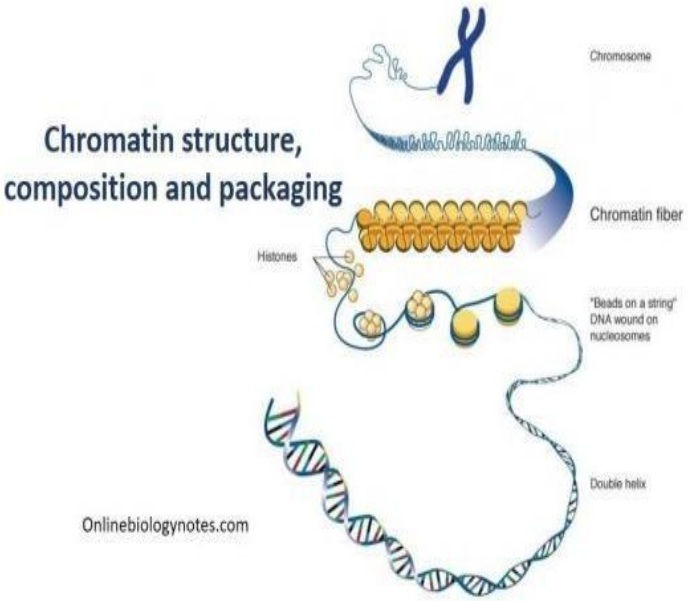


Fig.(3): Schematic drawing of the chromatin structure

3) Nucleolus

- It is round dense basophilic structure presents in most cells especially in active protein synthesizing one.
- By E.M. :it appears as a diffuse irregular mass not limited by a membrane composed of two areas :
  - **Pars fibrosa:** the central part which is formed of the nucleolar organizer chromosomes and filamentous RNA.
  - **Pars granulosa:** represents the accumulation of RNA protein in various stages of formation.
- **The function of nucleolus:** is the site of transcription of ribosomal RNA that is concerned with protein synthesis.

4) Nuclear matrix

- It is colloidal clear solution in which the chromatin is suspended and has fibrillar appearance. It provides a medium for chromosomal duplication, gene transcription and acts as mechanical support to the nuclear pores and the chromatin.

DNA and RNA

Structure of DNA (Fig. 4) :

- Each molecule consists of 2 long strands of polynucleotide chains wound together in double helix form.
- The backbone at each strand is formed of alternating phosphate and deoxyribose groups with side chains of nitrogen base, which is one of four bases, Adenine (A), thymidine (T), cytosine (C) and Guanine (G).

- The mode of attachment of the 2 strands is through obligatory base pairing between A & T and G & C respectively.
  - A single strand of DNA is made of letters:
    ATGCTCGAATAAATGTGAATTTGA
  - The letters make words:
    ATG CTC GAA TAA ATG TGA ATT TGA
  - The words make sentences:
    <ATG CTC GAA TAA> <ATG TGA ATT TGA>

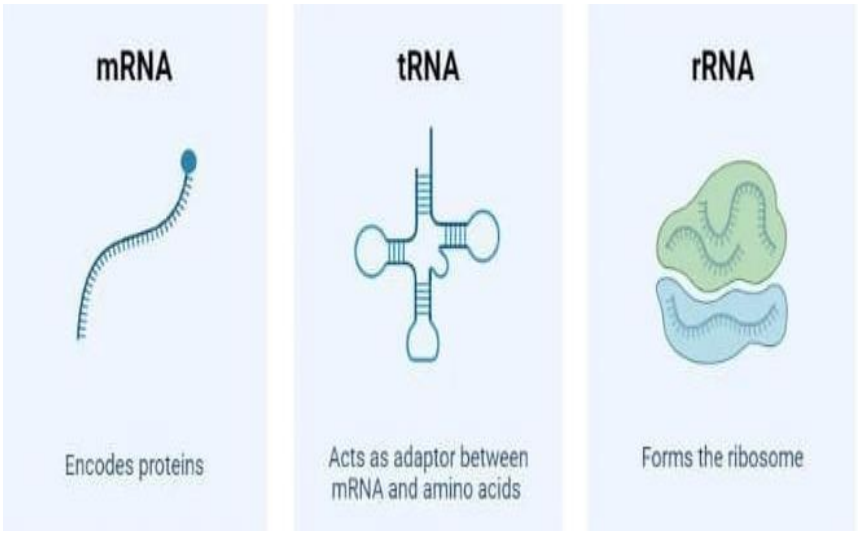
- These "sentences" are called genes. Genes tell the cell to make other molecules called proteins. Proteins are required for the structure, function, and regulation of the body's cells, tissues, and organs.
- We have approximately three billion base pairs (6 billion bases total) of DNA in most of our cells. This complete set of genes is called a genome. With the exception of identical twins, the sequence of the bases is different for everyone, which makes each of us unique.

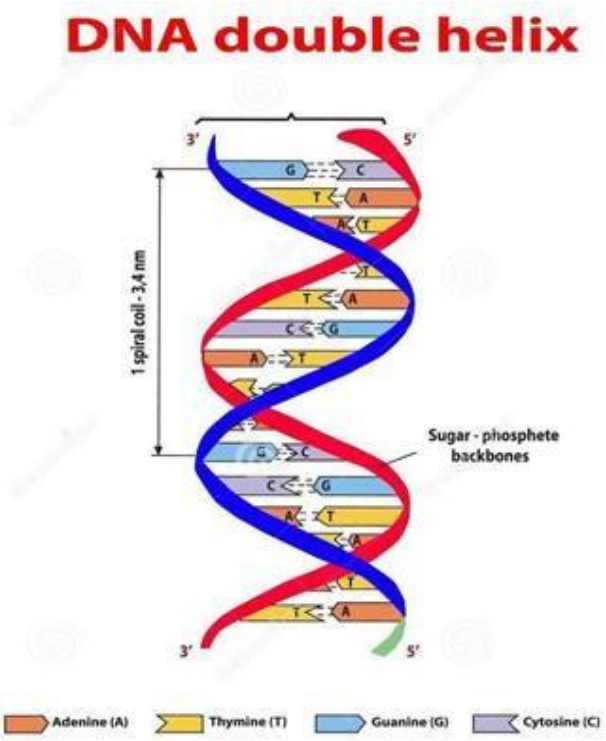
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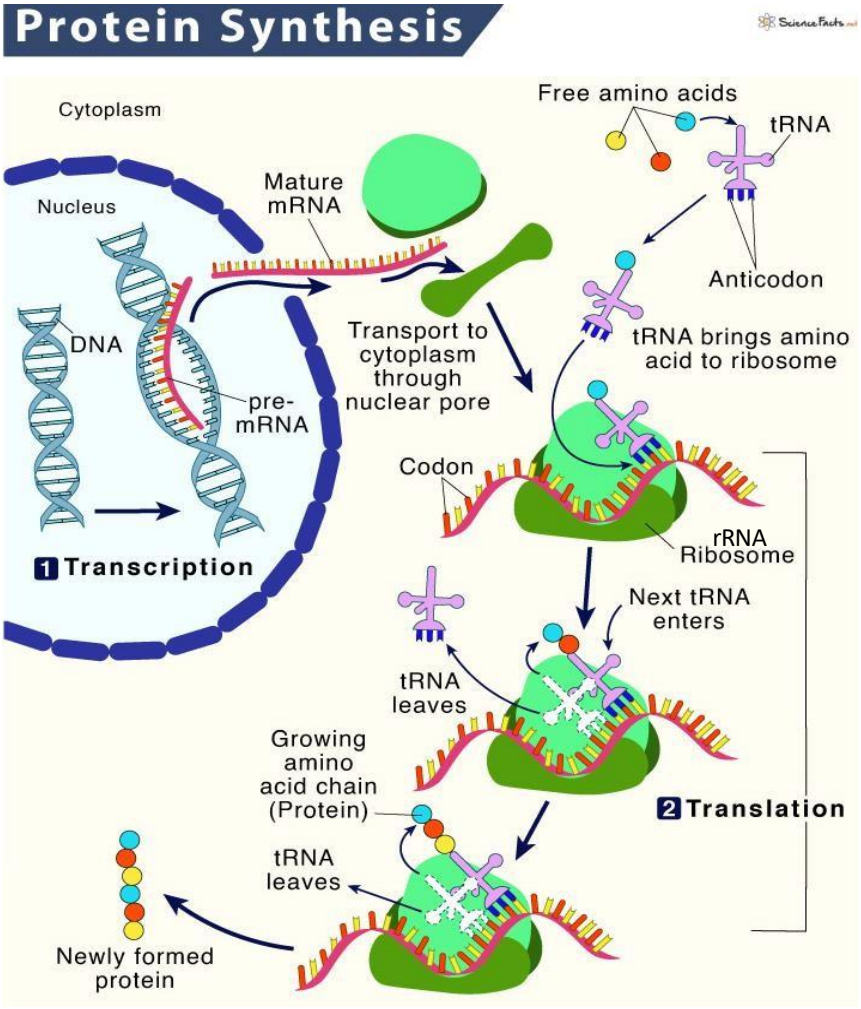
- **In RNA:** The nitrogen base is uracil instead of thymidine.
  The sugar group is ribose instead of deoxyribose.
  RNA is present in three forms, messenger, ribosomal and transfer.

Role of DNA and RNA in protein synthesis:

1. The genetic information (for synthesis of a certain protein) is present in inherited genes on DNA molecules.
2. The genetic code is represented in the form of codons (3 letter codes e.g. TGA, GAT, etc. ….).These codons are transcribed to mRNA . Sometimes the template contains non-coding sequences which are spliced out.
3. The mRNA enters the cytoplasm through a nuclear pore and binds the ribosomes (rRNA) and starts
4. protein synthesis by the aid of tRNA that transfers amino acids.
5. A polypeptide chain is formed.


mRNA
Encodes proteins
tRNA
Acts as adaptor between mRNA and amino acids
rRNA
Forms the ribosome


DNA double helix
3'
5'
1 spiral coil - 3,4 nm
Sugar - phosphete backbones
Adenine (A)
Thymine (T)
Guanine (G)
Cytosine (C)


Protein Synthesis
Cytoplasm
Nucleus
Mature mRNA
DNA
pre-mRNA
1 Transcription
Free amino acids
tRNA
Anticodon
tRNA brings amino acid to ribosome
Transport to cytoplasm through nuclear pore
Codon
rRNA
Ribosome
Next tRNA enters
tRNA leaves
Growing amino acid chain (Protein)
2 Translation
tRNA leaves
Newly formed protein

7 Lecture (7) Structure of chromosome & karyotyping

- A human being is formed of billions of cells; **these cells** are **of Two Types** :
  - **Somatic cells** which are present in the whole tissues of the body.
  - **Gametes** which are the sperms and ova.

1- Somatic cells

- Each somatic cell nucleus contains 46 chromosomes (23 pairs) which are as follows :
  - **22 pairs** of identical chromosomes known as **autosomes.**
  - **One pair of sex chromosomes** which differ in the two sexes.

    ★ **In female somatic cells:** the pair of **sex chromosomes** are identical and are called **XX chromosomes.**
    ★ **In male somatic cells:** the pair of sex chromosomes are quite different, one being long and is known as X chromosomes and the other is much smaller and is known as **Y chromosome.**

2-gametes

- **Each gamete** (ovum or sperm) contains 23 chromosomes
  - **In Females;** The sex chromosomes in the mature ovum are alike **X.**
  - **In Males;** The sex chromosomes in sperms are unlike, so that we have 2 types of sperms some with X and others with Y.

Karyotyping (Fig. 1)

- It is the study of the number, type and morphology of chromosomes of an individual. It is done by treating a number of dividing cells (**The white blood cells** are the most convenient cells for chromosomal studies) with **cholchicine** to arrest them in metaphase stage. They are photographed and homologous chromosomes are put together and the whole set is arranged to form a map.

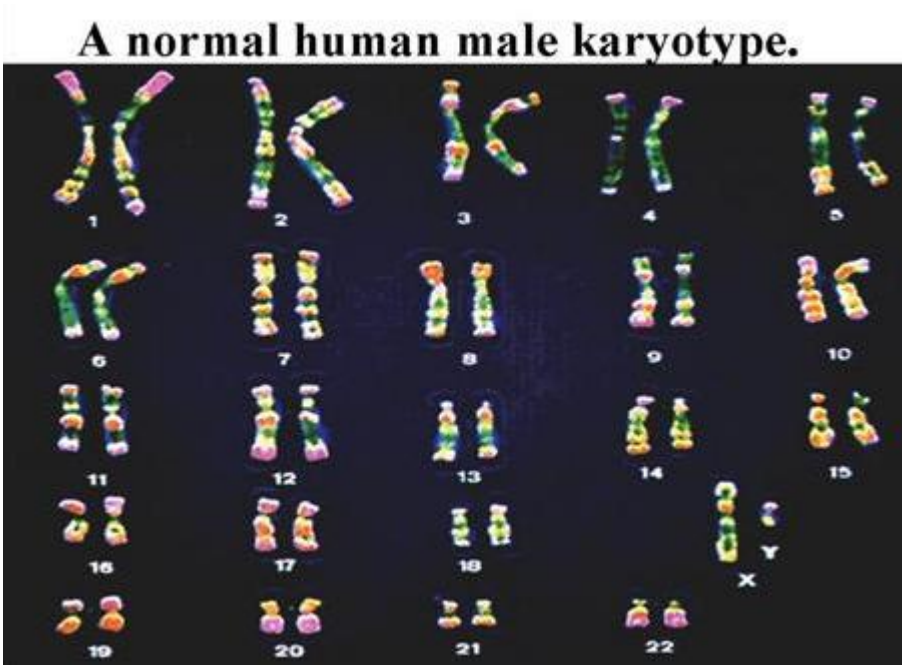


Fig. (1): karyotyping

The Structure of a Human Chromosome (fig. 2)

1. Each chromosome is formed of **two chromatids** which are connected together at one point known as **centromere**
2. The centromere appears as light- stained region except at its center, a dense small granule is present, the **spherule**.
3. **Kinetochore** is a disc shaped structure present in the centromere acts as microtubule organizing center during cell division.

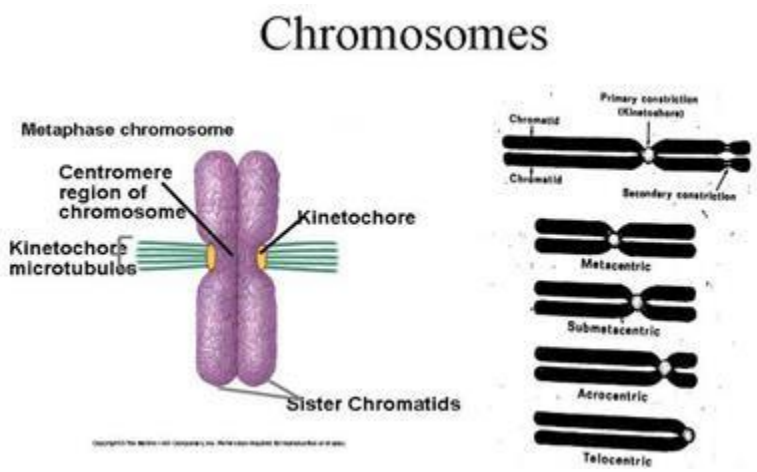


Fig. (2): Schematic drawing of the structure and types of chromosomes.

Types of Chromosomes

According to the position of the centromere, they are of 4 types.

1. **Metacentric**: centromere is located centrally.
2. **Submetacentric**: centromere is present between the centers of the chromatids and their ends.
3. **Acrocentric**: centromere is present near the terminal ends of the chromatids.
4. **Telocentric:** centromere is present at the terminal ends of the chromatids.

Chromosomal abnormalities

- **Abnormality in the number:**

1. polyploidy in which the cell contains more than 2 haploid number of chromosomes e.g. cancer cells.
2. Anuploidy e.g. the trisomy 21 results in Down syndrome.

- ***Abnormality in structure (Fig.3):***

1. Deletion: loss of an arm of a chromosome.
2. Duplication: addition of extra piece to a chromosome.
3. Inversion: occurs when a chromosome breaks and re-attach in a reverse position.
4. Isochromosome: occurs when a chromosome divides in a transverse manner instead of longitudinally.

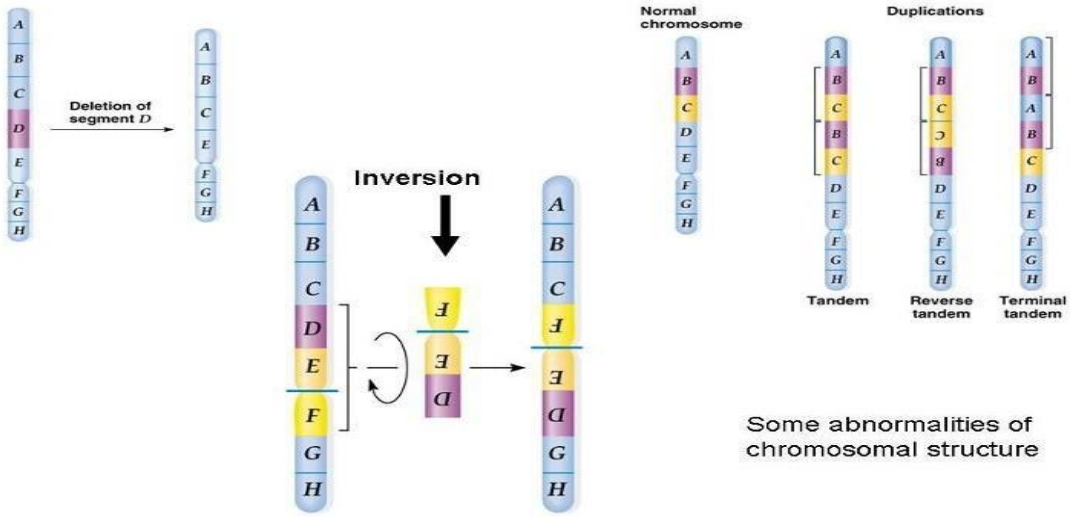


Fig 3 :Abnormality in chromosomal structure

8 Lecture (8) Cell cycle &cell death

- **The cell cycle:** It is the time from the end of mitosis to the end of the following mitosis. It is formed of 2 phases: interphase and mitosis **(Fig. 1).**

A- Interphase:

- this is the interval between 2 consecutive cell divisions during which the cell performs its specialized functions.
- It is divided into 3 phases.

✓ **Gap I (G1) phase:** during which the chromosomes are formed of single chromatids. The cell may enter **G0** stage (resting stage) for long time or for its life span as nerve cells and cardiac muscle.

✓ **Synthesis (S) phase:** during which DNA and histones replicate and the chromosomes are formed of double chromatids. Once the cell enters the S- phase, it must continue the cycle.

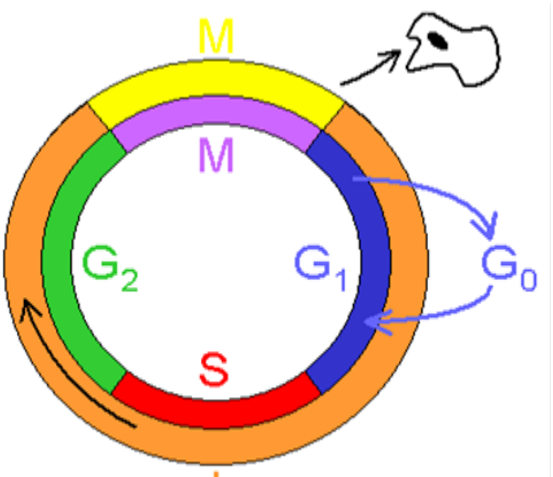


Fig.1: Schematic drawing of the cell cycle

✓ **Gap 2 (G2) phase:** during which, the cell synthesize a great amount of cytoplasmic material.

B- Mitosis:

- it is a type of cell division occurs in somatic cells.
  It is formed of 4 phases **Fig. (2)**:

1-prophase: the 2 centrioles migrate to the opposite side. Microtubules arise radially between the 2 centrioles. At the end of this stage, the nuclear envelope disappears and the chromosomes become short (coiled).

2-Metaphase: chromosomes continue to shorten and thicken and align in the equatorial plane of the cell. The microtubules attach to the centromeres from both sides of each chromosome to form the mitotic spindle.

3-Anaphase: each chromosome splits into two chromatids at the centromere as a result of shortening of the mitotic spindle. Each half of each chromosome moves to the opposite pole of the cell.

4-Telophase: constriction occurs in the equatorial plane which deepens leading to cleavage of the cell into 2 daughter cells. The nuclear envelope is reassembled

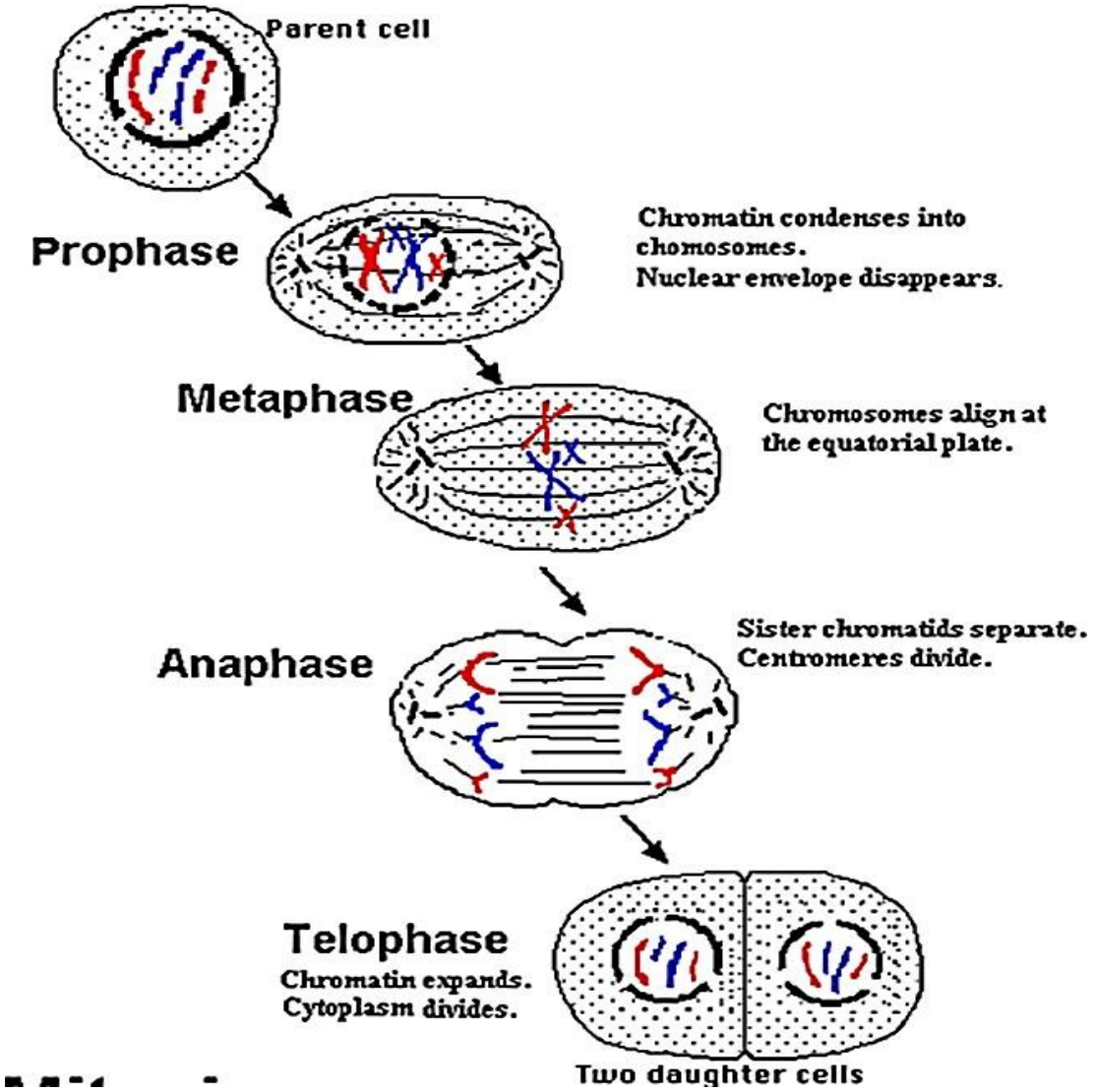


Fig.2: Schematic drawing of the different stages of mitosis:
1-prophase 2-metaphase 3-anaphase 4-telophase

Cell development:

- The fertilized ovum (one cell) forms a perfectly organized organism by the process of cell development which involves three aspects: growth, differentiation and integration.

1. **Growth:** increase in all mass through increase in their number by cell division (mitosis).
2. **Differentiation**: it is the process by which new features and new functions are achieved. As a result of this, specialized cells such as nervous, muscular, secretory, etc … arise from undifferentiated cells.
3. **Integration**: is the process by which growth and differentiation are kept in
4. harmony resulting in the formation of normal organism.

Cell death: Fig 3

Different cell types have different life span in the same organism. There are two types of cell death; necrosis and apoptosis.

- **Necrosis:**

is the cell death due to exogenous factors such as injury, toxins or hypoxia.

- The cell swells, the chromatin is clumped and the cytoplasm becomes weakly stained.
- The organelles are destroyed.
- The cell bursts and scattered fragments are removed by phagocytosis.

- **Apoptosis (programmed cell death):**

Cell death due to endogenous factor which is activation of certain genes.

- There is condensation and fragmentation of DNA
- The cell shrink but the organelles remain normal.
- The cell fragmented into membrane-bounded globules known as apoptotic bodies that are phagocytized by neighboring cells.

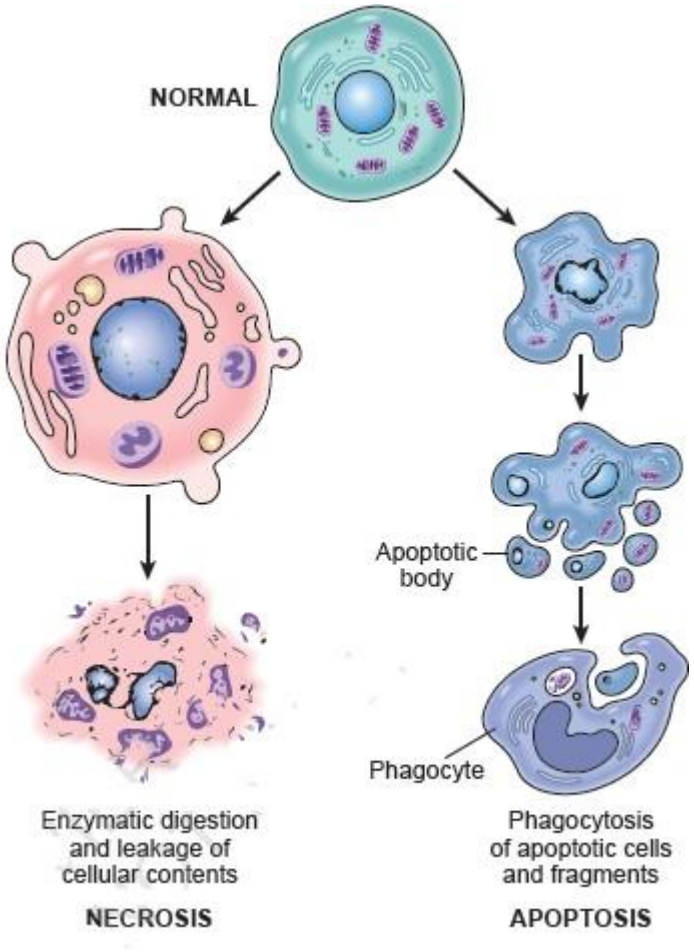


Fig.3: Necrosis & Apoptosis

9 Lecture (9) Stem cells and tissue renewal

Definition:

- Undifferentiated Cell has the ability to continuously divide and differentiate (develop) into various other kind(s) of cells/tissues

Characteristics

- **Undifferentiated:** unspecialized
- **proliferation and self-renewal**: Capable of dividing and renewing themselves for long periods of time
- **Differentiation:** can give rise to specialized cell types
- **plasticity:** cells of one type of tissue can give rise to cells of different type of tissue

Classification (stem cell types):

1. Types of stem cell according to differentiation capability:

Stem cell type	Description	Examples
Totipotent	Each cell can develop into a new individual (embryo and placenta	Fertilized egg, Cells from early embryos (1-3 days)
Pluripotent	Form (embryo without placenta): it can form the three germ layers of the embryo: ectoderm, mesoderm, endoderm.	cells of inner cell mass of the blastocyst (5 to 7 days)
Multipotent	can generate different cell types for the specific tissue or organ in which they live	blood-forming (hematopoietic) stem cells in the bone marrow can give rise to red blood cells, white blood cells and platelets.
Oligopotent	differentiate into a few cells.	lymphoid or myeloid blood forming cells in bone marrow

Bipotent	Into two types; biliary, hepatocyte	Hepatic progenitor cells
Unipotent	only produce cells of their own type, but have the property of self-renewal required	Epidermal stem cells. Muscle stem cell

2. Types of stem cell according to source:

1) Embryonic stem cells (Fig.1):

- ✓ During days 5-7 following fertilization and prior to implantation, the embryo (at this stage, called a (**blastocyst**), contains an **inner cell mass** that is capable of generating all the specialized tissues that make up the human body.
- ✓ ESCs are derived from the inner cell mass of an embryo that has been fertilized *in vitro* and donated for research purposes following informed consent.
- ✓ These pluripotent stem cells have the potential to become almost any cell type.
- ✓ Scientists hope to understand how these cells differentiate during development. As we begin to understand these developmental processes, we may be able to apply them to stem cells grown in vitro and potentially regrow cells such as nerve, skin, intestine, liver, etc for transplantation.

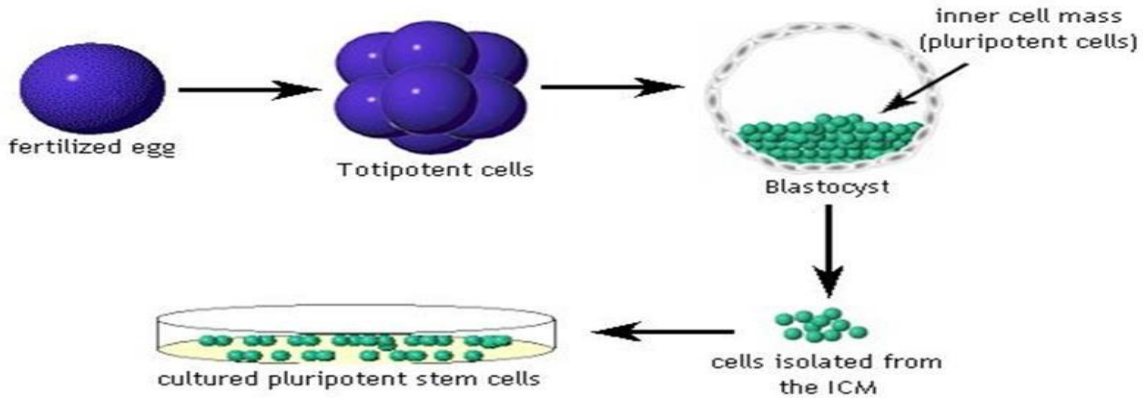


Fig.1: Embryonic stem cells

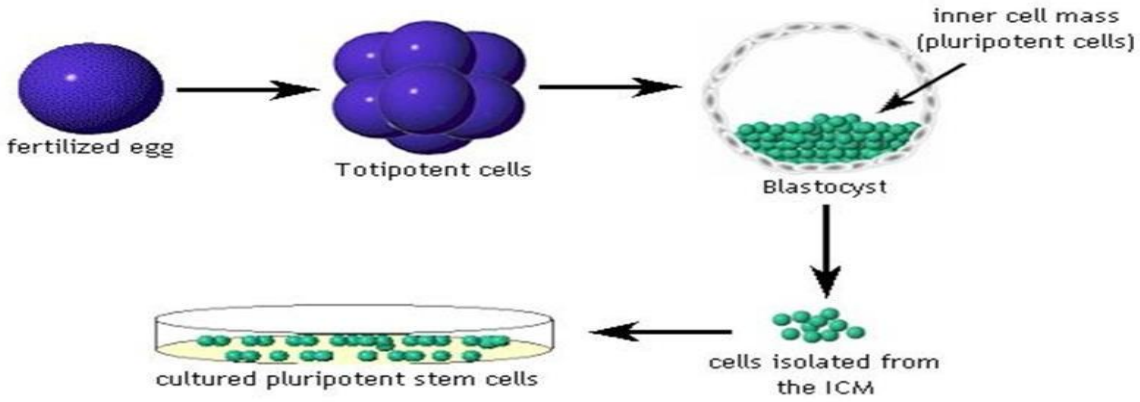


Fig.1:Embryonic stem cells

2) Adult stem cells :

- Found in our bodies all our lives for repair and tissue renewal (also referred to as somatic stem cells), and are more specialized than embryonic stem cells.

Sites:

1) Tissue-specific stem cells: undifferentiated cells found among specialized or differentiated cells in a tissue or organ after birth, these stem cells can generate different cell types for the specific tissue or organ in which they live as in skin, liver, or lung.
2) Bone Marrow.
3) Adipose tissue-derived stem cells (ADSCs) are mesenchymal cells and a source of multipotent stem cells
4) Umbilical cord stem cells
5) Human amniotic fluid and membranes.

3) Induced pluripotent stem cells

- (iPSCs) , are stem cells that are created in the laboratory.
- iPSCs are created through the introduction of embryonic genes into a somatic cell (a skin cell for example) that cause it to revert back to a "stem cell like" state, a process called genetic reprogramming.
- These cells, like ESCs are considered pluripotent Discovered in 2007 and needs many more years of research before use in clinical therapies.

Cell surface protein markers for identification of stem cell:

- clusters of differentiation (CD): CD45, and c-Kit (CD117).

Stem Cell Applications in medicine

- Theoretically and under research stem cell therapies could be applied in regenerative medicine in those with spinal cord injuries, type 1 diabetes, Parkinson's disease, Alzheimer's disease, myocardial infarction, burns, cancer and osteoarthritis.

Clinical Trials:

- Currently, the only stem cell-based treatment that is routinely reviewed and approved by Food and Drug Administration (FDA) is **hematopoietic stem cell (HSCs) transplantation**.
- Involves the intravenous infusion of hematopoietic stem cells in order to reestablish blood cell production.
- It is used in patients whose bone marrow or immune system is damaged or defective.
- These cells are derived from bone marrow, peripheral blood, or umbilical cord blood.
- The procedure can be autologous (when the patient's own cells are used), allogenic (when the stem cell comes from a donor), or syngeneic (from an identical twin).
- **Stem cell-based therapies for all other conditions are still experimental.**

Stem cells and tissue banks

- The umbilical cord is rich in mesenchymal stem cells. These cells can be successfully stored and used in therapies to prevent the future life-threatening diseases of a given patient.
- Stem cells of human exfoliated deciduous teeth (SHED) has the ability to develop into many types of body tissues and easily stored.

Cell renewal:

- Occur in all our lives by tissue specific stem cells for repair of damaged tissue and also maintain the pool of undifferentiated stem cells.

Examples :

- Skin epidermis: renewal in three weeks
- Intestinal epithelium: renewal in four days.
- Spermatogenesis in spermatogenic cells to maintain sperm formation.

10 Histological structure of the neurons

Objectives:

By the end of this unit, the student should be able to:

1- Mention the different types of neuron in the body

2- Describe the structure of neuron

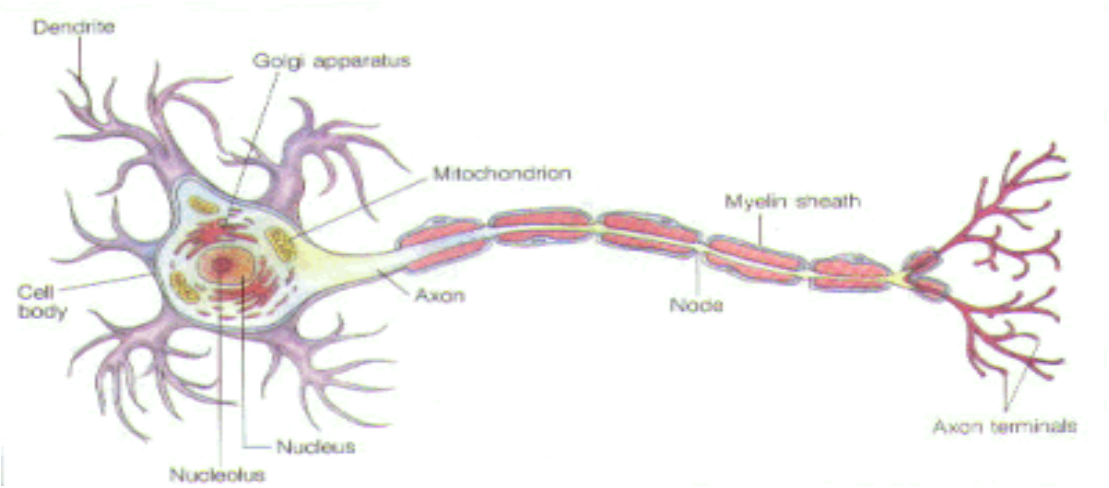
3- Identify the mechanism of myelination in peripheral and central nervous system

Contents:

- Structure of the neuron (LM & EM) - cell body - axon - dendrites.
- Myelination of peripheral nerve fibers.

NEURONS

- Neurons are highly differentiated cells, of various sizes and shapes with the highest degree of conductivity.
- The neuron consists of a cell body (Perikaryon) and processes axon and dendrites.



(Fig. 1). ***Fig.(1): Schematic drawing of neuron***

The Cell Body

- Includes the neuroplasm (cytoplasm) and nucleus. The processes attached to it are the axon and dendrite(s).
- Nerve cells consist of ***nucleus*** and ***neuroplasm***

L.M.:

- The nerve cell has different shape with pale acidophilic cytoplasm with deep basophilic masses of Nissl bodies and oval vesicular pale-staining nucleus with a large and clear nucleolus.
- The staining of nerve cell body by silver stain shows neurofibrils (aggregation of neurofilaments) that appears brown or black.

E.M.:

- The nerve cell is protein-producing cell
- Nucleus is large, euchromatic with distinct nucleolus.
- Cytoplasm contains abundant RER and polyribosomes (Nissl bodies), neurofilaments, microtubules, Golgi apparatus, mitochondria, lysosomes and lipofuscin pigment.
- The latter is more numerous in older individuals.

Neural Processes

- There are two types:
  1) **Dendrite**
  2) **Axon**

1) Dendrite

- One or many in each cell.
- Thick at the base, then branching repeatedly to increase area receiving impulse and becoming tapering toward the apex.
- Its cytoplasm contains mitochondria, RER, ribosomes and neurofibrils with no Golgi apparatus.

2) Axon (nerve fiber)

Structure:

- Each cell has only one axon.
- Straight, regular in diameter and smooth surface.
- The point of axon origin is called axon hillock that is free of Nissl bodies.
- Axoplasm contains mitochondria, transport or synaptic vesicles and large number of neurofilaments and microtubules.

Myelination in PNS (Fig. 2):

- As the plasmalemma of the Schwann cell (one of neuroglial cells) encircles the axon up to fifty times.
- It forms a membranous sheath enclosing it called the myelin sheath.
- So myelin is composed of lipoprotein material of Schwann cell membrane wrapped spirally around the axon.
- Myelin sheath is deficient (absent) at certain points along the axon known as nodes of Ranvier. They correspond to the spaces between adjacent Schwann cells.

- The distance between two nodes is the internodal segments.
- The thickness of myelin is directly proportional to the conductivity of the nerve as it faster with large diameter and thick myelin sheath.

Morphological Characteristics of Myelin Sheath in PNS:

- Myelin sheath is seen in H&E stained slide as an empty (vacuolated) layer between the axolemma and neurolemmal membrane and can be demonstrated using osmic stains.
- Myelin sheath is interrupted at periodic intervals by Ranvier's nodes.
- At the EM level, it displays many concentric lamellae with alternate light and dark lines.
- Unmyelinated fibers as in autonomic fiber.
- They are included inside Schwann cells and are surrounded by only one layer of the cell membrane. As many as 12 axons may be seen within one Schwann cell.

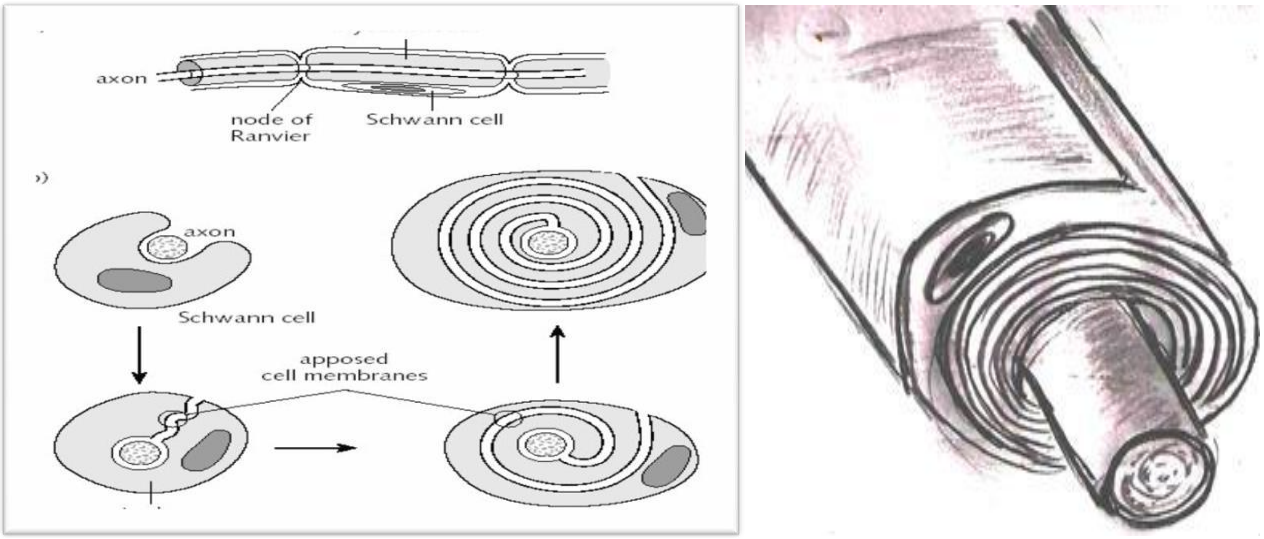


Fig 2: Myelination in PNS

11 Epithelium: Characteristic features, basement membrane & intercellular

Characteristic Features Of Epithelium

1) The epithelium is composed of cells close to each other and tightly packed with minimal inter cellular spaces and extracellular fluid.

2) It is derived from all three embryonic tissues depending on location e.g. the epithelium of the mouth, nose, anus and skin are of ectodermal origin, the respiratory and digestive tracts epithelia are of endodermal origin while that lining the blood vessels is derived from embryonic mesoderm.

3) The cells lie on a fibrous lamina called basement membrane or basal lamina. It is non-cellular and composed of carbohydrates and proteins which are secreted by the epithelial cells and connective tissue cells.

4) The epithelium is avascular and its nutrition occurs by diffusion from the underlying C.T. blood vessels.

5) Epithelial cells are traversed by numerous nerve endings.

6) Epithelial cells form sheets that cover the surface of the body and line cavities, tubular organs and blood vessels.

7) The cells are continually degenerate and in most epithelial cells, there are provisions for cell replacement at different rates.

8) Epithelial tissue is characterized by close cell-to-cell contacts by the presence of many types of cellular junctions e.g. tight junction, zonula adherence, macula adherence or gap junction.

9) The epithelial cells that are composed of one cell layer (simple epithelium) shows the property of epithelial polarity. This is the property of having their functions preferentially toward one end. **Their polarity is expressed in:**

a) Apical specialization that amplify the free surface (microvilli).
b) Supranucleur Golgi.
c) Secretory granules in the apical cytoplasm.
d) Biochemical differences between the apical and basolateral regions (domains) of the membrane e.g., glycolipids, H+, ATP ase in the apical domain, and Na / K+ ATP ase in the basolateral domain.

10) Protection, absorption, secretion, filtration and diffusion are all functions of epithelium according to its site.

11) Under certain conditions e.g. irritation, one type of epithelial tissue may undergo transformation into another type. This is called **metaplasia.**

Basement membrane Fig. (1)

- It was called basement membrane, but because it is not a lipid bilayer like the membranes of cells, the term basal lamina is preferred.

- **By L.M.** it appears as PAS positive layer beneath the epithelial cells.
- **By E.M.** it is composed of two zones; the lamina lucida and the lamina densa. The later is formed of a network of randomly oriented interwoven filaments. These are formed of collagen fibers, proteoglycans and laminin.

- Lamina lucida is traversed by very fine strands that extend from the cell membrane to the lamina densa. The basal lamina is attached to the underlying C.T. by anchoring

fibrils formed by a special type of collagen and microfibrils of elastic fibers. The components of basal lamina are secreted by epithelial cells.

- Reticular fibers are closely associated with the basal lamina forming the reticular lamina. These are produced by the underlying C.T. cells.

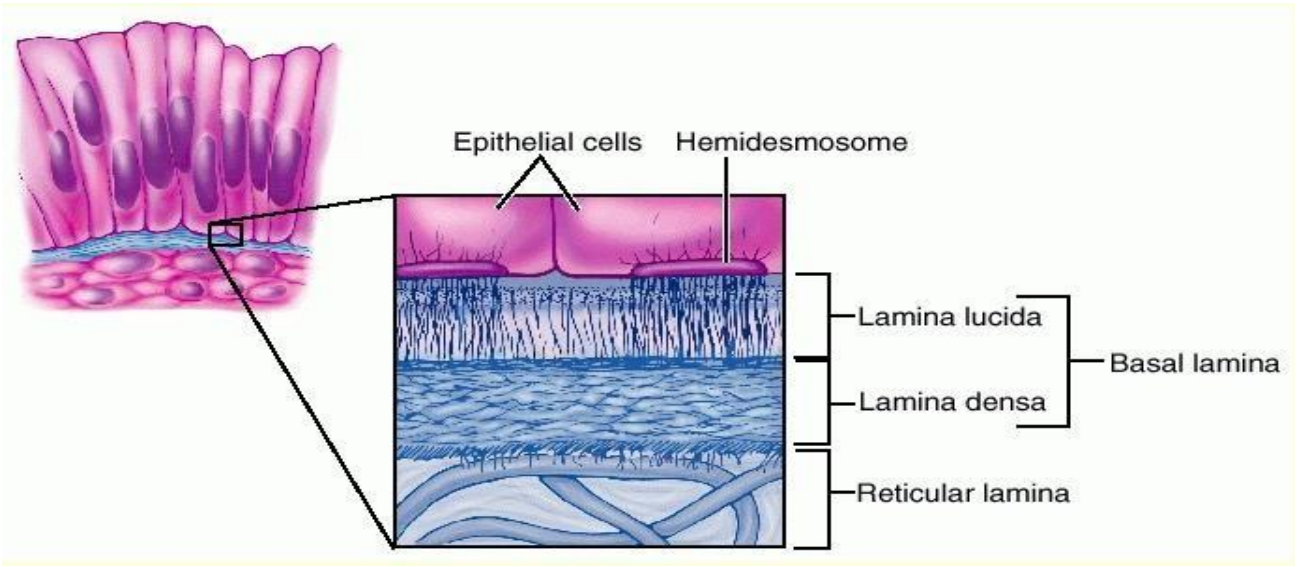


Fig (1): A drawing of the EM structure of the basement membrane

▪ Inter cellular junctions and apical modifications of epithelial cells

Inter cellular junctions

- **It is a modification of lateral cell membrane and it include :(Fig:2)**

1) ***Zonula occludens*** or **tight junction**:

- ✓ intercellular sealing is required to prevent intercellular diffusion.
- ✓ The outer leaflets of the adjacent cell membranes fuse leaving no intercellular space at points of fusion sites.
- ✓ It is the most apical of the junctional complex seen between epithelial cells.

2) ***Zonula adherens or intermediate junction*** :

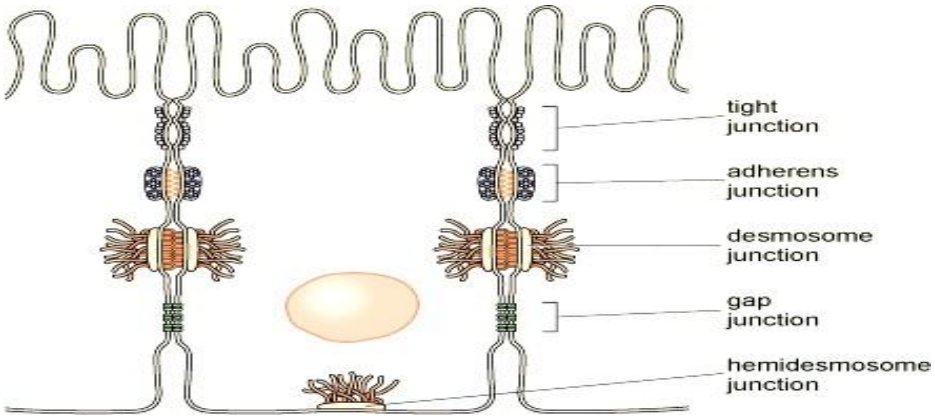
- ✓ the adjacent membranes thickened and come into close contact leaving a regular intercellular space filled with adhesive material mainly formed of glycoprotein.
- ✓ The adjacent cytoplasm is dense and contains tonofilaments and is called terminal web.

3) ***Macula adherens or deomosome***:

- ✓ it is found between cells exposed to mechanical stress e.g. cells of the epidermis. It is the strongest junction between cells.
- ✓ Numerous intermediate filaments come out of the PM of the two cells and loop in a dense discoid material found midway between the cells which is called attachment plaque.
- ✓ These looping filaments cause the strong attachment.
- ✓ The adjacent cytoplasm on both side of the junction is rich in tonofilaments.

4) ***Nexus, gap junction or electric coupling***:

- ✓ the PM of adjacent cells is separated by a space of only 2 nm.
- ✓ Trans membrane integral proteins of the facing PM form a continuous ionic channel known as connexons allow free movement of ions from one cell to other.
- ✓ These points represent regions of low electrical resistance helps to transmit impulse rapidly from one cell to another e.g. inbetween cardiac muscles.



12 Apical cell specializations and covering &lining epithelium

1) Cilia:

- ***L. M.:*** They are motile hair like processes extending from the free surface of some cells e.g. respiratory epithelial cells.
- ***E.M.:*** it is formed of three parts *Fig. (1):*

Shaft or axoneme :

- projecting from the surface of the cell and surrounded by the PM.
- It is formed of 9 peripheral bundles of microtubules (MT).
- Each one consists of double Mt. and 2 central single MT., all embedded in the ciliary matrix.

Basal body:

- organized from the centriole and have the same structure.
- It consists of 9 bundles of triplet MT.

Rootlet:

- Minute fibers which anchor the basal body to the cytoplasm.
- **Function:**
  - ✓ beat over the surface in one direction thereby pushing fluids and small particles towards the exterior of the body.

NOTE

Flagella of the sperms have the same structure of the cilia but they are longer and their movement is whip like

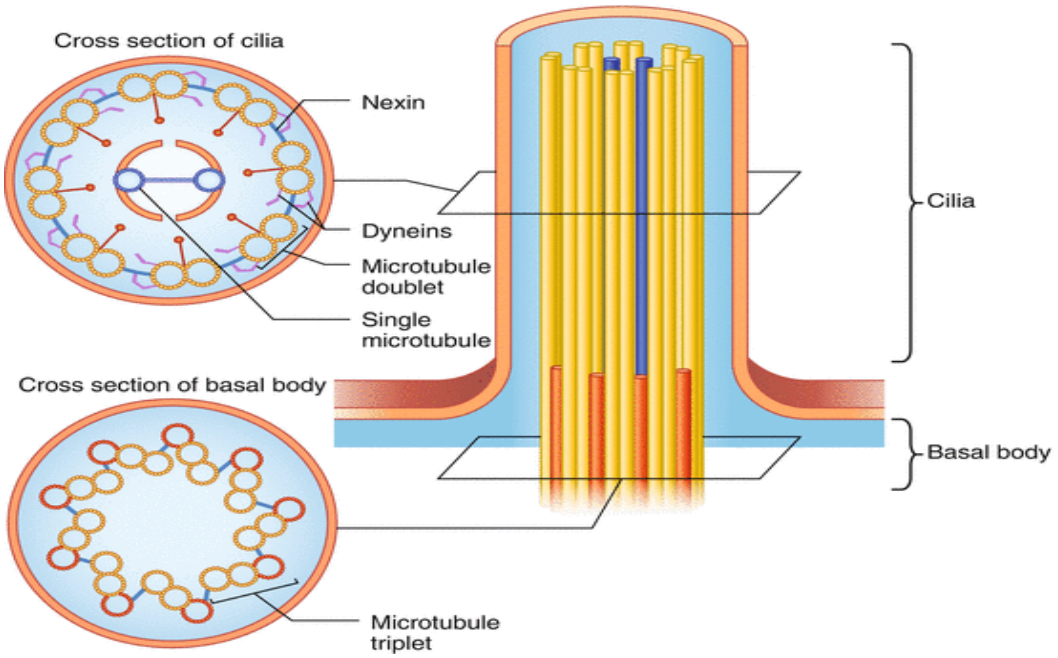


Fig. (1): Electronmicrograph of the cilia in longitudinal and cross sections .

2) Microvilli:

- Cylinder processes project from the surface of absorptive cells e.g. intestinal and kidney tubule lining cells.
- ***L.M.***: they give the absorptive surface brush like appearance.
- ***E.M.:*** the microvilli consist of fine filaments surrounded by the PM which is covered by thick glycocalyx.

- ***Function:*** they increase the surface area of the absorptive surface.

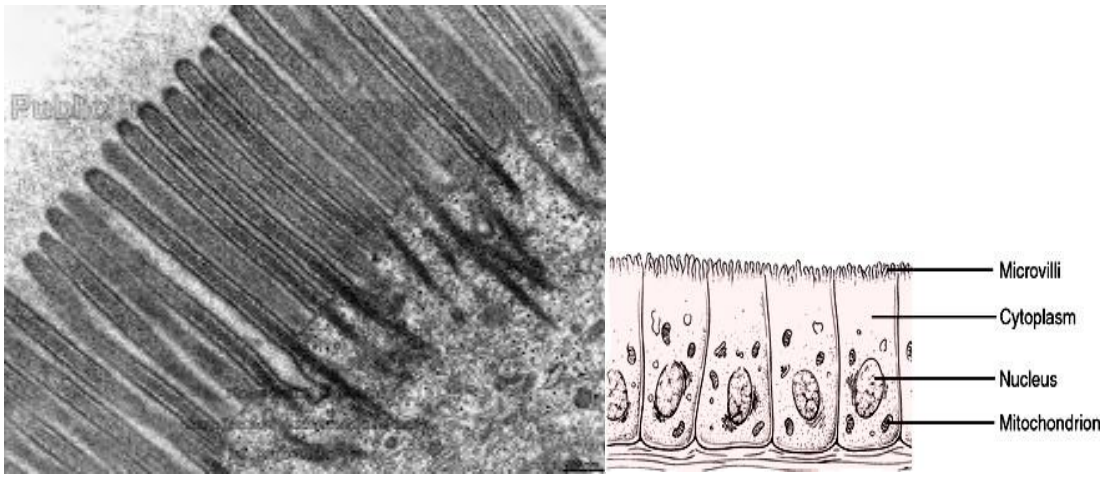


Fig. (2): Light and electron microscope of microvelli

3) Caveoles:

- little caves are small invaginations on the cell membrane resulted from the process of endocytosis and exocytosis e.g. endothelial lining of blood capillaries.

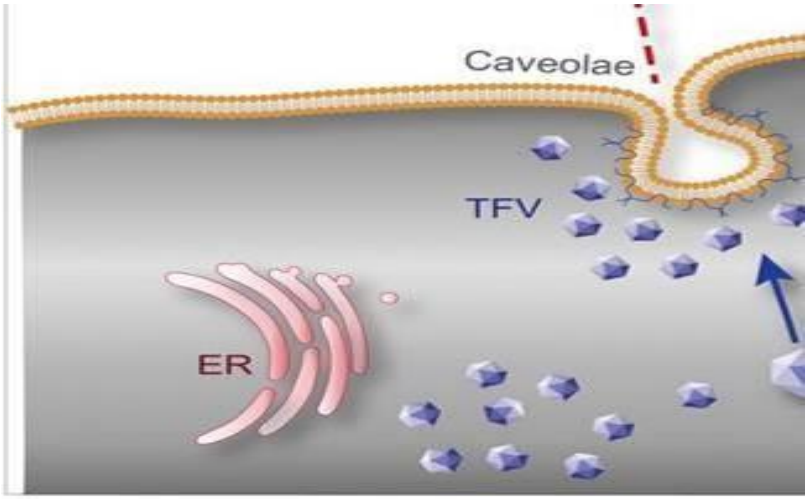


Fig. (3) : **caveole of apical phagocytic cell**

4) Cuticle:

- This is rich protein substance deposited on the outer lamina of the PM.
- present in cells exposed to harmful substances providing protection to them e.g. cells lining the urinary passages.

Covering And Lining Epithelium:

- The epithelium is named according to the number of cell layers, the shape of cells and the specializations of their free surface.

Types Fig. (4):

1- Simple epithelium: an epithelium made of a single layer of cells. These are:

(a) Simple squamous epithelium.

(b) Simple cuboidal epithelium.

(c) Simple columnar epithelium.

(d) Pseudo stratified epithelium.

2- Stratified epithelium: an epithelium made of multiple layers of cells. These are named according to the shape of the upper layer of cells. These are:

(e) Stratified squamous epithelium

(f) Stratified columnar epithelium.

(g) Transitional epithelium.

a) Simple squamous epithelium:

- The cells are polygonal in outline. In sections perpendicular to the plane of the epithelium, they appear thin and fusiform with flattened nuclei.
- They are present in the parietal layer of Bowman's capsule of the kidney, in the thin segment of loop of Henle's, in the rete testis, on the inner aspect of the tympanic membrane and in the endothelium lining the blood and lymph vessels.
- Endothelium is a type of simple squamous epithelium which lines blood vessels. Mesothelium is a type of simple squamous epithelium which lines the body cavities.

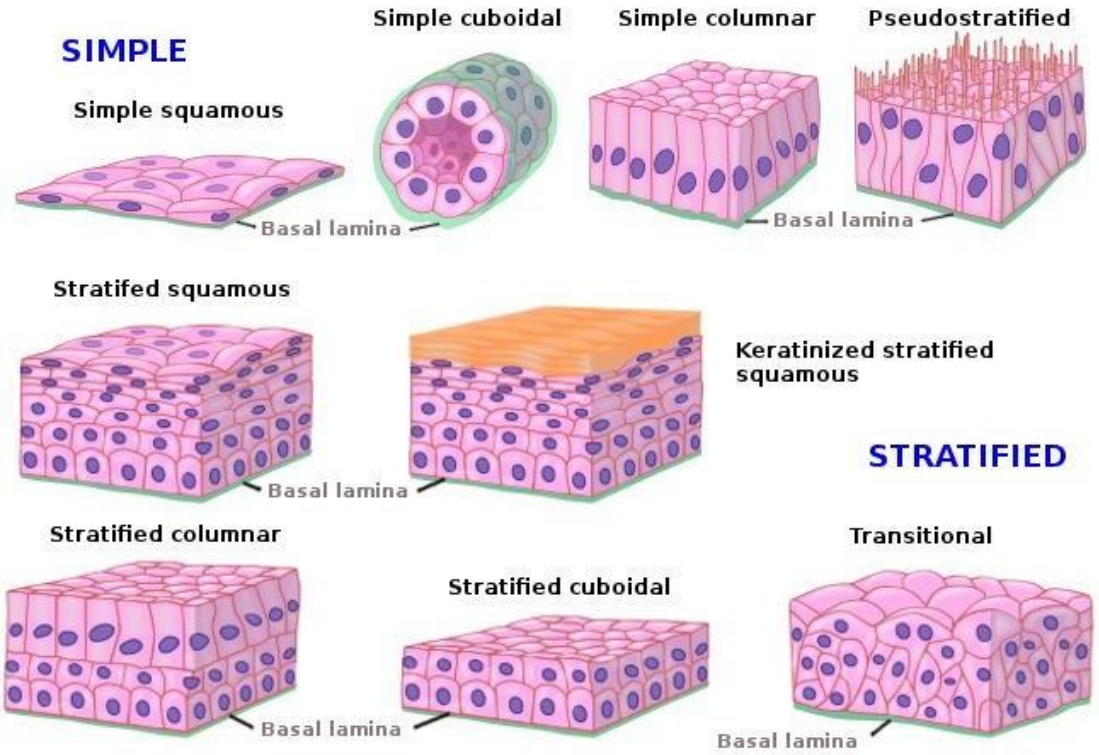


Fig (4): Different typesofepithelium

b) Simple cuboidal epithelium:

- In vertical section, this epithelium appears as a row of square or rectangular cells and in surface view, as a mosaic of polygonal cells with rounded nuclei.
- It is found lining the follicles of the thyroid gland and the kidney tubules, in the choroid plexus, on the capsule of the lens, in the pigmented epithelium of the retina and in the lining of ducts of many glands.

c) Simple columnar epithelium:

- The cells are rectangular in shape with their long axis perpendicular to the basal lamina.
- The nuclei are elongated aligned at the same level.
- It is found in lining of the digestive tract from the cardia of the stomach to the anus and in the large excretory ducts of some glands and in lining of the gall bladder.

- Simple columnar ciliated epithelium is found lining the uterus, the paranasal sinuses and in the central canal of the spinal cord.

d) Pseudo stratified columnar epithelium:

- All the cells lie on th e basal lamina but only a portion of them reach the surface.
- The cells are quite variable in shape.
- The nuclei of these different cells are present in different levels creating a false impression of cell stratification hence its descriptive term.
- It is found lining the greater part of the trachea and primary bronchi of the respiratory tract, the auditory tube, part of the tympanic cavity and lacrimal sac.

e) Stratified squamous epithelium:

- The cells vary in shape from the base to the free surface.
- The cells on the basal lamina are cuboidal or even columnar.
- Those above this layer are irregularly polyhedral, become increasingly flattened toward the surface and in the superficial layers they are thin squamous.
- It is found in the lining of the oral cavity, the epiglottis, oesophagus, conjunctiva, cornea, and vagina and in the distal portion of the urethra.
- On the surface of the body (skin), the superficial layers of cells lose their nuclei and the cytoplasm is largely replaced by the scleroprotein, keratin. Such epithelium is called keratinized stratified squamous epithelium.

f) Stratified columnar epithelium:

- The superficial cells are columnar.
- The basal cells are cuboidal and one or more rows of polygonal cells are interposed between the basal and the columnar cells.
- It is found in the fornix of the conjunctiva, in the cavernous urethra, in the pharynx and in large excretory ducts of some glands.
- Ciliated stratified columnar epithelium is found in the nasal surface of soft palate, in the larynx and is the fetal oesophagus.

g) Transitional epithelium:

- It occurs mainly in the urinary passages (uroepithelium).
- Its structure varies depending on its distension at the time of fixation.
- In the empty state, it has many cell layers.
- At the base the cells are low columnar or cuboidal, above there are several layers of polyhedral cells and the superficial cells are very much larger and may be binucleated with a characteristic rounded free surface (domed shape).
- During filling of the organ, the relationships of the cells change, there are usually only two or three layers and the superficial one is squamous.
- It is found throughout the urinary tract from the renal calyx to the urethra.

13 Secretory epithelium & glands

Bone growth, remodeling &repair:

- This is a tissue formed by cells specialized to produce a fluid secretion.
- Glands arise from covering epithelia by means of proliferation and invasion of subjacent C.T. followed by further differentiation .

Classification Of Glands:

1-Exocrine Glands: they retain their connection with the surface epithelium from which they originated. They have secretory portion which contains the cells responsible for the secretory process and ducts, which transport the secretion to the exterior of the gland e.g. salivary glands.

2-Endocrine Glands: these are ductless glands in which their connection with the surface from which they originated is obliterated. Their secretion is transported by blood stream to the target organ, e.g. pituitary gland.

3-Mixed Glands: they consist of the two previous types at the same time e.g. pancreas (exocrine pancreas and Islet's of Langerhans).

Classification Of Exocrine Glands :

A-According to the number of cells :	B-According to the nature of secretion
1) **Unicellular glands:**composed of one cell only e.g. goblet cell of the intestine Fig, (1). 2) **Multicellular glands:** consist of numerous cells e.g. salivary glands.	1) **Serous glandse**.g. parotid gland. 2) **Mucous glands** e.g. oesophageal glands. 3) **Seromucous glandse**.g. submandibular and sublingual. 4) **Glands of special nature**: e.g.mammary glands, ceruminous glands, sebaceous glands, lacrimal glands, and sweat glands.

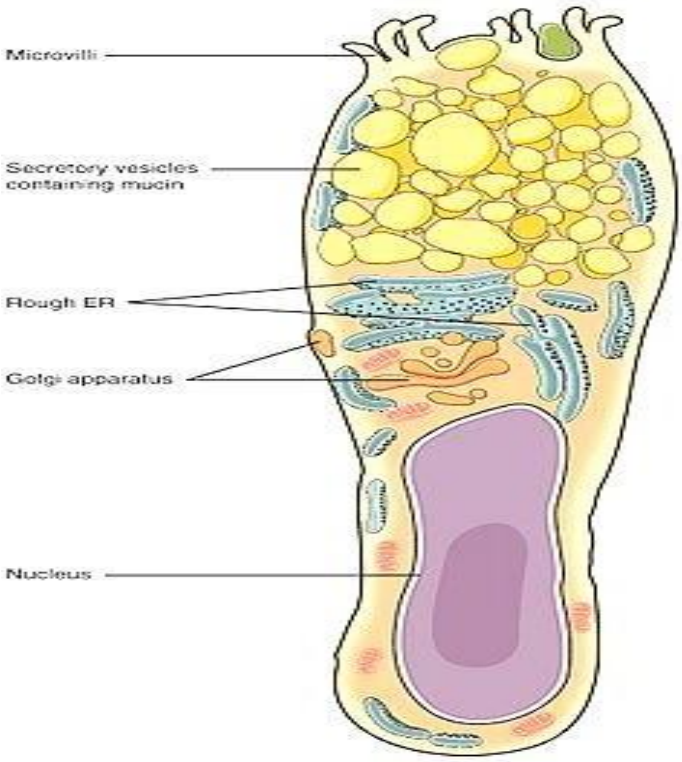


Fig. (1): Drawing of intestinal goblet cell, a unicellular gland

C-According to the mode of secretion:

1) **Merocrine glands:**

- in which the product is excreted by exocytosis without discontinuity in the plasma membrane.
- This occurs by fusion of the limiting membrane of secretory granules or vesicles with the cell membrane followed by outflow of their content e.g. eccrine sweat glands.

2) **Apocrine glands:**

- the mammary gland is a classical example.
- One or more lipid droplets are surrounded by a thin film of cytoplasm and a portion of the cell membrane project from the apex of the cell.
- This is pinched off by constriction at its base.

3) **Holocrine glands:**

- sebaceous gland is a classical example.

- The cells break down with an outpouring of cytoplasm and accumulated lipid-rich secretion.

D-According to the shape of secretory part and branching of ducts(Fig. 2):

1) Simple exocrine glands:

- The secretory portion is connected to a surface epithelium by an unbranched duct.
- **They are further classified according to the shape of secretory portion to**:

A-Simple tubular glands:
- are usually a straight tubule opening directly onto an epithelial surface without a duct e.g. glands of intestinal mucosa.

B-Simple coiled tubular glands:
- the secretory protion is a coiled tubule connected to the surface by unbranched duct e.g. sweat glands.

C-Simple branched tubular glands:
- the secretory portion bifurcate and there is a short unbranched excretory duct as in some glands of the oral cavity, oesophagus and duodenum.

D-Simple acinar (alveolar) glands:
- in which the secretory portion is expanded into a bulbous-shaped structure connected to the surface by unbranched duct e.g. glands of penile urethra.

E-Simple branched acinar :
- several acini are clustered around the duct e.g. sebaceous glands.

F-Simple branched tubulo-acinar:
- the multiple tubuloalveoli attached to the duct e.g. minor salivary glands.

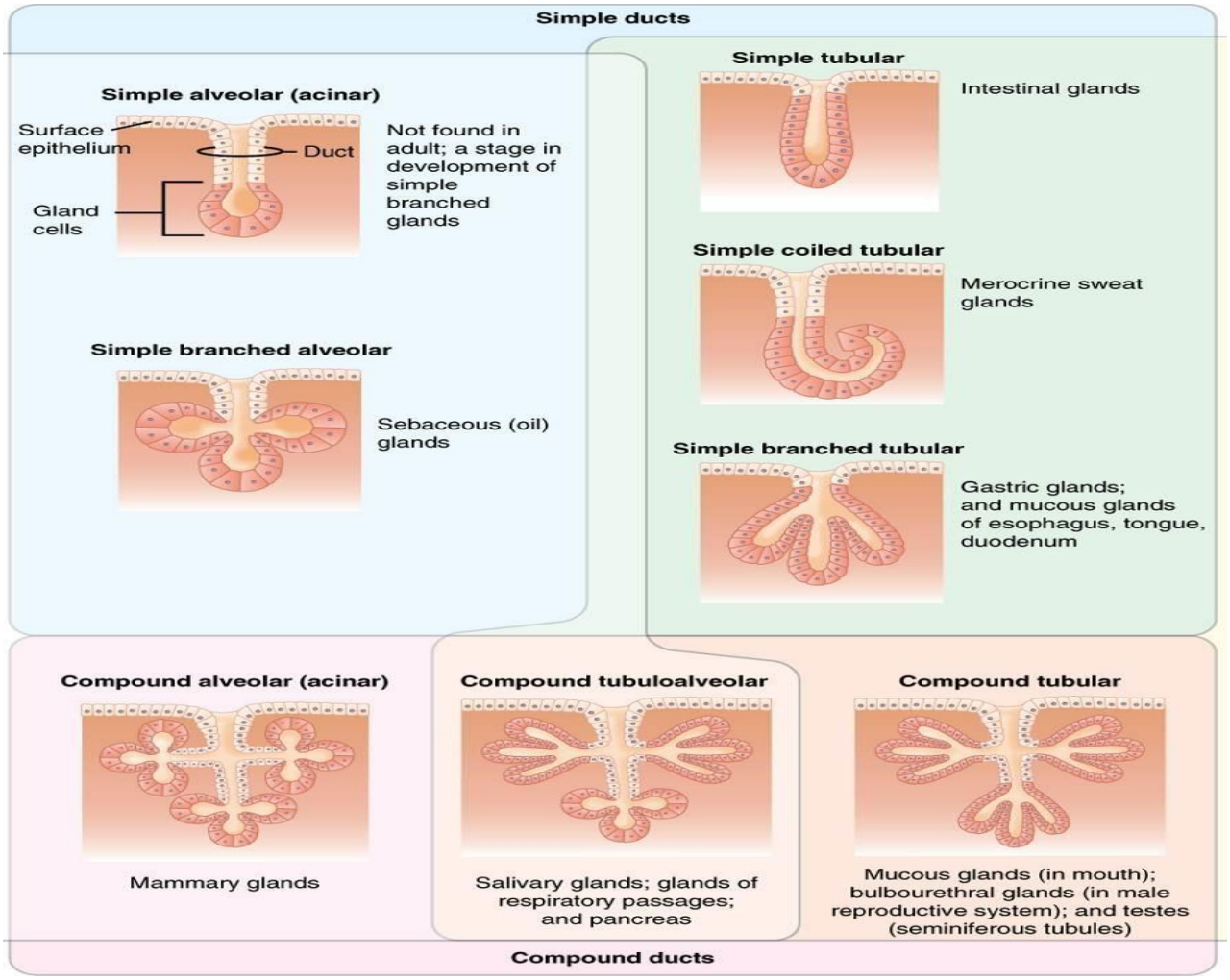


Fig. (2): Diagram of classification of multicellular exocrine glands.

2) Compound Exocrine Glands:

- The duct branches repeatedly.
- **They are further classified into:**

A-Compound tubular:
e.g. gastric glands and bulbourethral glands.

B-Compound acinar: e.g. pancreas.

C-Compound tubulo-acinar glands: e.g. major salivary glands.

D-Compound saccular:
e.g. mammary glands and prostate. Because of the greater size and relatively large lumens of their secretory units, the duct system doesn't converge on a single main duct; instead, several medium-sized ducts open separately on a restricted area of an epithelial surface.

Endocrine glands

- These glands synthesize signaling molecules (hormones) that are carried in the blood to act on distant target organs.
- They can produce sustained effects than signals carried by nerves.

Depending on the basis of chemical nature of their hormones, these glands are:

1) Protein secreting endocrine glands.
2) Polypeptide secreting endocrine glands.
3) Glycoprotein secreting endocrine glands.
4) Steroid secreting endocrine glands.

- A common feature of endocrine glands is their abundance of vascularity.
- The capillaries are fenestrated with large irregular lumen, occupying the plates or cords that make up the parenchyma of the gland.

Polypeptide secreting endocrine glands:

- ***e.g*** beta cells of pancreas, **Fig. (3).**
- These cells contain a few profiles of rER, a small Golgi complex and numerous secretory granules. The latter tend to accumulate at the vascular pole of the cell.

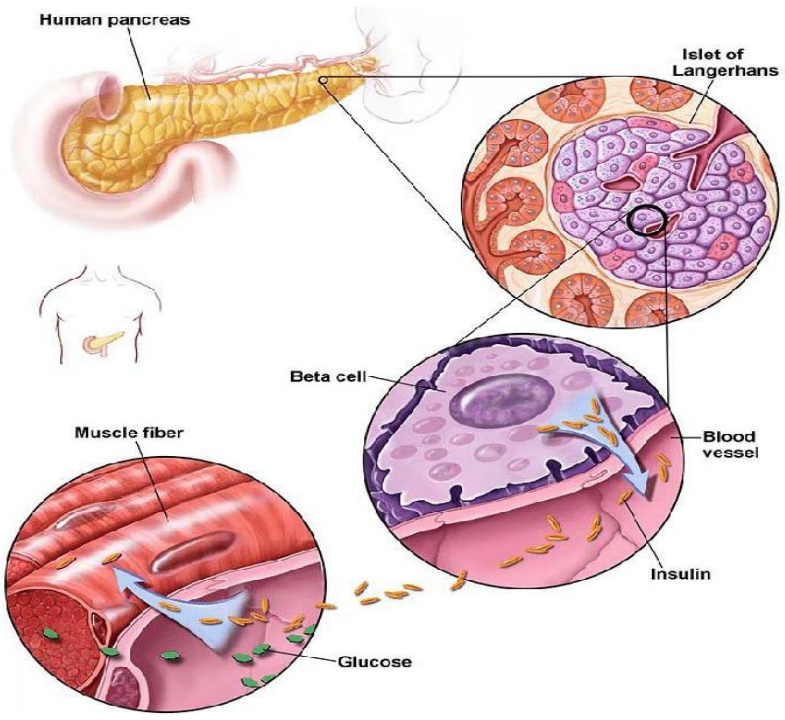


fig. (3): Drawing of a polypeptide-secreting endocrine cell.

Steroid Secreting Endocrine Glands:

- e.g endocrine cells of the ovary, testes and adrenal glands, **Fig. (4).**
- These cells have little rER, few free ribosomes, very extensive SER.
- This forms meshed network of branching and anastomosing tubules.
- The Golgi is juxtanuclear and large.
- There are numerous mitochondria and lipid droplets

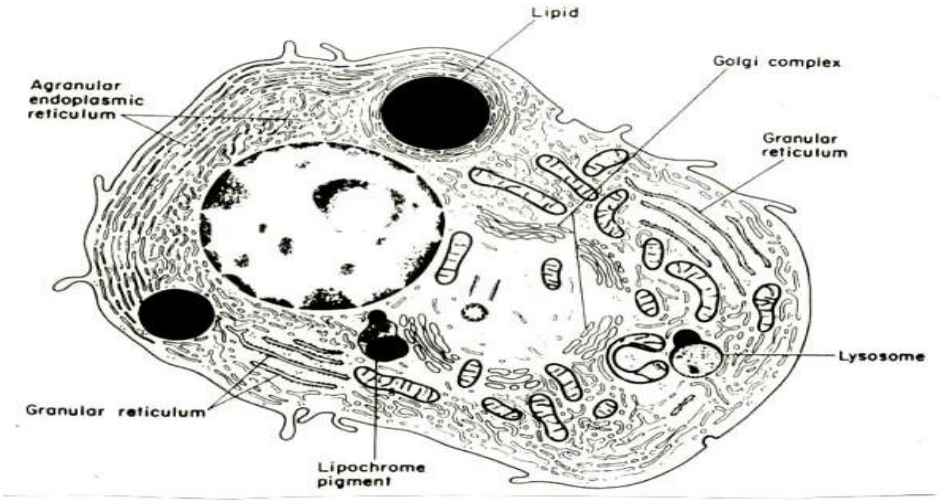


Fig. (4): Drawing of a steroid-secreting endocrine cell.

14 Connective tissue: Cells

Connective tissue cells can be classified into two groups:

A) Fixed cells :(fig.1)	B) Free cells :(Fig.1)
• These are stable long-lived cells and include the fibroblasts, fat cells, mesenchynal cells and pericyres.	• These are short-lived cells and are continually replaced from cells circulating in the blood. • This population includes mast cells, plasma cells, macrophages and leukocytes. • Some of the free cells participate in short-term responses to injury or bacterial invasion, whereas other participates in the long-term immunological defenses of the body.

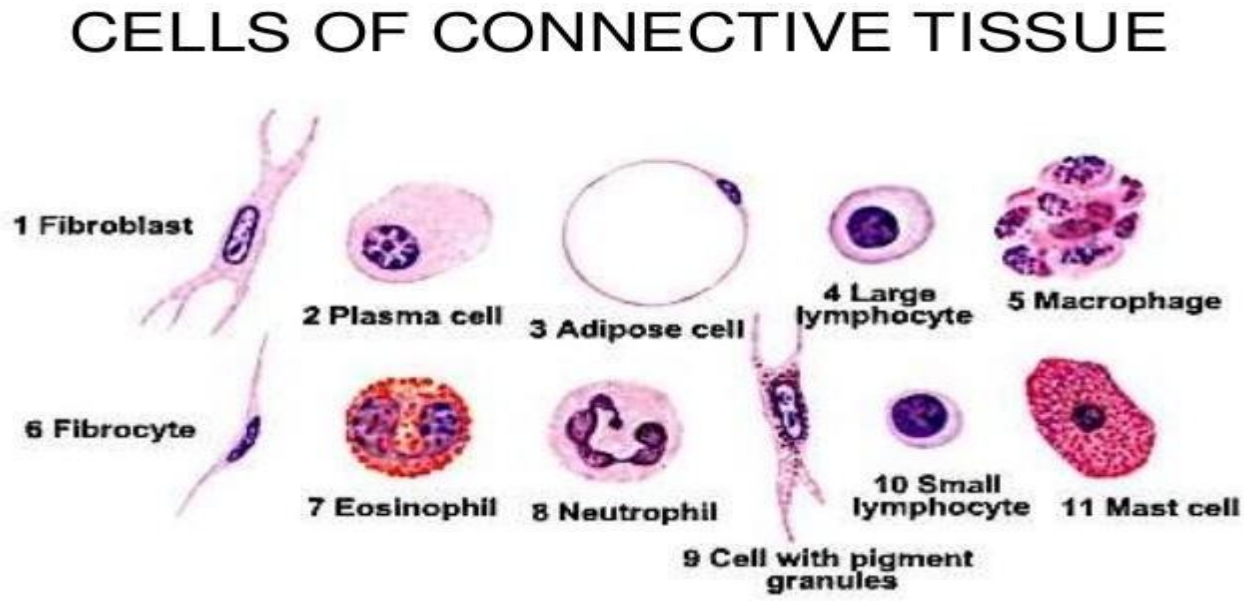


Fig.1 : Fixed and free cells of connective tissue .

1) Fixed cells :

a) Mesenchymal cells(Fig. 2):

- These are branched cell with pale basophilic cytoplasm and oval nucleus with prominent nucleoli and fine chromatin.

Function:

- Represent the origin **of** all types of connective tissue cells

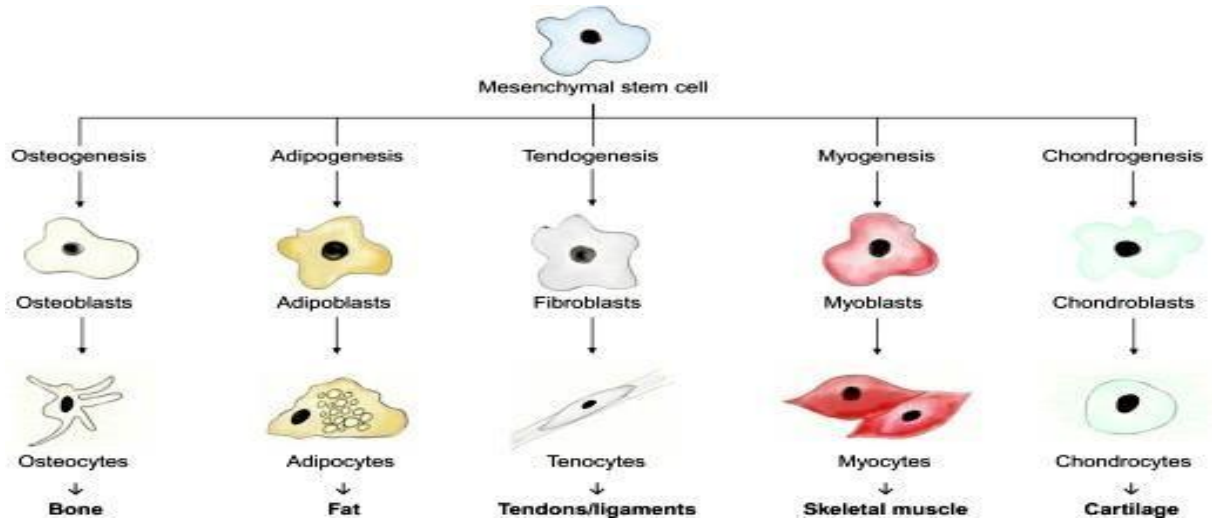


Fig.2: Mesenchymal cell

b) Fibroblast : (Fig. 3)

- Fibroblasts are the dominant cell type of connective tissue

Origin :

- Mesenchymal cells divide and differentiate into fibroblasts which are active young cells. Old fibroblasts are called fibrocytes.

A) L.M	B) E.M
• Their shapes vary depending on their location. • They are fusiform in between the bundles of collagen fibers. • In other situations, they may be flattened stellate cells with several processes.	**Since the fibroblast is a protein secreting cell ,It contains:** • Numerous rough endoplasmic reticulum. • Well developed Golgi stacks.

• The active young cells appear as branched cells with abundant basophilic cytoplasm and oval pale nucleus with prominent Nucleolus. The basophilia is due to the presence of numerous ribosomes and RER.	• Long, slender mitochondria are found mainly in the perinuclear cytoplasm. • The nucleus has small clumps of chromatin.

Functions:

- Formation of the intercellular substance.
- Secretion of tropocollagen whish assembly to form collagen fibers.
- Secretion of elastin which forms the amorphous non-flbrillar material of elastic fibrils and secretion of microfibrils which give it the shape of elastic fibers.
- Repair of wounds.

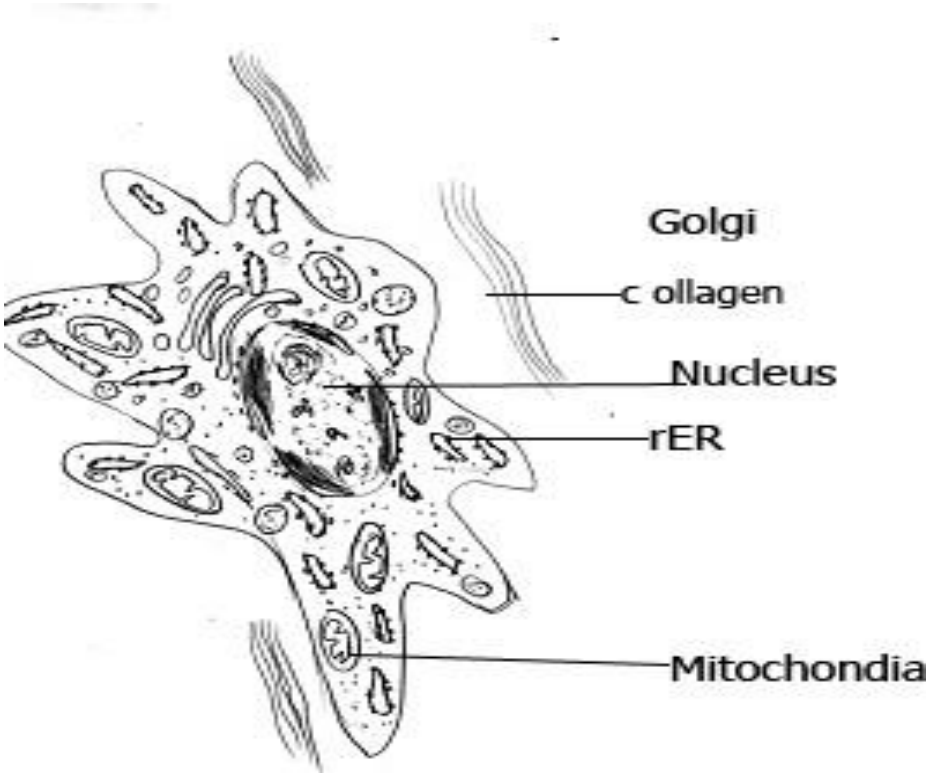


Fig.3: A fibrolast at EM level.

c) Fat (adipose) cells or adipocytes: (Fig. 4)

- They are fixed C.T cells that synthesize and store large quantities of lipid.

Origin

- They arise from mesechymal cells .

L.M

- The cell is spherical in shape
- The nucleus is flattened and displaced to the periphery by the lipid droplets.
- The cytoplasm is reduced to a thin film around a single large droplet of lipid.
- Using H & E stain, fat cells appear as empty spaces which have signet ring appearance .

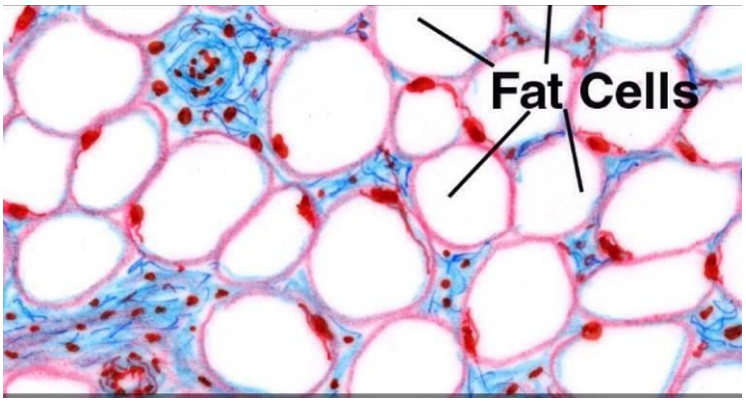


Fig. 4: Adipocyte

d) Pericytes (perivascular cells): (Fig. 5)

Site:

- **Pericytes are present associated with the endothelium of:**
  1. blood capillaries.
  2. metarterioles

L.M:

- Pericytes possess pale-staining narrow cytoplasmic processes that are similarly wrapped around the vessel.

Function:

- It is capable of division, constituting a potential source of new fibroblasts and smooth muscle cells as necessary in the repair of connective tissues and damaged blood vessels.

2) Free Cells of Connective Tissue

[a] Macrophages (Histiocytes) (Fig 6) :

Site:

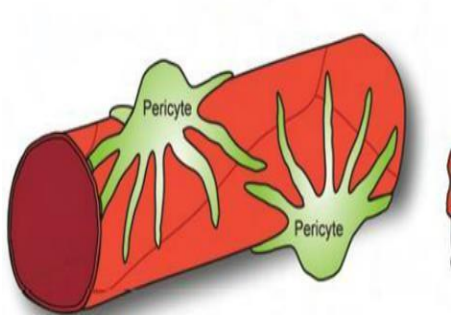
- Throughout the connective tissue of the body.
- In the liver macrophages are named kupffer cells
- In the central nervous system, microglial cells
- and in bone tissue. osteoclast

Origin:

- They arise by differentiation of monocytes of the blood, and constitute the mononuclear phagocyte system.
- The monocytes migrate into the connective tissue, where they mature and are called macrophages.

L.M:

- It has oval shape when present free and elongated or angular when it is compressed by other tissue.
- It has eccentric, kidney shaped and deeply stained nucleus.
- Macrophages can be identified with vital stains such as Trypan blue by the appearance of phagocytosed particles of the dye in their cytoplasm.



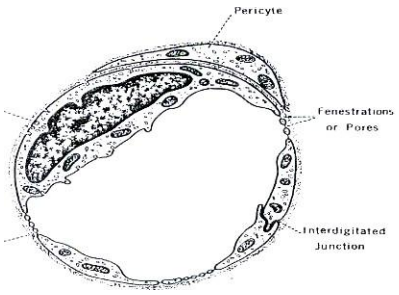


Fig.5 :Pericytes

EM:

- Macrophages are characterized by the numerous phagocytosed particles (Phagosomes), dense bodies and multivesicular bodies together with numerous lysosomes as well as rER, Golgi apparatus and mitochondria.

Function of macrophages:

1) Phagocytosis of foreign materials.
2) Release a variety of signaling molecules, called cytokines. One of these, interleukin-1, is a chemo-attractant that cause large numbers of lympoocytes to attract at the site and stimulates their proliferation and antibody production.
3) It is also chemotactic for neutrophils, attracting them to the site to participate in phagocytosis of bacteria.
4) Macrophages also have an essential role in initiating the body's immune response to an infection, by their intracellular processing of antigen and its presentation to p-lyrnphocytes in a form that is more irnmunogenic.

[b] Foreign Body Giant Cells:

Origin

- Originate when a foreign material is too large to be phagocytosed by the macrophages.
- The fusion of two or more macrophages or monocytes .
- It is a large cell with two or more nuclei which can engulf the large foreign particles.

[c] Leucocytes:

- Loose C.T. may contain some leucocytes such as Neutrophils, Eosinophils and lymphocytes.
    - They migrate through the walls of capillaries and venules from the blood to connective tissues.

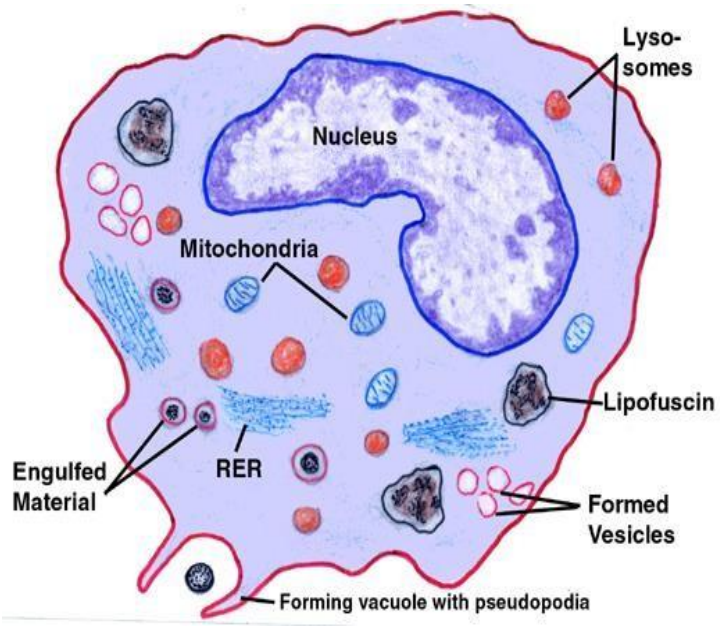


Fig.6: EM of macrophage.

[d] Plasma Cells (Fig .7) :

Origin

They Plasma cells arise in the connective tissues by further differentiation of B lymphocytes.

L.M:

- They are large, spherical or ovoid cells having an eccentrically placed nucleus.
- The cytoplasm is deeply basophilic with negative Golgi region close to the nucleus.

E.M:

- The cytoplasm is filled with closely spaced cisternae of rough endoplasmic reticulum.
- Their product, immunoglobulin (antibody), is secreted continuously by exocytosis of small vesicles that shuttle between the Golgi complex and the cell surface.
- The condensed chromatin of the nucleus is distributed peripherally giving it a characteristic clock face or cart wheel appearance

Function:

- Secretion of immunoglobulin (antibodies) in response to the stimulation of their mother cell (B lymphocytes) by specific antigens.

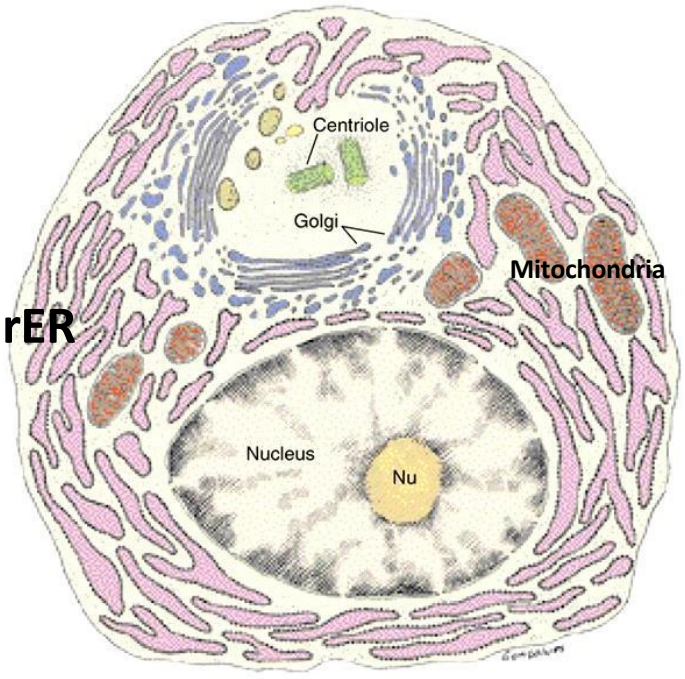


Fig.7: EM of plasma cell

[e] Mast Cells(Fig. 8). :

Origin:

- They originate from stem cell in the bone marrow.

L.M:

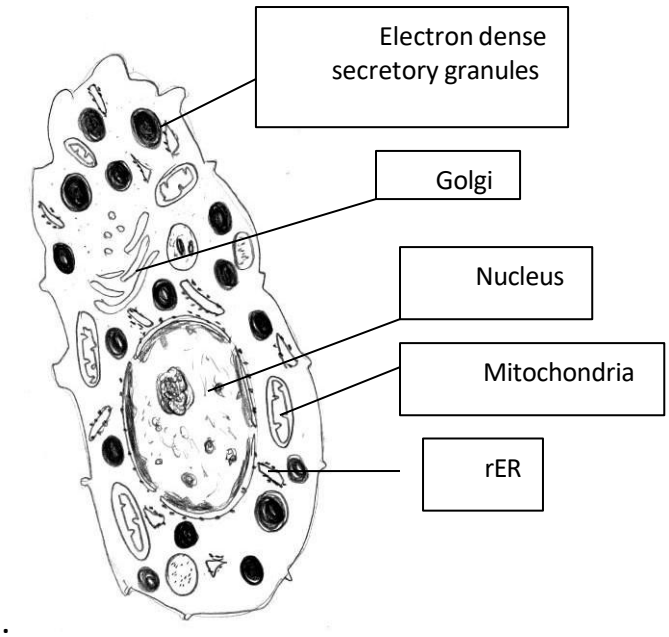
- Mast cells are the largest of the connective tissue cells.
- They are easily identified by the intensely basophilic granules that contain histamine and heparin.
- Each cell contain single central, rounded and vesicular nucleus.

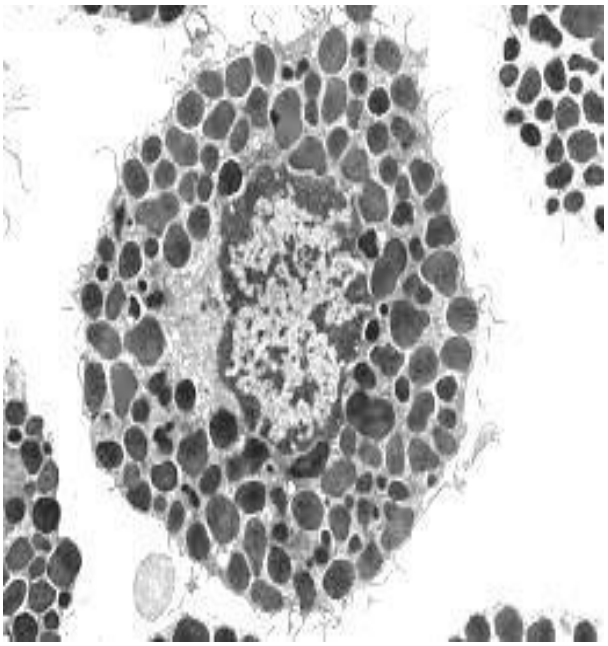
E.M

- Mast cell contains profiles of endoplasmic reticulum, a small Golgi, and a few mitochondria.

Function of Mast Cells

1) Secretion of heparine and histamine:





Connective tissue: fibers

- It consists of widely separated cells in an abundant intercellular matrix.

❑ Functions of the C.T.:

- It is found throughout the body, supporting and binding other structural elements together.
- It forms the capsules of organs.
- It is the tissue through which the blood vessels enter and leave the organ.
- The intercellular matrix act as a medium through which all nutrients and waste products diffuse between the blood and the parenchymal cells of the organs.
- It has a role in defense mechanism and immunity.

❑ Classification of C.T.:

- According to the amount of the intercellular substance and the type of cells, the C.T. is classified into:

- ***Embryonic Connective tissue including:***
  - Mesenchymal connective tissue.
  - Mucous connective tissue

❑ Connective tissue proper including:

- Loose (areolar) C.T.
- Dense regular and irregular C.T.

❑ Connective tissue with special properties including:

- Adipose tissue
- Blood
- Hematopoietic (Lymphatic and myeloid) tissue.
- Cartilage.
- Bone.

Connective Tissue Proper

Loose (Areolar) C.T. : (Fig.1)

- Loose connective tissue is the most widespread of the connective tissues.
- ***Site:***
  - It is present in sites where relatively little resistance to stress is required.
  - It fills spaces between groups of muscle cells.
  - Supports epithelial tissue.
  - Forms a layer that sheathes the lymphatic and blood vessels.
  - It is also found in the papillary layer of the dermis.
  - In the serosal linings of peritoneal and pleural cavities.
  - In glands.
- ***Structure of loose C.T. :***
  - The loose C.T. is composed; fibers, amorphous matrix and cells.

C.T. Fibers

- There are three types of fibers: Collagenous, elastic and reticular fibers.

Collagenous (white) fibers

- ***General characters***
  - Are elongated or wavy bundles which are formed of fibrils.
  - Have longitudinal striations.
  - Composed of the protein collagen (which gives gelatin on boiling).
  - Collagenous fibers are acidophilic thus staining pink with H&E.

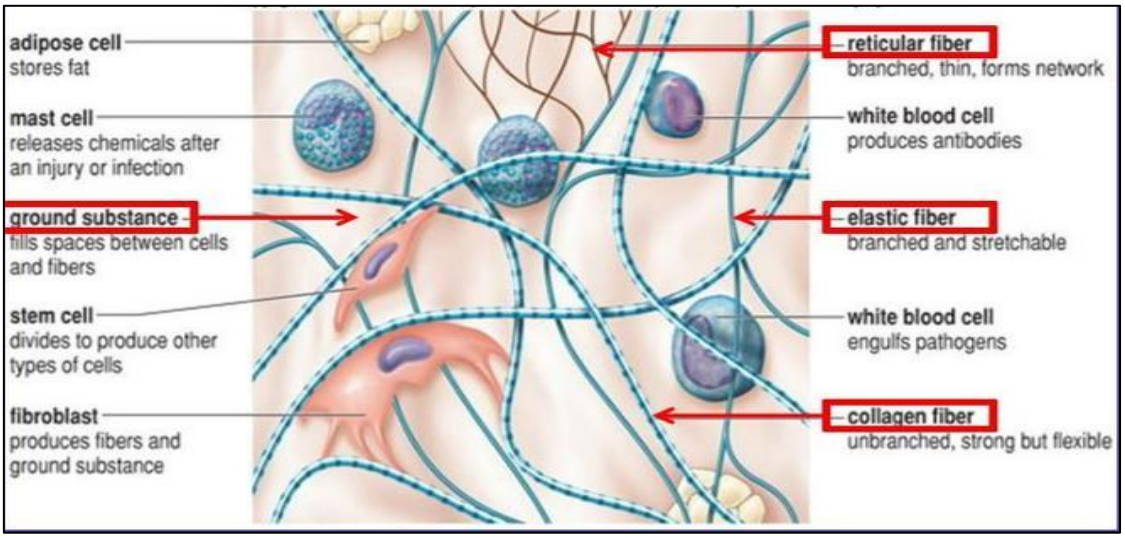


Fig .1: Loose connective tissue

- ***LM:***
  - The collagenous fibers in spreaded C.T. appear as long, wavy, pink fiber bundles (acidophilic). The fibers do not branch, while the bundles can.
- ***EM:***
  - The collagenous fibers demonstrate periodicity alternating dark and light segments that are repeated at regular intervals of 67 nm along the length of the fibril (Fig. 2). Each collagen fiber is composed of parallel fibrils. Each fibril is formed of parallel aggregates of micro-fibrils (tropcollagen)
- ***Function of collagenous fibers :***
  - They provide strength of the tissue.
- ***Types of collagen:***
  - Collagen type 1: Occurring in the dermis, bone, tendon and in the capsule of organs.
  - Collagen type П: In hyaline and elastic cartilage.
  - Collagen type Ш: It forms the reticular fibers.
  - Collagen type IV: In the basal lamina of epithelia.
  - Collagen type V: It is associated with the external lamina of smooth and striated muscle fibers.

Elastic (yellow) fibers

- ***LM***
  - They are long, thin fibers which have no longitudinal striations.
  - They branch and may unite and form irregular networks.
  - They are mainly composed of the protein elastin.
  - Elastic fibers are weakly acidophilic. They cannot be demonstrated well with H & E stain. However, they are well demonstrated with orcein stain where they take a dark brown color.
- ***EM***
  - Elastic fibers consist of an amorphous central region of elastin surrounded by a sheath of fibrils. There is no periodicity in elastic fibers (no cross striations).
- ***Site***
  - Connective tissue through the body

- The connective tissue in the alveolar septa
- The large blood vessels as the aorta

- ***Function of elastic fibers :***
  - They provide elasticity and strength to the tissue.

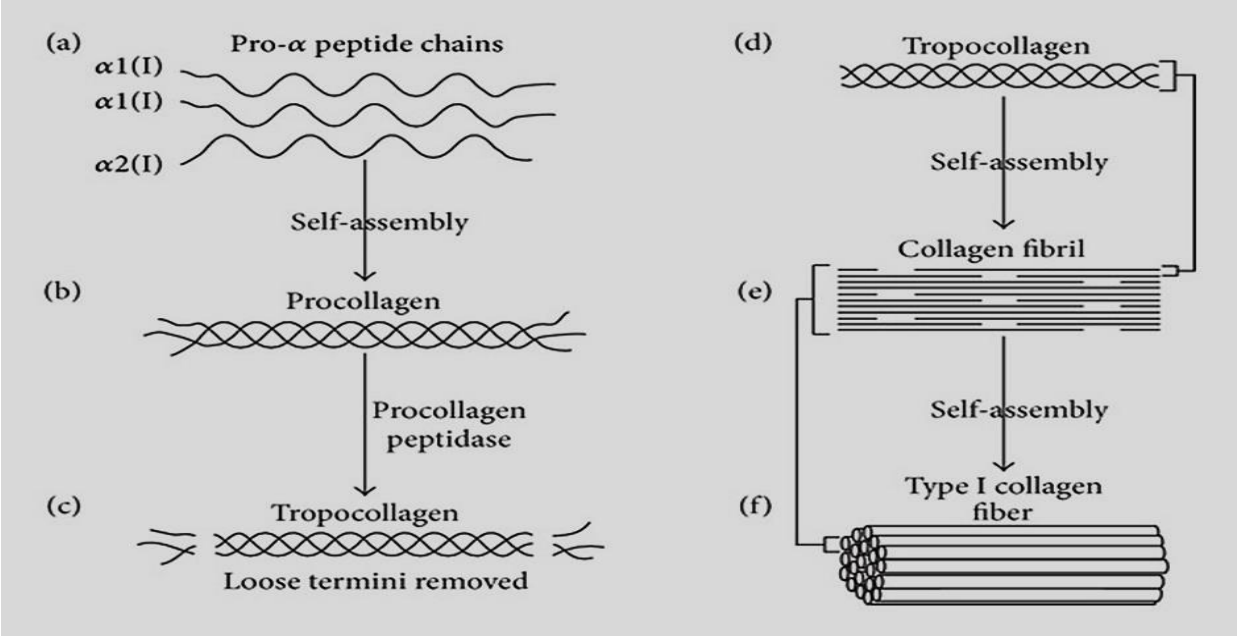


Fig.2: Schematic drawing of collagen molecules (tropocollagen) , fibrils, fibers and bundles .

Reticular fibers

- ***General characters***
  - They are fine fibers which are arranged in the form of networks.
  - They are mainly formed of the protein collagen in addition to carbohydrates.
  - They are considered to be immature collagenous fibers.
  - They are formed of fibrils 60 nm in diameter and have periodicity typical of collagenous fibrils.

- ***LM***
  - They are not visible in H &E preparations but they can be demonstrated well with **sliver** stains where they take a black color.

- ***Function of reticular fibers :***
  - They provide a network for cells which form together with the fibers the framework of the

C.T. proper: Ground substance &Types

❑ Connective Tissue Proper

Loose (Areolar) C.T.

- Loose connective tissue is the most widespread of the connective tissues.
- It is composed; fibers, amorphous matrix and cells.

Amorphous material (fig.1

- Is called amorphous because it has no particular form.
- It is formed of a semi fluid material.
- it is either : -
  - Watery and soft (loose C.T.)
  - Firm and flexible (cartilage)
  - Hard and calcified (bone)
  - Liquid (blood).

Function of the amorphous material:

- Transport of nutrients and oxygen from capillaries to tissues by diffusion through the intercellular substance of C, T. which contains a considerable amount of fluid important for nourishment of epithelial tissue.

The components of the amorphous

- It is a hydrated, amorphous gel-like material that is composed of:

❑ three families of macromolecules:

- ***Glycosaminoglycan:***
  - are long un-branched polymers of repeating disaccharides of two groups :
    - None sulfated mucopolysaccharides in the form of hyaluronic acid.
    - Sulphated mucopolysaccharids

- ***Proteoglycans :***
  - protein cores to which various glycosaminoglycans are covalently linked
- ***Adhesive glycoproteins:***
  - functions in attaching various components of the extracellular matrix to one another and to the cell membrane

❑ *Soluble proteins of the matrix:*

- They are the precursors of collagen fibers and the plasma proteins that leak from blood capillaries. It is a transparent semi fluid, stains weakly basophilic.

❑ *Water (tissue fluid):*

- It is the medium through which cells take nutrients and oxygen. Water is removed during preparation so its sites appear as empty spaces not stained with H&E thus loose C.T. is called areolar C.T.
- The water part of the amorphous substance contains some soluble materials that are diffusing from and to the blood capillaries and cells. This tissue fluid or extracellular fluid is a simple solution of gases and nutrients. The increase in the amount of tissue fluid than normal results in a condition called **edema**.

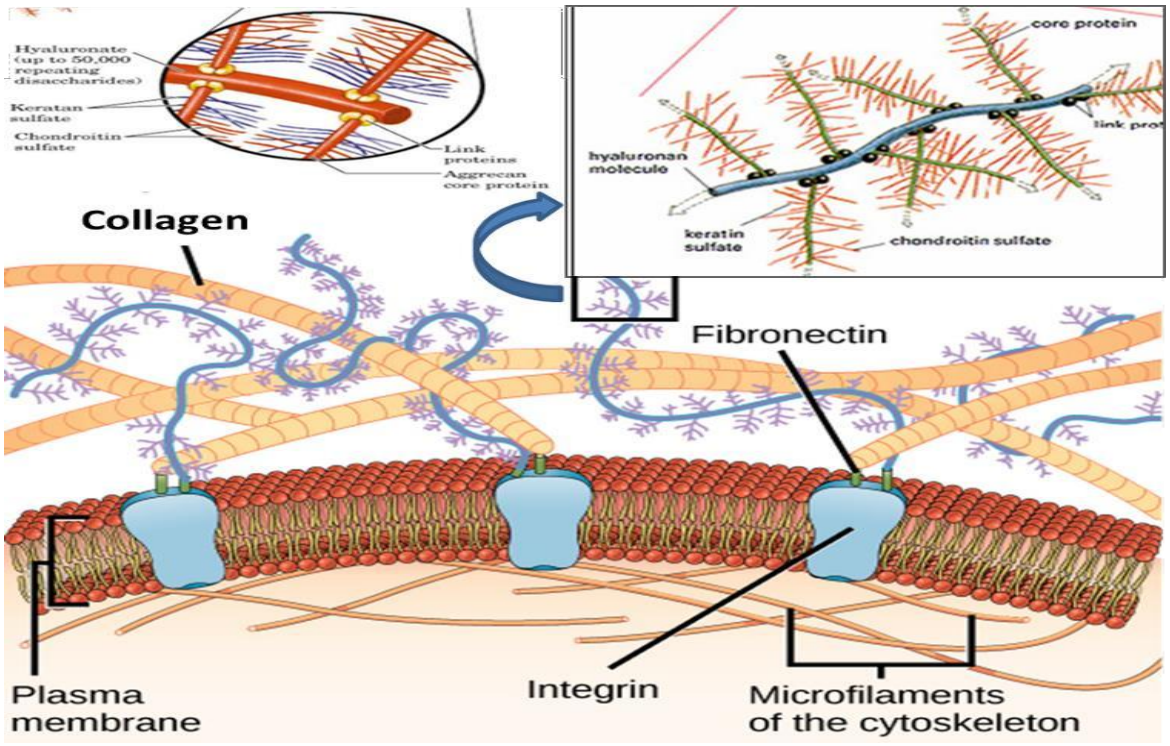


Fig.(1):The molecular structure of proteoglycans and glycoproteins

Dense Connective Tissue

Dense Connective Tissue

- Dense connective tissue consists of the same components found in loose connective tissue.
- But there are fewer cells and a clear predominance of Collagen and elastic fibers.
- It includes different subtypes :

❑ *Dense white fibrous connective tissue:*

A. Dense regular white fibrous C, T. (e.g., tendon):

- It consists of intercellular substance and fibroblasts. The intercellular substance is composed of an amorphous material and collagenous fibers which are numerous and arranged regularly in the form of bundles in the same plane and the same direction.

B. Dense irregular white fibrous C. T. (e.g., sclera of eye):

- In this type, the collagenous fibers run in various directions and are irregularly arranged.

❑ *Dense elastic (yellow) connective tissue:*

- ***Dense regular elastic C. T (e.g. in the ligamentum nuchae):***
  - In this type elastic fibers are predominant and they are regularly arranged in the same plane.
- ***Dense irregular elastic C. T (e.g. in the media of the aorta):***
  - The elastic fibers form fenestrated elastic membranes.

C.T with Special Prosperities:

❑ *Adipose connective tissue (FIG. 2):*

- There are two types of adipose tissue that have different locations, structures, colors, and pathologic characteristic

- ***Unilocular (common, or white adipose tissue):***

- It is composed of cells that contain one larg central droplet of yellow fat in their cytoplasm.

- ***Multilocular (brown) adipose tissue:***

  - It is composed of cells that contain numerous lipid droplets and abundant brown mitochondria. It is abundant in animals. Both types of adipose tissue have a rich blood supply.

❑ *Function of adipose tissue :*

- Act as a source of energy.
- Give the body surface its shape.
- Act as thermal insulation of the body.
- Adipose tissue also fills up spaces between other tissues.

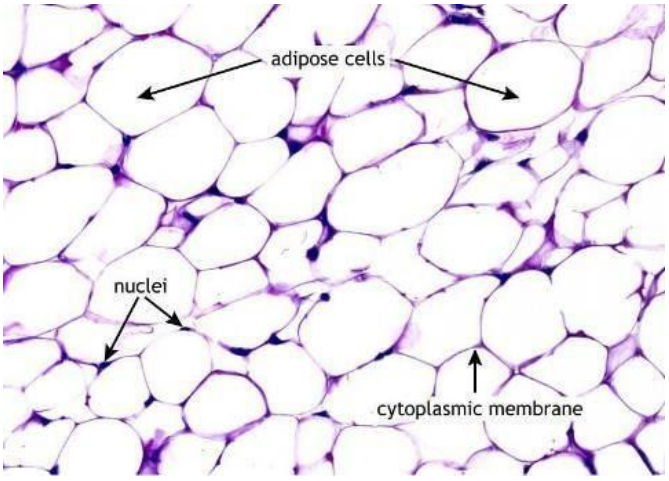


Fig .2: Section from liver stained with silver showing reticular fibers

Reticular Connective Tissue (FIG.A

- Is a specialized loose connective tissue that provides the framework of the myeloid (bone marrow) and lymphoid (lymph nodes, spleen) organs.
- It is formed of reticular cells and reticular fibers
- Reticular cells are fibroblasts specialized for secreting the constituents of reticular fibers and are present dispersed along the matrix and sheath the reticular fibers and ground substance with cytoplasmic processes.
- The resulting cell-lined trabecular system creates a sponge-like structure within which cells and fluids are freely mobile

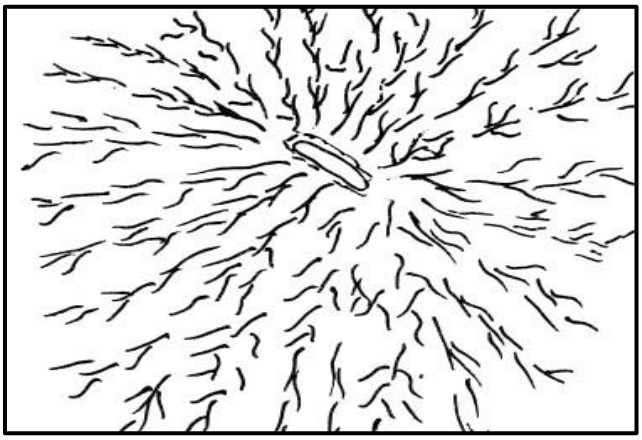


Fig .3: Section from liver stained with silver showing reticular fibers.

Mucoid Connective Tissue (FIG 4):

- Mucoid tissue has an abundance of ground substance composed chiefly of Hyaluronic acid.
- It is a jelly-like tissue containing very few fibers.
- The cells in this tissue are mainly fibroblasts.
- Mucoid tissue is the principal component of the umbilical cord and it is also found in the pulp of young teeth.

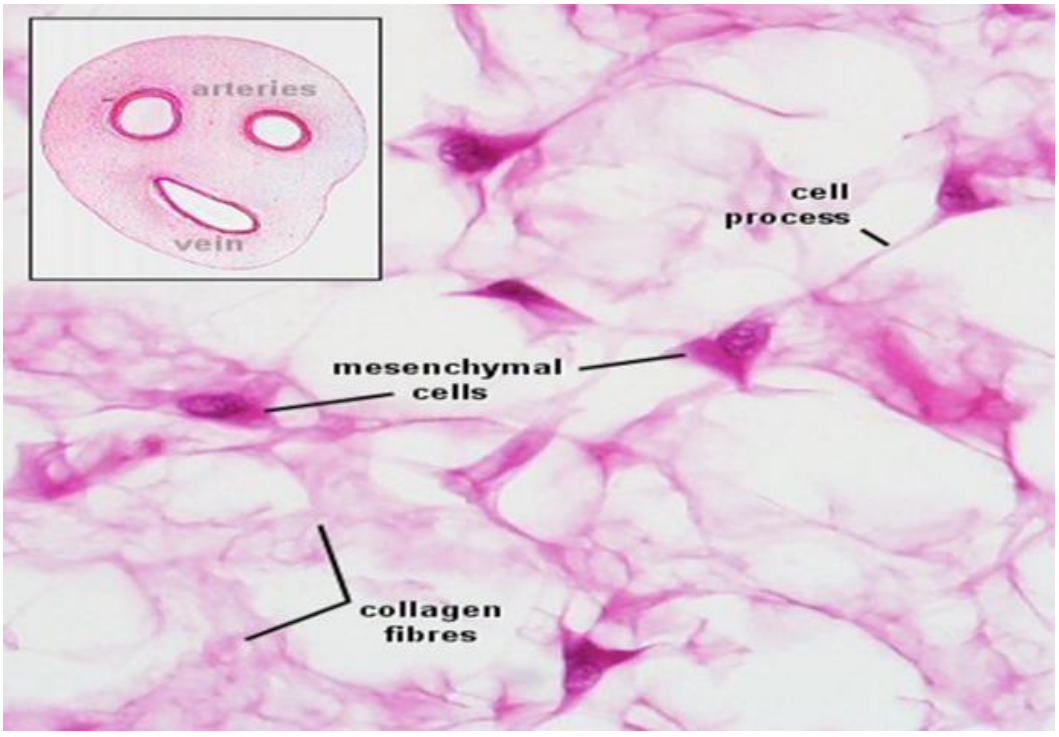
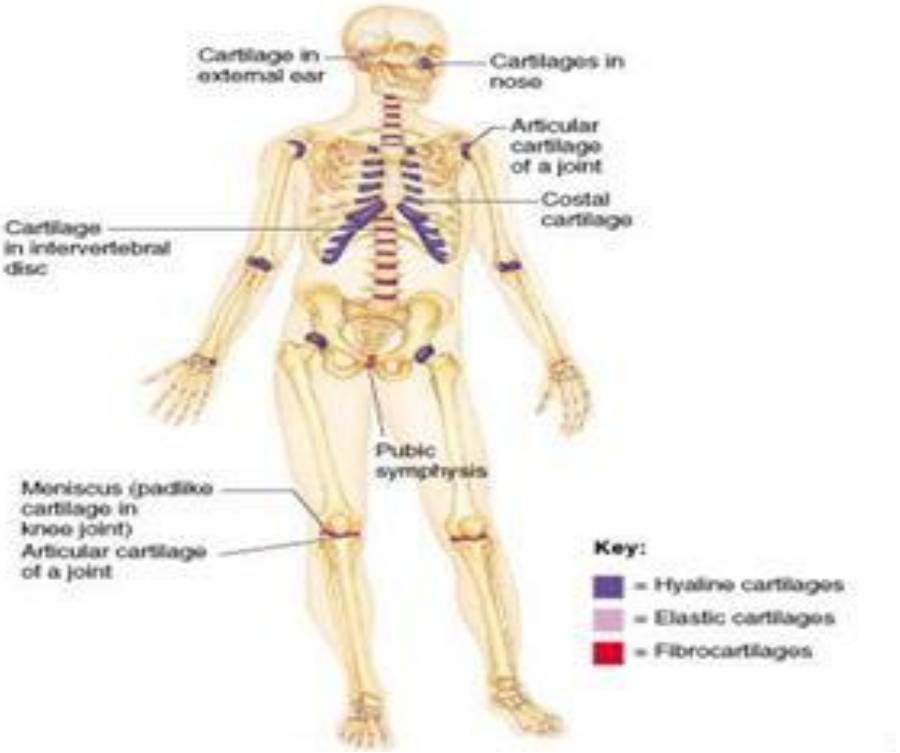


Fig .4: Section from umbilical cord (mucoid C.T.) stained by Hx&E .

CARTILAGE

Definition

- Cartilage is a connective tissue in which the amorphous intercellular substance is hardened to provide rigidity, support and attachment for tissue. So it is characterized by glassy, rigid, rubbery, non vascular matrix with some degree of flexibility



➢ Histogenesis of cartilage :

Origin;

- The mesenchymal cells which then form chondroblasts.

➢ Types of cartilage: (Fig.1)

1. Hyaline cartilage
2. Yellow elastic cartilage
3. White fibrocartilage

➢ Histological structure of cartilage

- Formed of; cartilage cells, matrix and perichondrium

➢ Cartilage cells

- Chondroblasts
- Chondrocytes

1) Chondroblasts

Origin

- Chondroblasts develop from embryonic mesenchymal cells

L.M

- Shape: chondroblasts are oval or spindle in shape with basophilic cytoplasm. The nucleus is rounded, central and vesicular

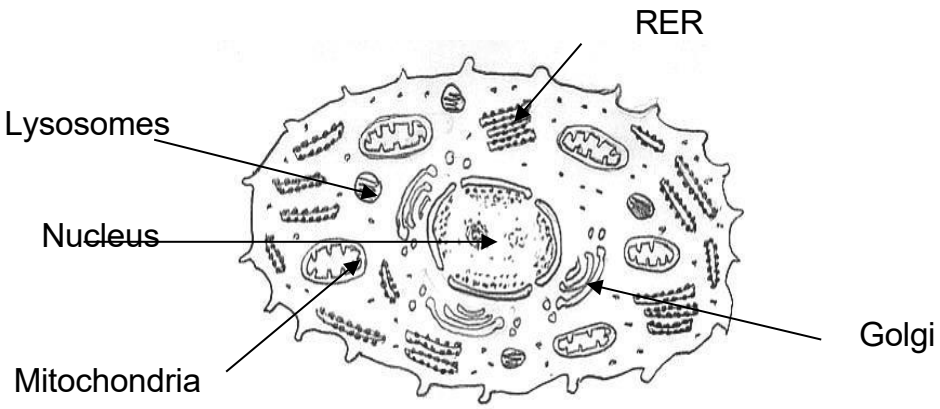


Fig.2: A diagrm showing the ultrastructure of chondroblast

Site

- They are located in the chondrogenic layer of perichondrium

E.M

- They are (protein secreting cells), contain abundant rER, ribosomes, prominent Golgi apparatus and many mitochondria . The nucleus is euchomatic

Function

- They are cartilage forming cells:
    1. Secretes type II collagen and other ECM components
    2. After maturation they are surrounded with lacunae and then they are called chondrocytes

2) Chondrocytes

- Mature cartilage cell

Site

- present in the deeper part

L.M

- They are round if single, or semicircular (if in groups)
- Reside in a space in the matrix called the **lacuna.** Each lacuna is surrounded with a condensed matrix called **capsule**
- Each lacuna contains one cell or it may contain group of cells (2,4,8) , then it is called **cell nest**
- In routine histological preparation, the cartilage cells and the matrix shrink, resulting in both irregular shape of the chondrocyte and retraction from the capsule
- However,the chondrocytes, fill the lacunae completely in properly prepared section

E.M

- Chondrocytes completely fill their lacunae
- Protein secreting cells: secrete matrix
- Well-developed RER, Golgi, mitochondria and euchromatic nuclei
- Irregular outline due to the presence of cytoplasmic processes

Function

1. Synthesize and secrete sulfated glycosaminoglycans and collagen fibrils of the matrix
2. Growth of cartilage (interstitial growth)

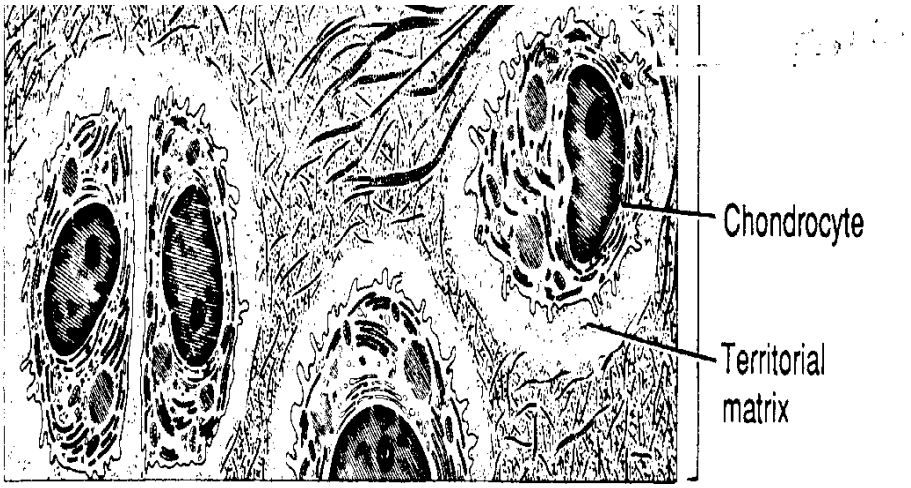


Fig.3: Diagram showing the E.M structure of the chondrocyte

1 Hyaline cartilage

- Hyalos; means glass
- It is the most common form and has a pearly white translucent appearance like frosted glass

Site

- In the embryo, it forms the skeleton prior to its replacement by bone
- **In the adult, it is located in;**
  - Articular surfaces of joints
  - Walls of the large respiratory passages
  - Costal cartilage (ventral ends of ribs, site of articulation with the sternum)
  - Epiphyseal plate, it is responsible for the longitudinal growth of bone

Function

- Support tissue and organs
- Model for bone development

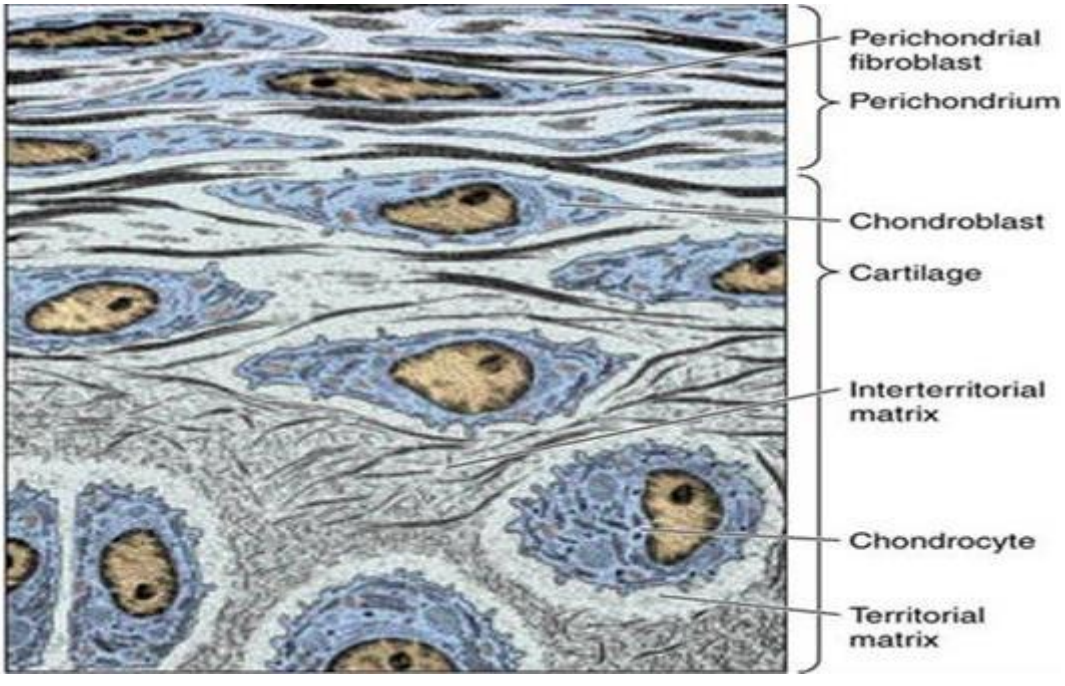


Fig.(4) :Hyaline cartilage

➢ **Histological structure of hyaline cartilage:**

- Formed of; cartilage cells, matrix and perichondrium

1) Cartilage cells

- Chondroblasts
- Chondrocytes

2) Perichondrium

- Dense irregularly arranged connective tissue (type I collagen)
- Ensheaths the cartilage
- Contains the blood vessels that nourish chondrocytes in addition to nerves and lymphatic vessels

➢ **Consists of 2 layers**

- Outer fibrous
  - Formed of collagen fibers, fibroblasts and blood vessels
- Inner cellular (chondrogenic)
  - Formed of chondroblast which can be differentiating into chondrocytes

3) Matrix

➢ **Non vascular**, rigid and flexible matrix. The matrix appears homogenous and basophilic. It is formed of:

a. Ground substance: formed of;
   - Tissue fluid (75%) to facilitate diffusion of oxygen and nutrient
   - amorphous substance (25%), which includes;
     - ➢ glycosaminoglcan
       - sulphated (chondrotin sulphate)
       - non sulphated (hyaluronic acid)
     - ➢ Protoglycan
       - formed of protein core and side chain of glycosaminoglycans
     - ➢ Glycoprotein
       - chondronectin, fibronectin and anchorin
     - ➢ Non collagenic proteins

stains

- Hx&E: basophilic, due to the presence of acidic chondrotin sulphate
- Metachromatic, due to the presence of glycosaminoglycan
- PAS: due to the presence of glycoprotein
- The cartilage matrix surrounding each chondrocyte is rich in glycosaminoglycan (sulphated groups) and poor in collagen
- This zone, called **territorial**, or capsular matrix exhibits an intense basophilia, metachromasia, and PAS positive that is greater than does the **interterritorial** matrix (between the capsules).

b) Fibers; form a network of collagen fibers (type 2)

➢ The fibers are not seen by Hx&E. because:

- The matrix and collagenous fibrils have the same refractive index
- The collagen fibrils present in submicroscopic dimension

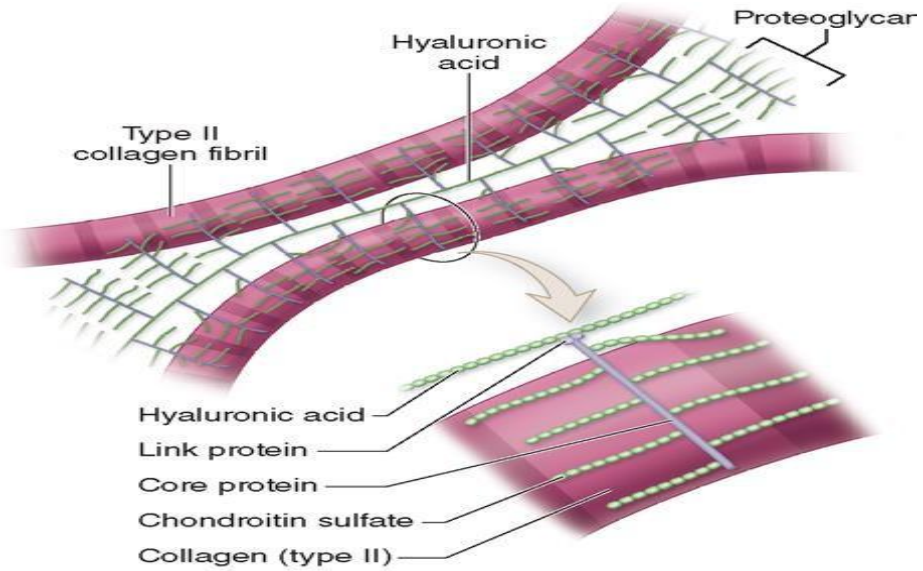


Fig.5: the molecular organization of cartilage matrix

2 Elastic cartilage

- It appears yellow in fresh state

➢ The histological structure:

- similar to hyaline cartilage; however its matrix contains a threadlike network of elastic fibers

Site

- in tissues that need stiffness and elasticity such as;
    1. Ear pinna
    2. Epiglottis
    3. External auditory meatus
    4. Eustachian tube

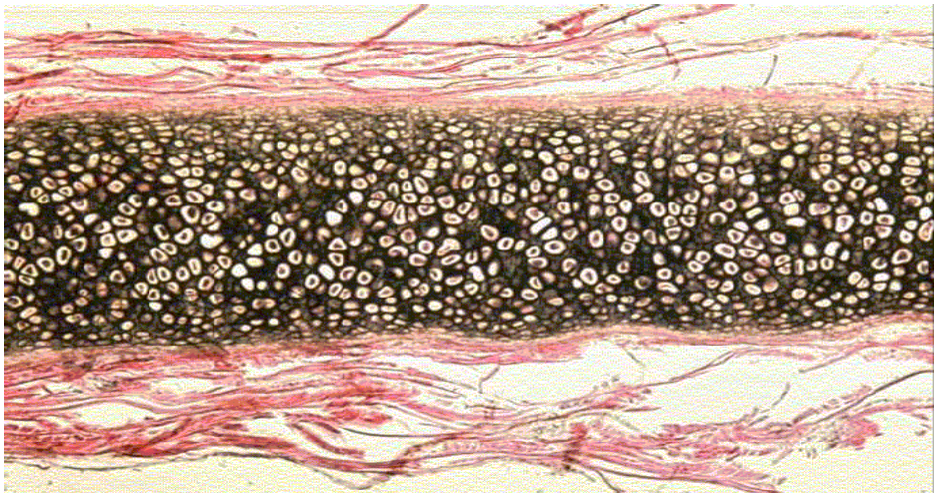


Fig.6: Elastic cartilage by Orcein stain

3 White Fibrocartilage

- It appears white in fresh state

➢ The structure :

- characterized by the presence of groups of chondrocytes arranged in rows, separated by bundles of collagenic fibers **(collagen type 1)**
- It does **not have a perichondrium**

Site

- Symphysis pubis
- Intervertebral discs
- Semilunar cartilage of knee joint
- Around the acetabulum at the tip of glenoid cavity

➢ Growth of cartilage

1. *Interstitial growth "length"*

- Occurs in moderately young cartilage (in which intercellular substance is not too stiff)m within the substance of the cartilage
- Chondrocytes divide and Produce matrix

2. *Appositional growth "in width"*

- From surface perichondrium
- Occurs by the inner chondrogenic layer of the perichondrium



Fig.7: Growth of cartilage

Bone cells, bone matrix, periosteum & endosteum

Bone

- is a specialized dense connective tissue composed of intercellular calcified material, the bone matrix (ground substance and fibers), and four cell

❖ **types**: Osteogenic cells; Osteoblasts, Osteocytes and Osteoclasts.

- In most areas of the body, bone is covered by periosteum and lined from inside by endosteum.

❖ **Bone matrix**: It is composed of organic and inorganic materials.

The organic material	The inorganic material	A layer of water and ions (hydration shell)
• It is composed of: 1. Collagen (type I) bundles. 2. Ground substance in the form of proteoglycan, glycosaminoglycans, osteonectin, and osteocalcin	It is composed of : • Hydroxyapatite, which is essentially identical in composition to a component of geological deposits of calcium phosphate. • Citrate. • Carbonate	• It surrounds each hydro¬xyapatite crystal and is believed to facilitate the exchange of ions between the crystals and the body fluids.

❖ **Bone cells: (Fig. 1)**

1- **Osteoprogenitor cell**
2- **Osteoblast**
3- **Osteocytes**
4- **Osteoclast**

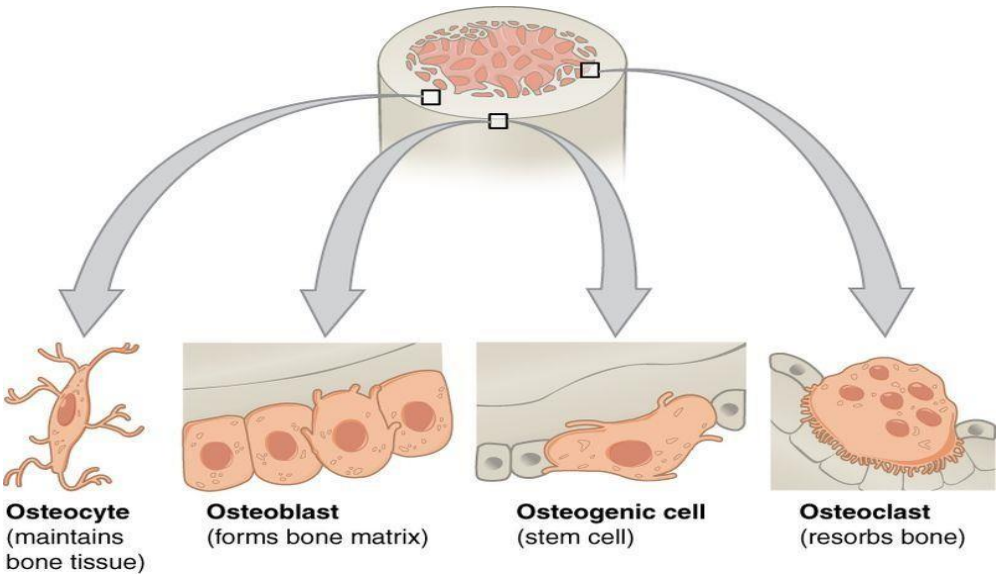


Fig. (1): The four types of bone cells.

Osteoprogenitor cells

- **Origin:**
  - from undifferentiated mesenchymal cells
- **Site:**
  - They are found on or near all the surfaces of bones in the endosteum, the inner layers of periosteum and in the trabeculae of calcifying cartilage at the metaphysis of long bone.
- **L.M:**
  - They are oval cells with pale oval nuclei and scanty, faintly basophilic cytoplasm.
- **E.M:**
  - cytoplasm is rich in free ribosomes and contain centriole
- **Function**
  - Fracture healing

Osteoblasts

- They are lining the layer of bone matrix that they are producing.
- **Origin:** from osteoprogenitor cells
- **L.M:**
  - ❖ They are arranged as a layer of oval or fusiform cells.
  - ❖ Their cytoplasm is highly basophilic.
  - ❖ In well fixed cells, the nuclei appear irregularly rounded or fusiform and their nucleoli are eccentric in position.
  - ❖ The cytoplasm shows negative Golgi region close to the nucleus.
- **E.M (Fig.2)**
  - ❖ It has the appearance of protein secreting cells.
  - ❖ It contains numerous RER, well developed Golgi and secretory vesicles filled with procollagen.
  - ❖ The nucleus is present at one side and the RER is at the other side and the Golgi is present between them.

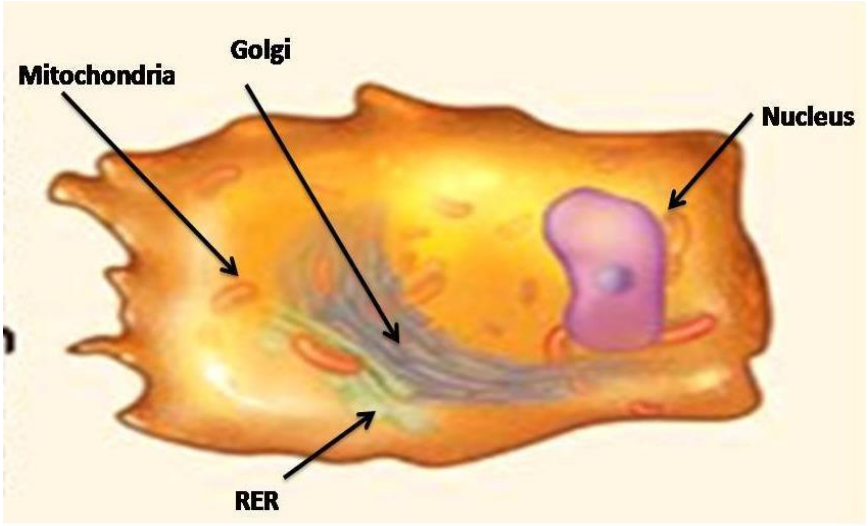


Fig. (2) EM of osteoblast

- ❖ These cells are able to proliferate and differentiate into osteoblasts.
- ❖ Osteoprogenitor cells are active during the growth of bone and repair of bone after its Matrix vesicles contain alkaline phosphatase enzyme

- ❖ The outline is irregular due to the presence of cellular processes. They communicate with each other by gap junctions.

- **Function:**
  - ❖ They are bone forming cells of developing and mature bone. It secrets the component of bone matrix and has a role in the mineralization. Matrix vesicles cleave the phosphate groups and acts as the foci for calcium and phosphate deposition.

Osteocytes

Once the osteoblast becomes surrounded with intercellular substance it is called osteocyte.

- **Origin:** from osteoblast
- **L M:**
  - ❖ The spaces they occupy are known as lacunae
  - ❖ They are oval branched cells with pale basophilic cytoplasm.
  - ❖ As the bone matrix is hard, osteocytes extend cytoplasmic processes that traverse the matrix through canaliculi that reach to osteoblasts and other osteocytes for communication by gap junction.
- **E.M: (Fig. 3)**
  - ❖ A young osteocyte has most of the ultrastructural contents of the osteoblast, but less in the organelles involved in protein synthesis (rough endoplasmic reticulum, Golgi).
  - ❖ An older osteocyte, located deeper within the calcified bone, shows a further decrease in cell volume and organelles with accumulation of glycogen in the cytoplasm.
- **Function:**
  - ❖ Bone maintaining cells (organic and inorganic matrix)

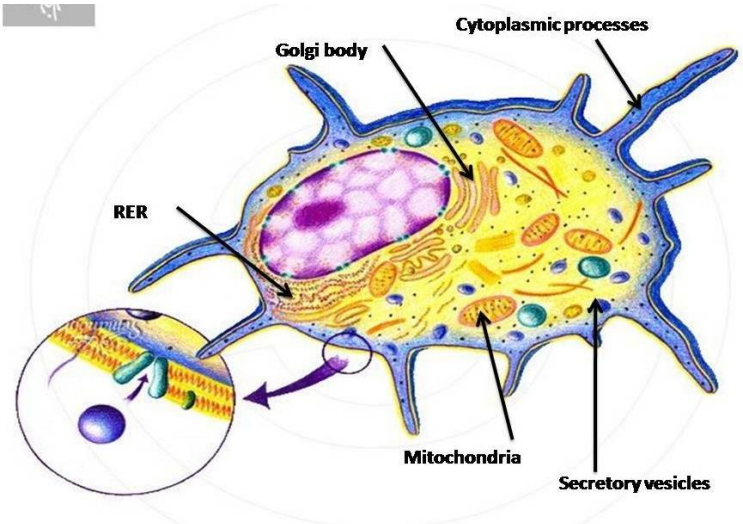


Fig. (3): A diagram of osteocyte with its cytoplasmic processes

Osteoclasts

- **Origin:**
  Fusion of monocytes
- **Site:**
  occupy pits called Howship's lacuna at the free surface of periosteum and endosteom
- **L.M:**
  - They are large multinucleated cells containing 6-12 nuclei.
  - The cytoplasm is acidophilic with frothy appearance and the acidophilia is increased with age.
  - Osteoclasts have a striated border which is at the surface exposed to bone and appears as hair like processes.
- **E.M:- Fig.(4)**
  - The striated or ruffled border of osteoclasts demonstrates irregular villi-like processes.
  - The cytoplasm at the striated or ruffled border is free from organelles but between this and the nuclei numerous mitochondria, well developed Golgi stacks, RER, numerous lysosomes, large pale vesicles and small coated vesicles are present.

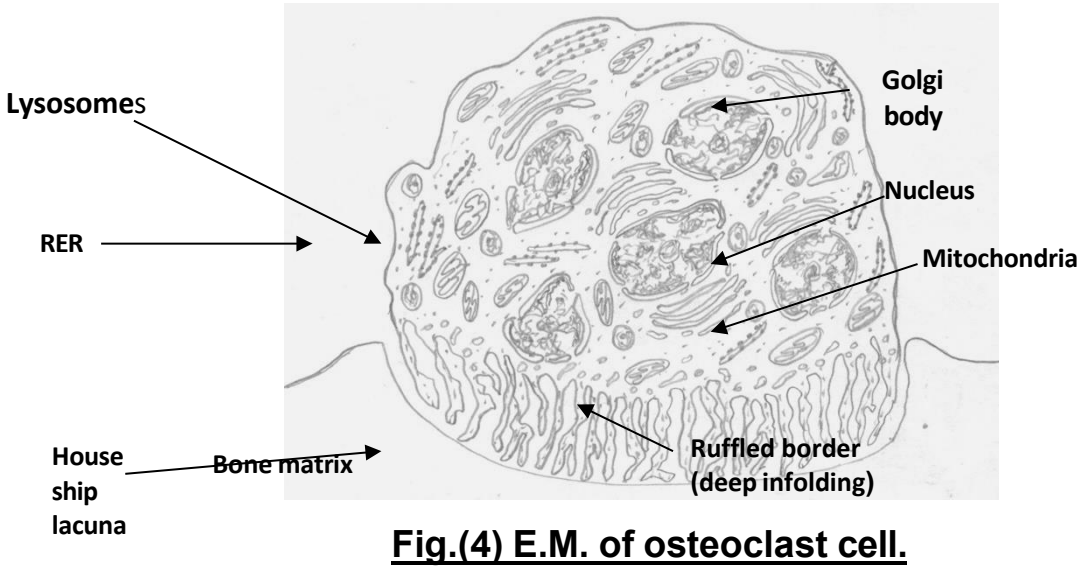


Fig.(4) E.M. of osteoclast cell.

- **Function of osteoclast:**

1-Bone resorption

- Osteoclast creates acidic media in the adjacent matrix substance to dissolve calcium (decalcification).
- The hydrolytic enzymes in lysosomes destroy the organic part of matrix.

2-Bone remodeling and repair

- The remodeling means the maintenance of the shape of the bone through the removal of old bone tissue, as well as fractured one by osteoclast followed by adding new bone matrix by osteoblast that subsequently become mineralized.
- This serves to adjust bone architecture in response to the mechanical needs and also helps to repair damaged bone. It also plays an important role in maintaining plasma calcium homeostasis.

Periosteum and Endosteum

:

The outer and inner surfaces of bone are covered by layers of bone-forming cells and connective tissue called periosteum and endosteum.

The periosteum	The endosteum
It lines the outer surface of the bone and consists of: • An outer fibrous layer is dense regular C.T composed of collagen fibers type I, fibroblasts and blood vessels. • Inner cellular layer composed of osteoprogenitor cells.	• It lines inner cavities within the bone and is composed of a single layer of flattened osteoprogenitor cells and a very small amount of connective tissue.

Types of bone ,Bone growth, remodeling &repair

Types of bone: Bone is organized in two histological forms:

1. Compact bone. 2-Cancellous bone

Compact bone

Sites:

-Shaft of long bones
-The outer part of the epiphysis
-The outer part of the short and flat bones.

Structure of compact bone (Fig.1):

- ✓ The outer surface of compact bone is covered with periosteum and bone marrow cavity is lined with endosteum.
- ✓ It consists of osteons or Haversian systems.
- ✓ Each Haversian system is composed of a Haversian canal surrounded by regularly arranged concentric lamellae of bone. Between concentric lamellae, there are interstitial lamellae. Under the periosteum and endosteum the lamellae of bone are called the outer and inner circumferential lamellae respectively.

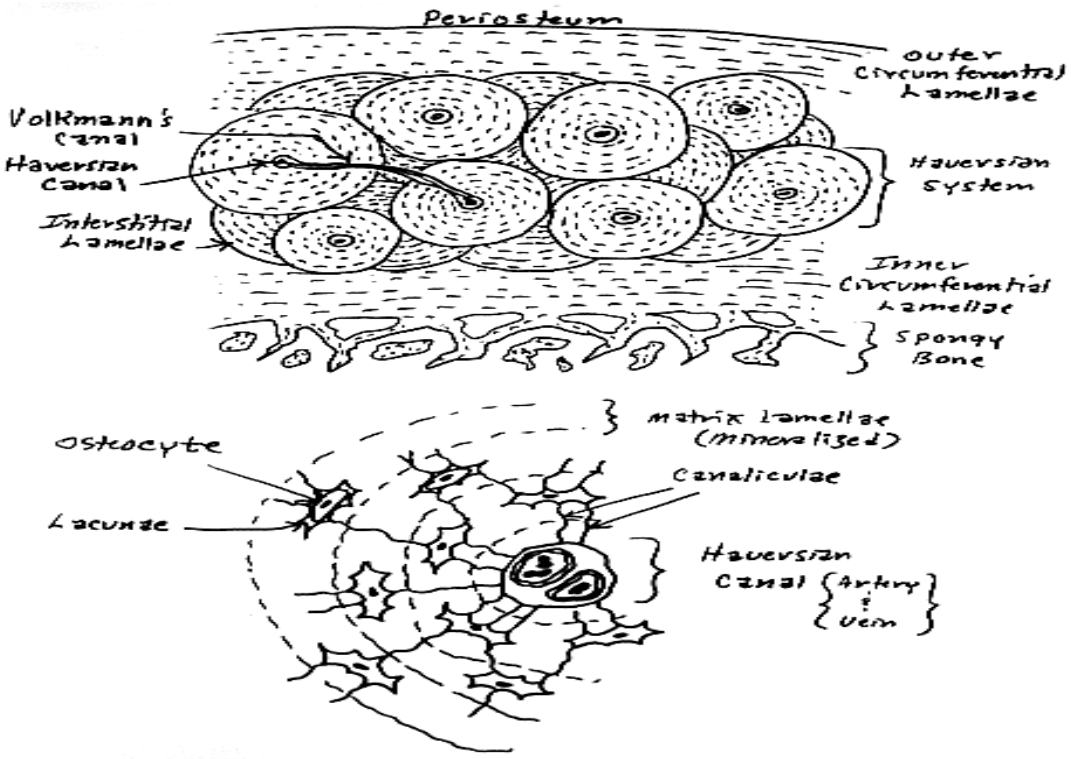


Fig.(1): Histolgical structure of compact bone

Cancellous bone

Sites:

- In the center of flat bone and irregular bones.
- At the end of long bones.

Structure of cancellous bone (Fig.2)

- ✓ Cancellous bone is composed of irregular trabeculae of bone separated by bone marrow spaces.
- ✓ The bone trabeculae are composed of osteocytes and intercellular substance and covered with endosteum.
- ✓ These are no Haversian systems.

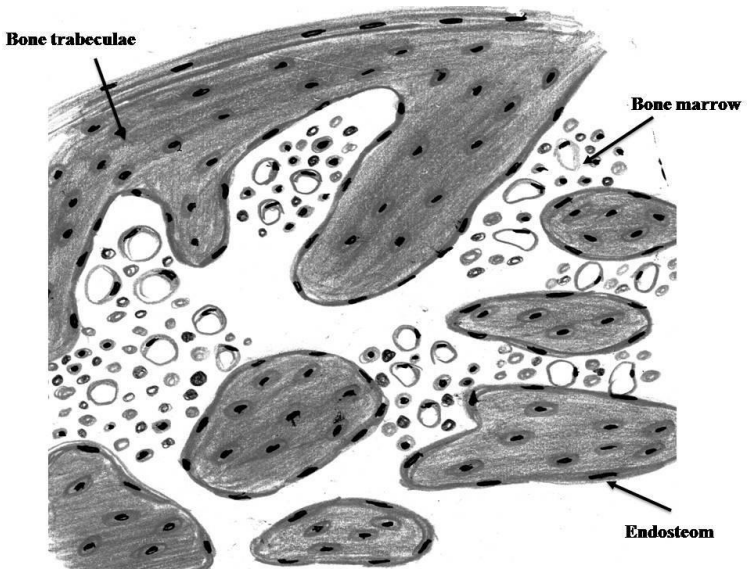


Fig.(2):Cancellous bone

Histogenesis of Bone (Bone formation): Bone can be formed in two ways:

1- Direct mineralization of matrix secreted by osteoblast **(Intramembranous ossification).**
2- Deposition of bone matrix on a preexisting cartilage matrix (**Endochondral ossification).**

Intramembranous Ossification: (Fig.3)

Site

Most of the flat bones as skull, clavicle and most of cranial bone)

Steps

1- A primary center of ossification appears in the mesenchymal C.T. membrane rich in blood supply.
2- Mesenchymal cells differentiation into osteoprogenitor cells that give osteoblasts.

3- Osteoblasts produce osteoid (bone matrix) and calcification follows, gradually it encircles osteoblasts and become osteocytes and the bone formed is of spongy type.
4- The connective tissue that remains among the bone walls is penetrated by growing blood vessels and the undifferentiated mesenchymal cells, giving rise to the bone marrow cells.
5- The portion of the connective tissue layers that covers the trabeculae gives rise to the endosteum and the periosteum.

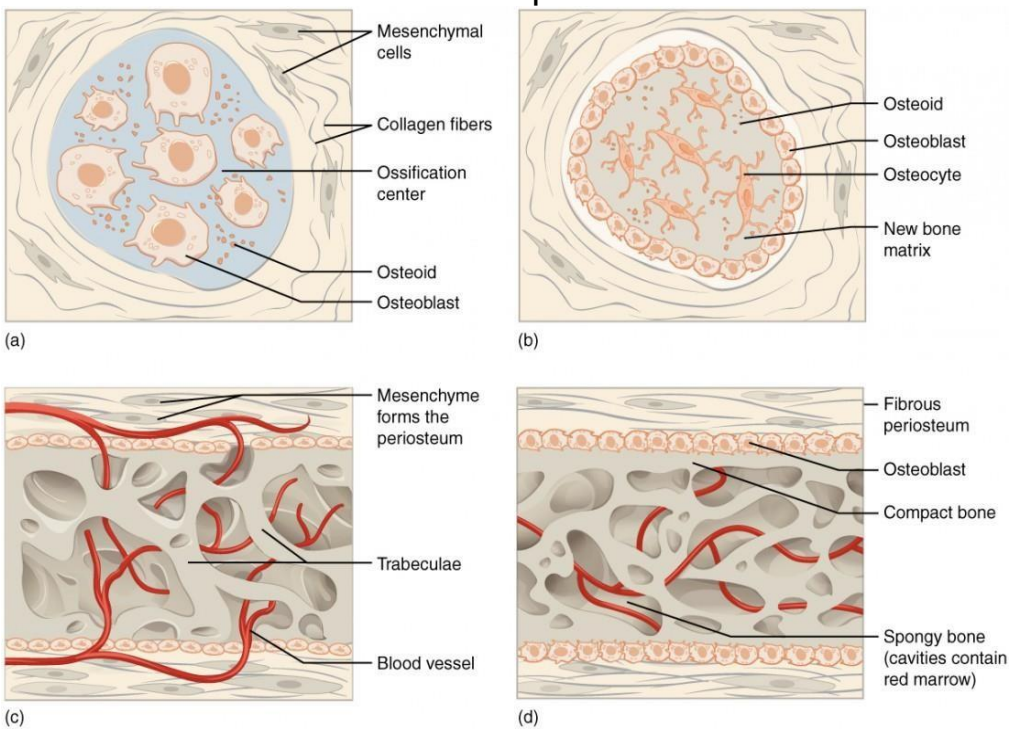


Fig. (3): Steps of intramembranous ossification

Endochondral Ossification: (Fig. 4)

Site

Endochondral ossification takes place within a piece of hyaline cartilage which has a shape resembling a model of the bone to be formed. This type of ossification is principally responsible for the formation of short and long bones.

Steps

The cartilage model consists of two **epiphyses** and a **diaphysis**.

1- A primary center of ossification arises in the middle of the shaft of the cartilage model (diaphysis).
2- Initially, the first bone tissue appears as a hollow bone cylinder that surrounds the mid portion of the cartilage model forming the bone collar.
3- The cartilage undergoes a degenerative process and cell death resulting in the formation of calcified cartilage matrix.
4- The nutrient artery from the periosteum invade the calcified cartilage through holes made by osteoclasts in the bone collar.

5- The mesenchymal cells give osteoprogenitor cells that differentiate into osteoblasts forming bone plate and also give bone marrow cells then cancellous bone develops.
6- Later on secondary centers of ossification occur after birth at the epiphyses. The ossification continues of whole bone except at two sites where hyaline cartilage persists; one is articular cartilage covering the articular surface of bone. The second site is at the junction between the epiphysis and diaphysis called epiphyseal plate.

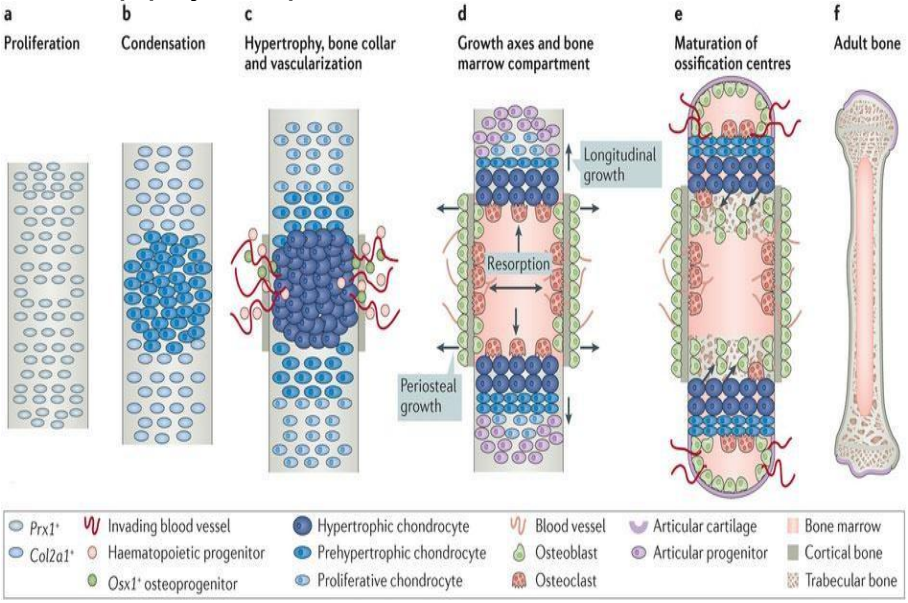


Fig.(4) Endochondral bone growth

Growth in length of bone: - (Fig. 5)

- Before the adulthood, the interstitial growth of the cartilage of epiphyseal plate continues to increase the length of long bone until the age of 20 years in males and 17 years in females. After that age the epiphysel cartilage disappear (closure of epiphysel line)
- The chondrocytes undergo changes similar to the early events in the primary center of endochondral ossification. Five zones become recognizable along their length.

(1) **Zone of resting cartilage**.
(2) **Zone of proliferation**: The chondrocytes divide several times and they are arranged in rows parallel to the longitudinal axis of the long bone.
(3) **Zone of Hypertrophy**: The chondrocytes increase in size and become rich in alkaline phosphatase enzyme vesicle.
(4) **Zone of Calcification**: Calcium salts are deposited (in the matrix) around the chondrocytes by the activity of phosphatase enzyme. The calcified matrix cuts off the nourishment, to the chondrocytes which die and degenerate leaving empty spaces.

(5) **Zone of Ossification**: Blood capillaries and mesenchymal cells penetrate the spaces left after the destruction of chondrocytes. Then bone matrix and bone marrow cells develop.

NOTE

Notice that the bone tissue is formed at the site of cartilage. Thus cartilage .is not transformed into bone but is rather replaced by bone

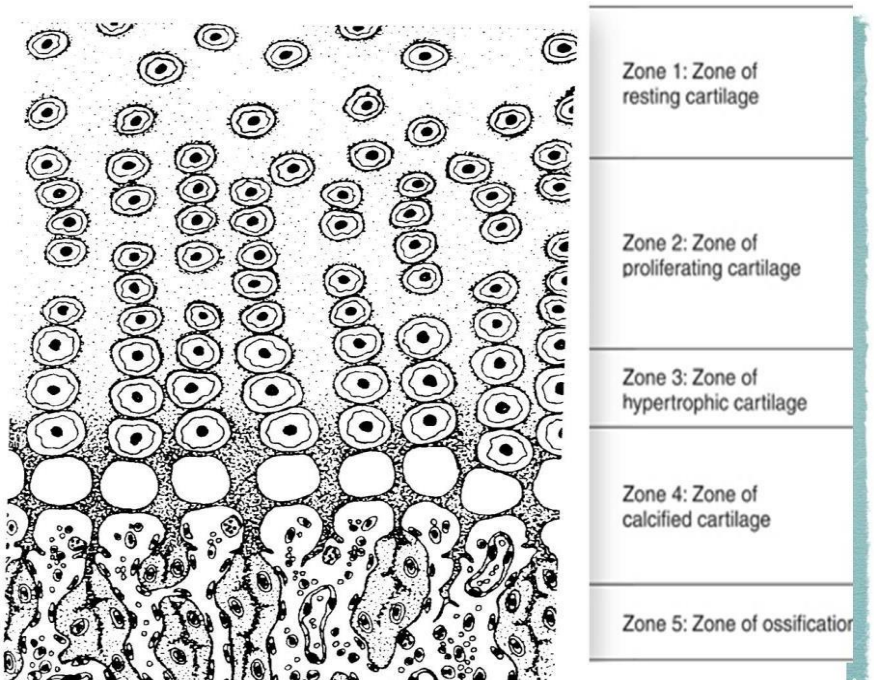


Fig. (5) Stages of ossification in the epiphyseal plate

- **Bone remodeling and repair: (previously discussed)**

References

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- Wheater's Functional Histology: A Text and Colour Atlas
- Textbook of Histology: Gartner